

A	23-06-2015	Final issue – As Commissioned	ABB AG	PPa
-	21-10-2014	FIRST ISSUE – For Installation	ABB AG	PPa
Rev.	Date	Description	Prepared	Approved

TP Aeolian Dynamics Wind Farm, 10.8 MW, at Agia Anna

Design, Manufacturing, Supply, Supervision of Erection and Commissioning of the MV Switchgear and its associated Control Equipment, Telecommunication Equipment, Load Flow & Short Circuit Study and Earthing Study of the 10.8MW Wind Farm in Ag. Anna, Larnaka, Cyprus.

Project: TP Aeolian Wind Farm 10,8MW				
Order No: P181114016	Contract No: -	Substation: Agia Anna S/S	Format: A4P	
Prepared: ABB AG 23-06-2015	Responsible Department: KE	Doc.Kind: Circuit Diagram	Language: en	
Approved: PPa 23-06-2015	Take Over Department: -	Title: MV Switchgear drawings	DCC: & EFS	
	Derived from: Replaces:	Item: =J01,=J02,=J03		
 ABB SA		Doc.No: 3VGR111416F003	Rev.: A	Sh/No.Sh 1/205

REVISION INDEX

Rev. ind.	Page (P) Chapt.(C)	Description	Date Dept./Init.
-		For Installation	21-10-2014 ABB AG
A		As Installed	23-06-2015 ABB AG

Prepared: ABB AG 23-06-2015	Responsible Department: KE	Doc.Kind: Circuit Diagram	
Approved: PPa 23-06-2015	Take Over Department: -	Title: MV Switchgear drawings	Language: en
Revision: A	Derived from: Replaces:	Item: =J01,=J02,=J03	DCC: & EFS
 ABB SA		Doc.No: 3VGR111416F003	Rev.: A
			Sh/No.Sh 2/205

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Prepared: ABB AG 23-06-2015	Responsible Department: KE	Doc. Kind: Circuit Diagram	
Approved: PPa 23-06-2015	Take Over Department: -	Title: MV Switchgear drawings	Language: en
Revision: A	Derived from: Replaces:	Item: =J01,=J02,=J03	DCC: & EFS
 ABB SA		Doc.No: 3VGR111416F003	Rev.: A
			Sh/No.Sh 3/205

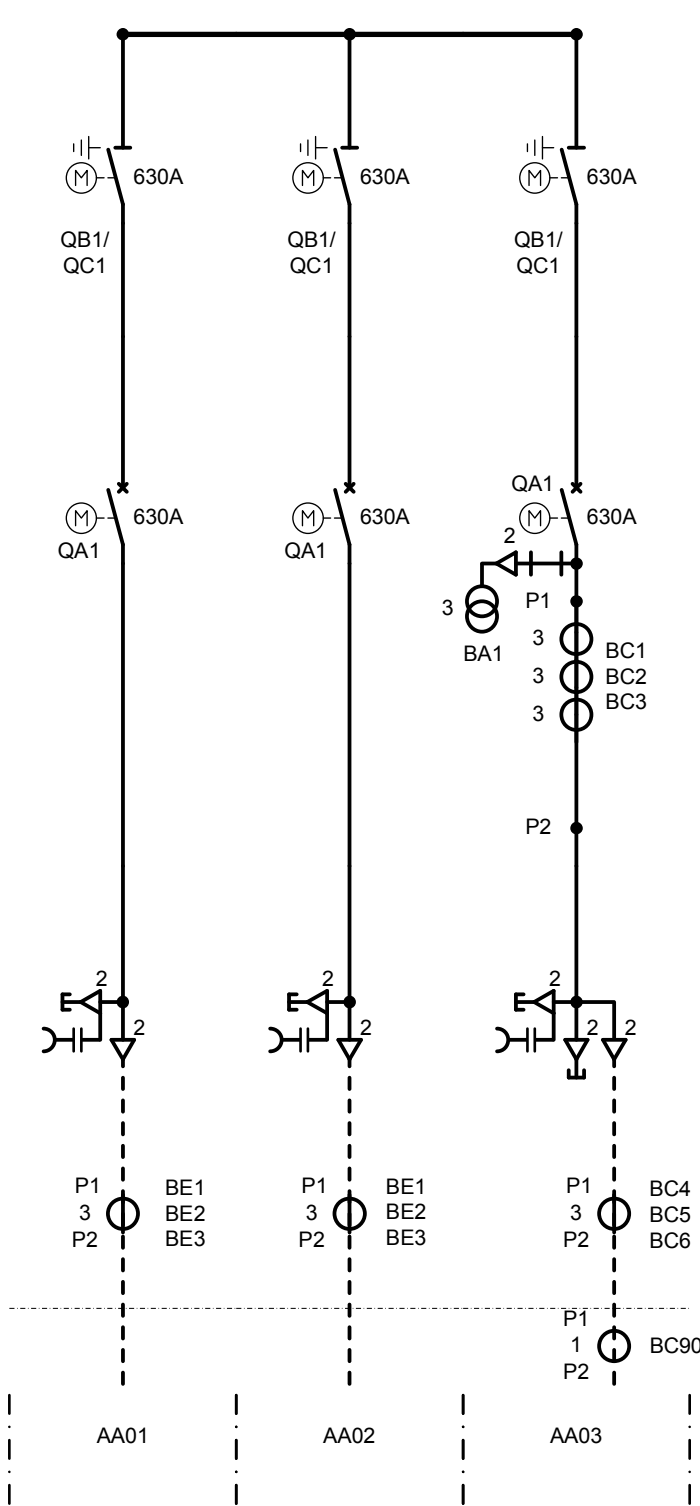
1. General

This document contains the MV Switchboard drawings for TP Aeolian Dynamics Wind Farm, 10.8 MW, at Agia Anna according to the specifications of the client.

2. Reference Documents

- 1VB2383201_SWD_001_03: Single line diagram
- 1VB2383202R0101_SWD_001_00: Front view
- 1VB2383203R0101_SWD_001_00: Civil work instruction
- 1VB2383211R0101_SWD_001_00: Assembly pressure relief duct
- 1VB2383216: BUS WIRES AC / DC VOLTAGE SUPPLY
- 1VB2 3832 14: BB-VOLTAGE
- 1VB2 3832 18: KEY OF VARIANTS
- 1VB2383221: UNDERGROUND FEEDER =J01
- 1VB2383222: UNDERGROUND FEEDER =J02
- 1VB2383223: INCOMER =J03

Prepared: ABB AG 23-06-2015	Responsible Department: KE	Doc. Kind: Circuit Diagram		
Approved: PPa 23-06-2015	Take Over Department: -	Title: MV Switchgear drawings		Language: en
Revision: A	Derived from: Replaces:		Item: =J01,=J02,=J03	DCC: & EFS
 ABB SA		Doc.No: 3VGR111416F003	Rev.: A	Sh/No.Sh 4/205



BASEMENT

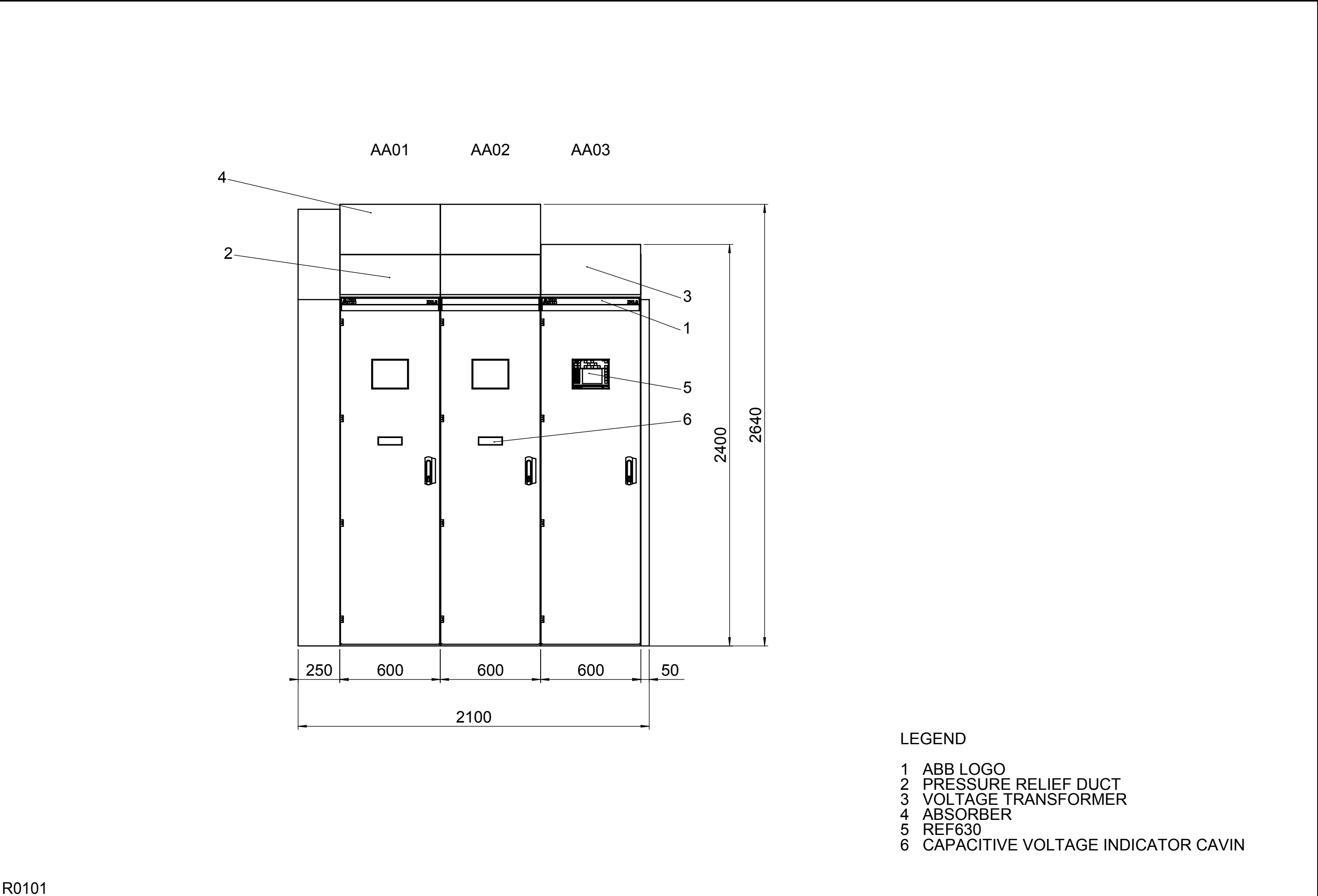
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PANEL NO.	VOLTAGE TRANSF.	RATIO	OUTPUT	TYPE
AA03	---	$\frac{22}{\sqrt{3}} / \frac{0.11}{\sqrt{3}} / \frac{0.11}{\sqrt{3}} / \frac{0.11}{\sqrt{3}}$ kV	15VA, CL. 0,5 15VA, CL. 0,5 10VA, CL. 3P	VEG 24-20

ABB		IEC 62271-200	
ZX1.2	GIS Switchgear	2014	
U_f	24 kV	U_s	22 kV
U_d	50 kV	U_p	125 kV
I_{busbar}	1250 A	at	40 °C
I_k	25 kA	t_k	3 s
I_p	63 kA	f_r	50 Hz
U_a	110 V DC	Altitude	< 1000 m
Pressure for insulation/abs.			
P_{re}	130 kPa		
P_{ae}	120 kPa		
P_{me}	120 kPa		
IAC	AFLR	25 kA	1 s

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- LEGEND
- 1 ABB LOGO
 - 2 PRESSURE RELIEF DUCT
 - 3 VOLTAGE TRANSFORMER
 - 4 ABSORBER
 - 5 REF630
 - 6 CAPACITIVE VOLTAGE INDICATOR CAVIN

Prep./Erst.	14.10.2014	Zenses	ABB S.A. TP Aeolian 24kV GIS TYPE ZX1.2	ABB AG Calor Emag Mittelspannungsprodukte Scale/Maßstab : 1:20	Resp. Dept./Zust. St.	FRONT VIEW					=	A3
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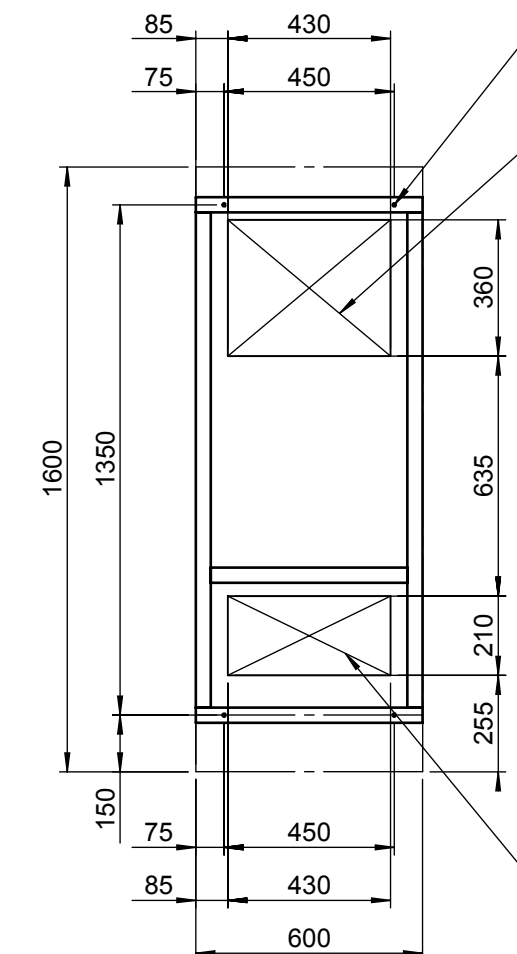
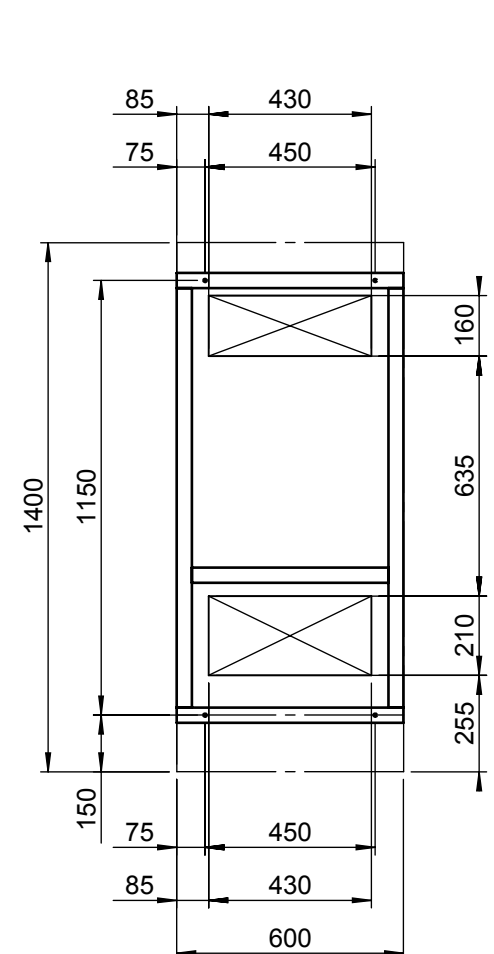
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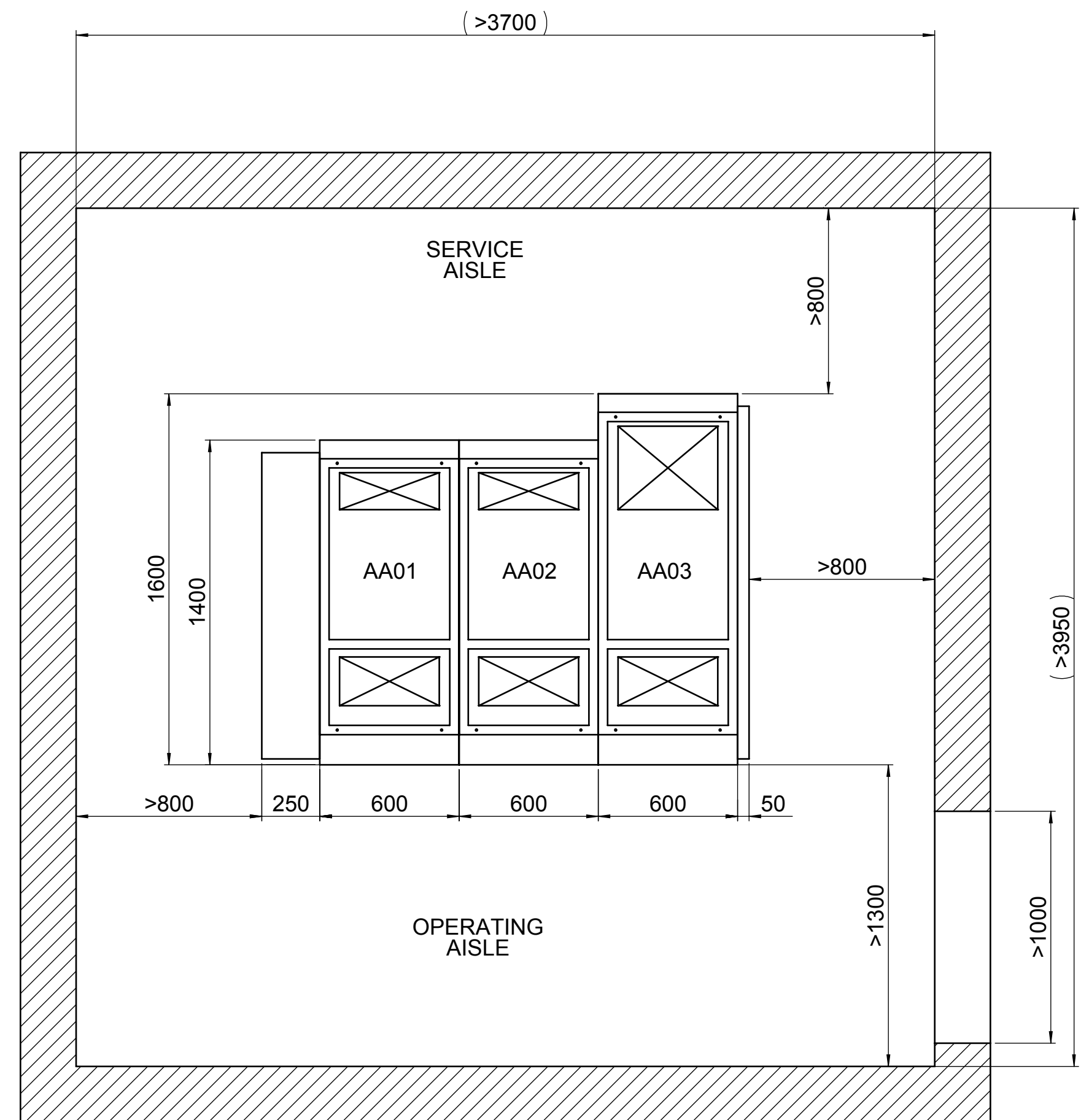
R0101



FIXING POINT M8

OPENING FOR POWER CABLE

OPENING FOR CONTROL CABLE



MIN. HEIGHT OF THE SWITCHGEAR ROOM: 2950MM

Consult the attached file for further information:
General Building Information ZX1.2 english.ZIP
► Technical catalogue TK 501-09 E.PDF / Chapter 10 Building planning
► Manual for installation and operation HB 601-03 E.PDF / Chapter 2.2.1 Installation of the foundation frame

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Rev./Änd./Ind.	00	

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24kV GIS TYPE ZX1.2

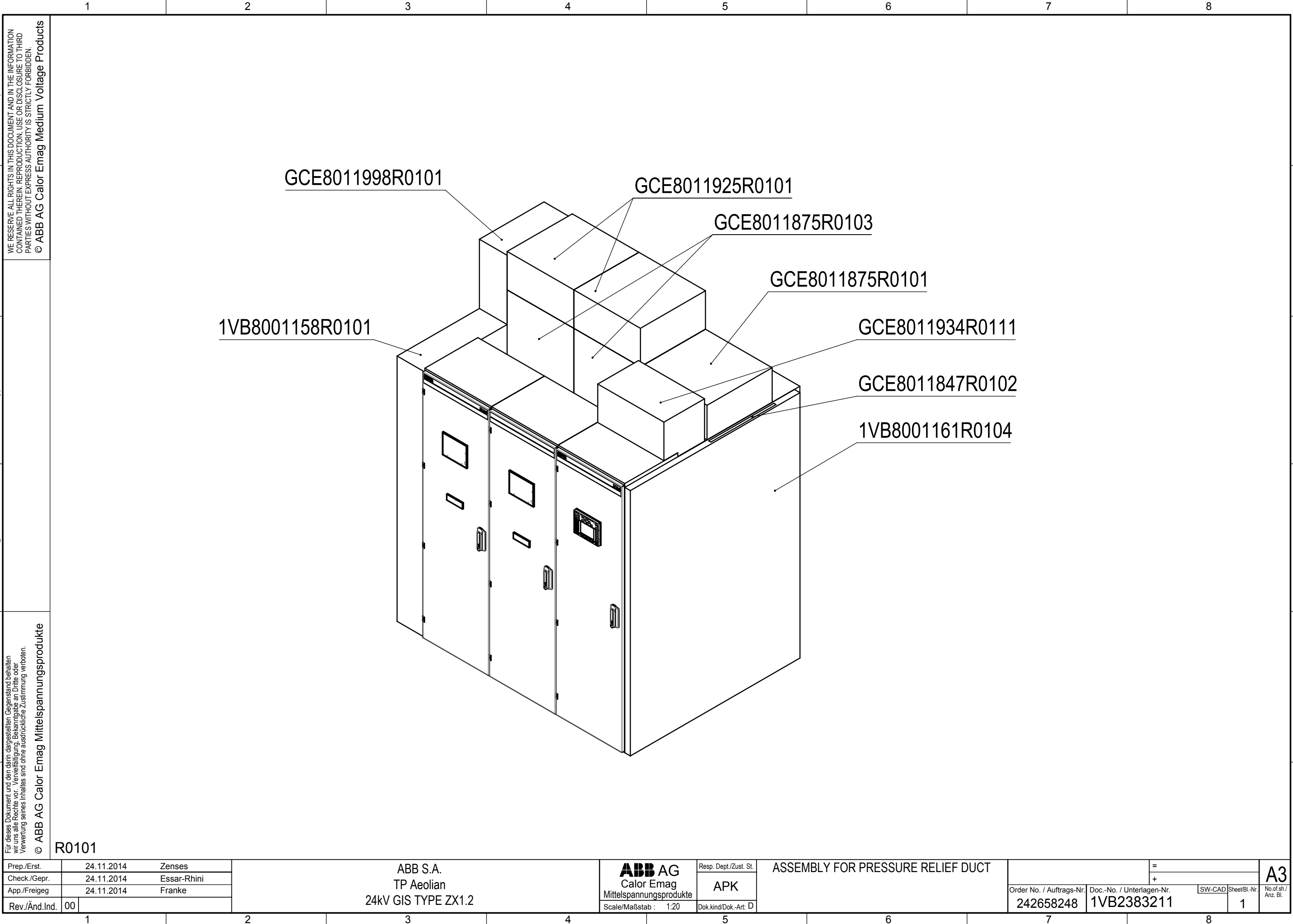
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Calor Emag
Mittelspannungsprodukte
Scale/Maßstab : 1:20

Resp. Dept./Zust. St.
APK
Dok.kind/Dok.-Art: D

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Rev./Änd.Ind.	00						242658248	1VB2383211	1	



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Fax: (02102) 12-1777

ABB SA

T.P. AEOLIAN DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

BUS WIRES AC/DC VOLTAGE SUPPLY

DEACE CONTRACT NO. : 242658248

DEACE DRAWING NO. : 1VB2383216

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Check :	HAMPEL
Repl.:	

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No. Revision

19. 1.15
Date

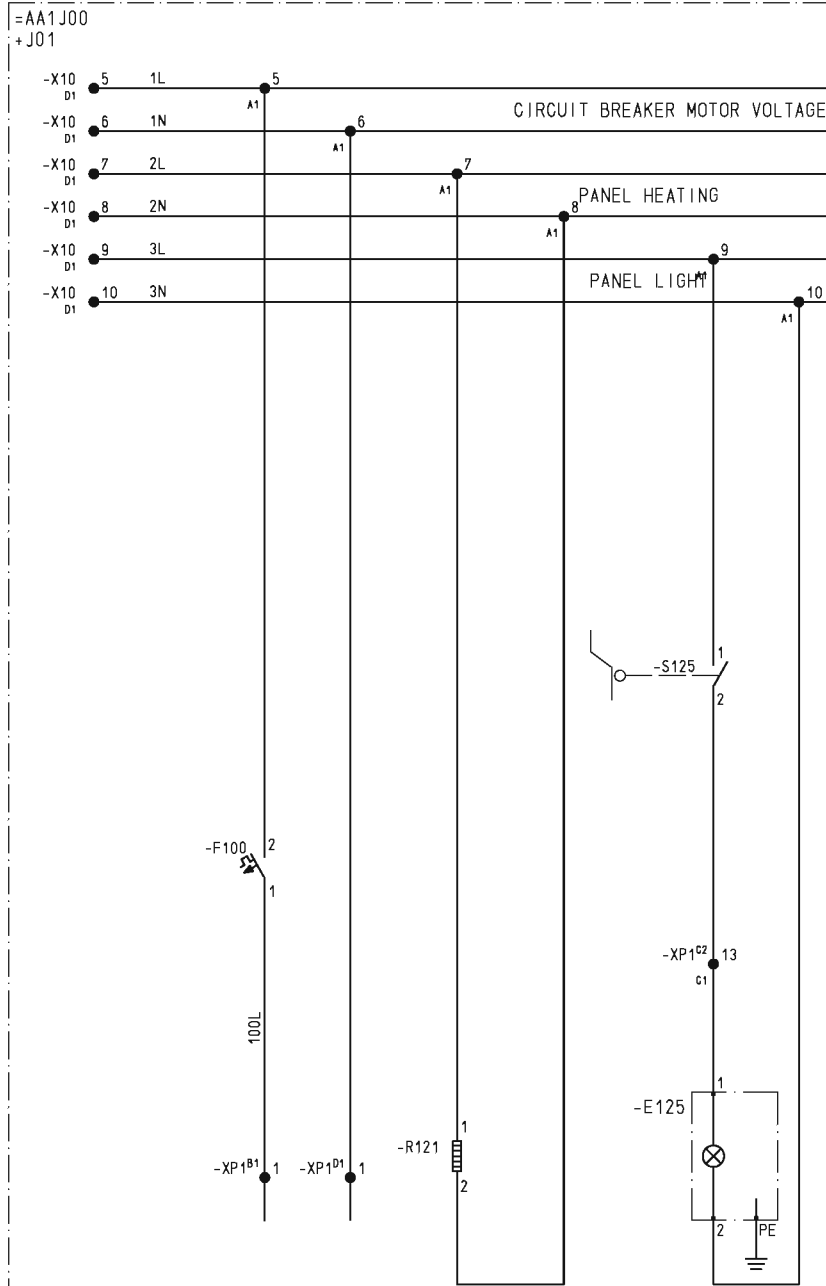
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AP22
RUPLAN 4.61

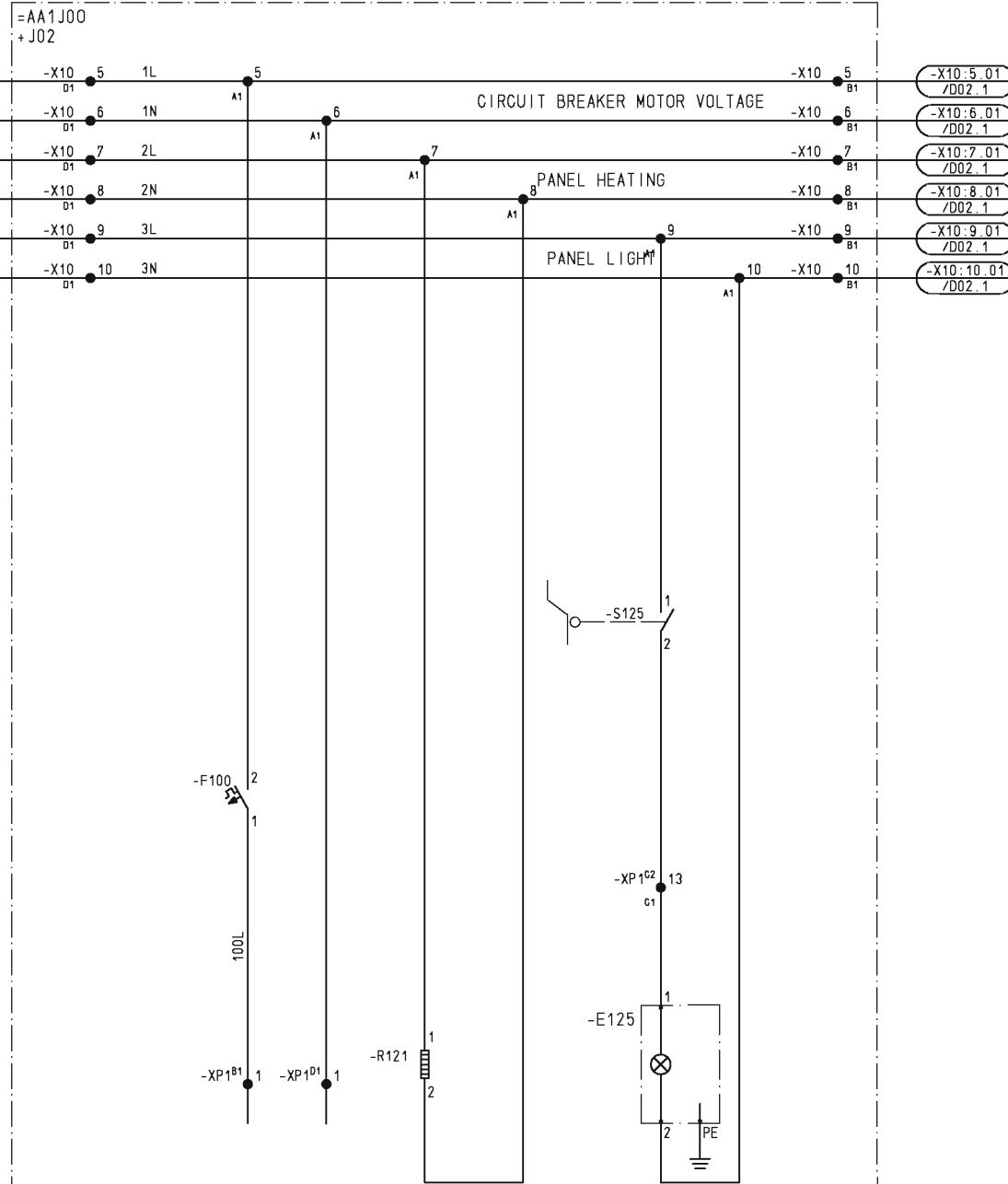
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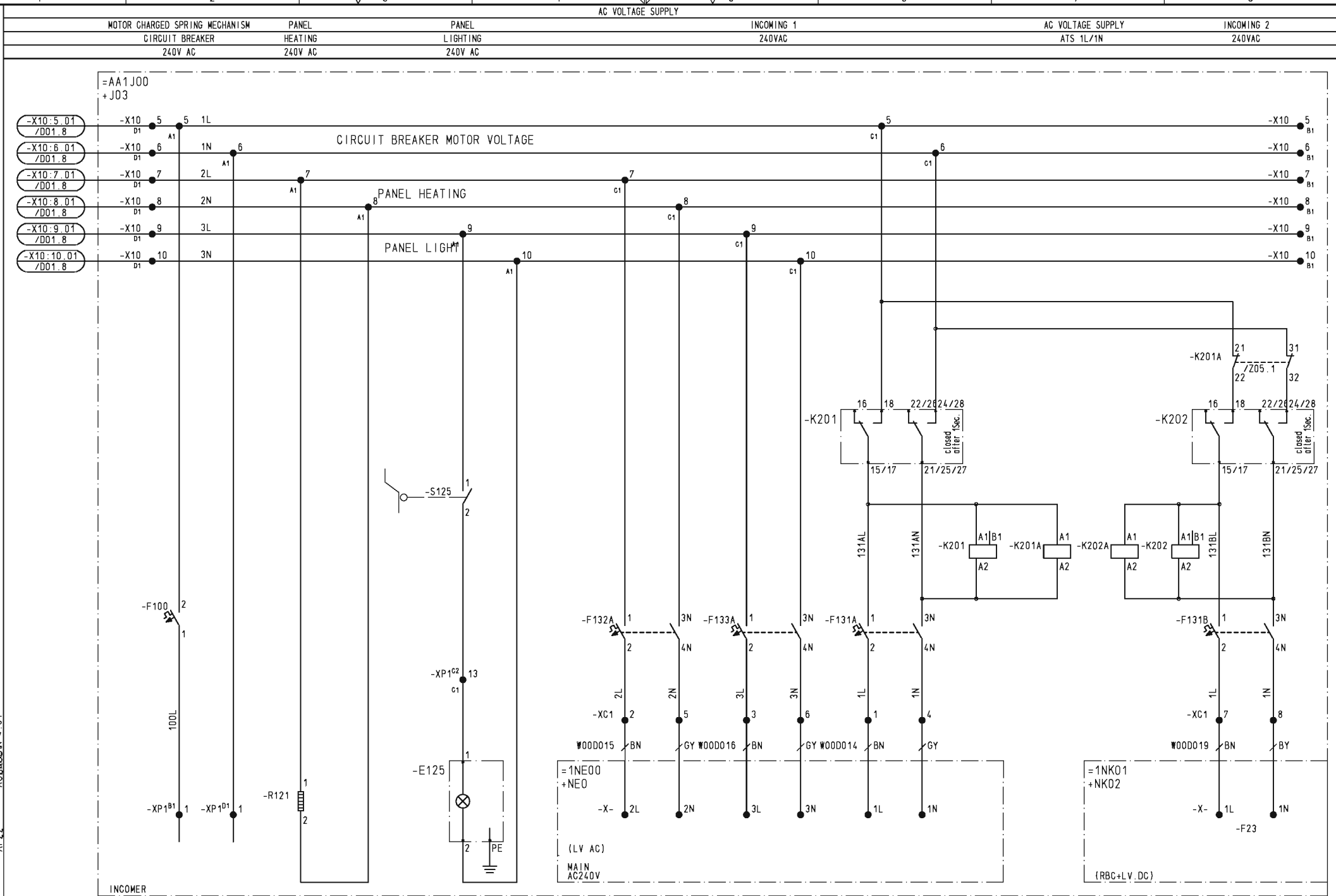


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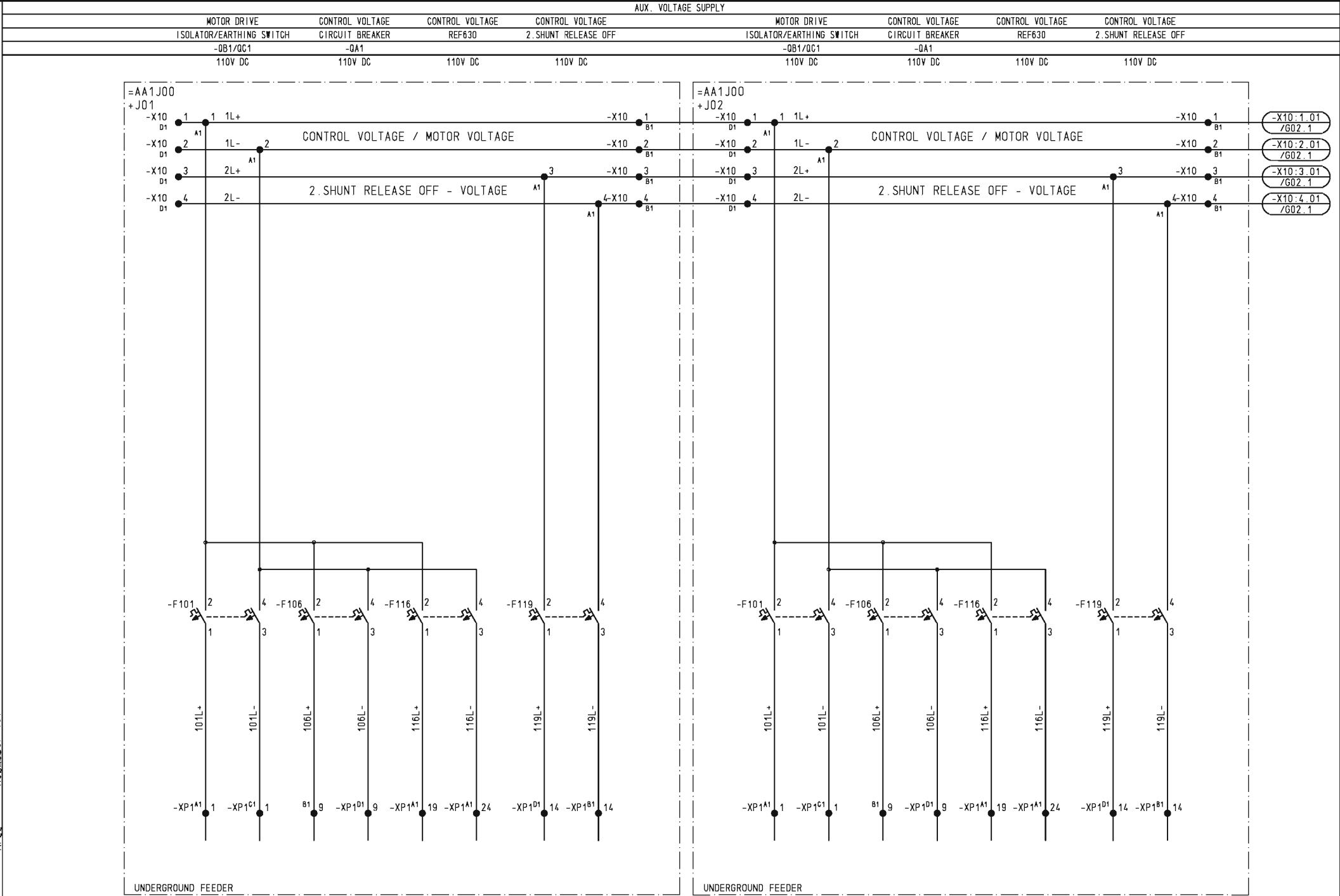


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ACBDDCN 4.61



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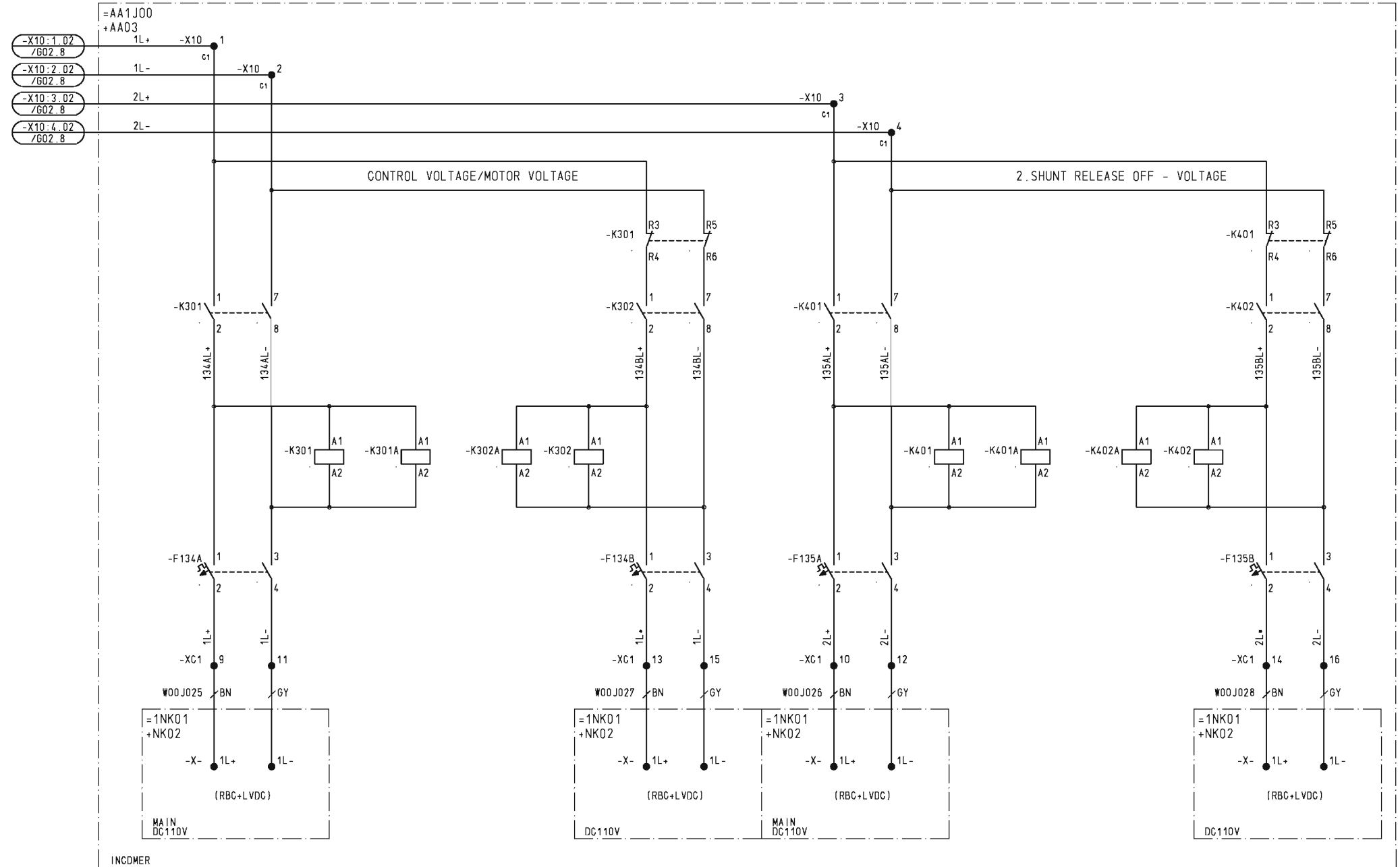


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DC VOLTAGE SUPPLY

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/ G03



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T.P. AEOLIAN

DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

BB-VOLTAGE

DEACE CONTRACT NO. : 242658248

DEACE DRAWING NO. : 1VB2383214

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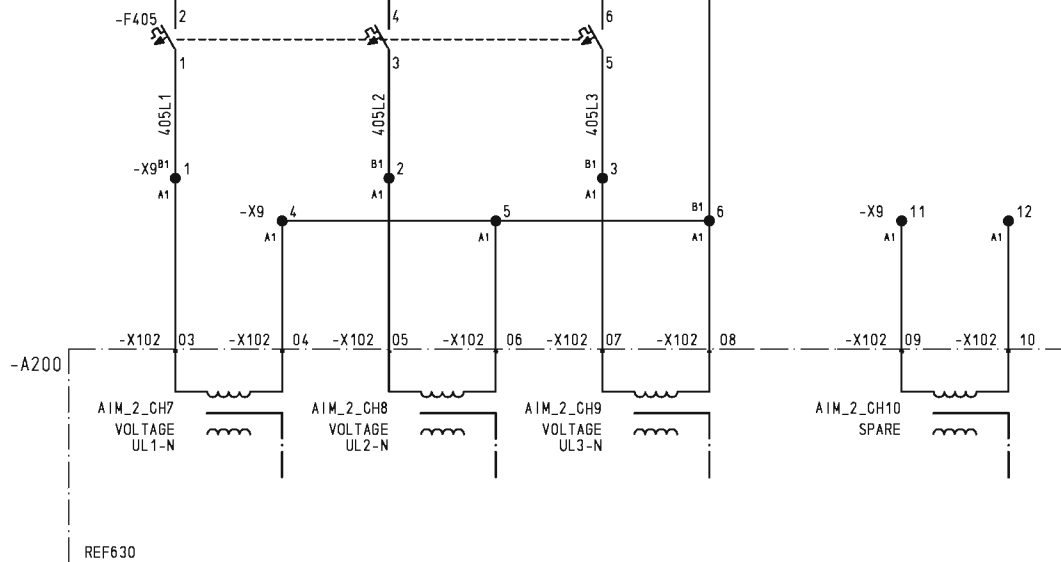
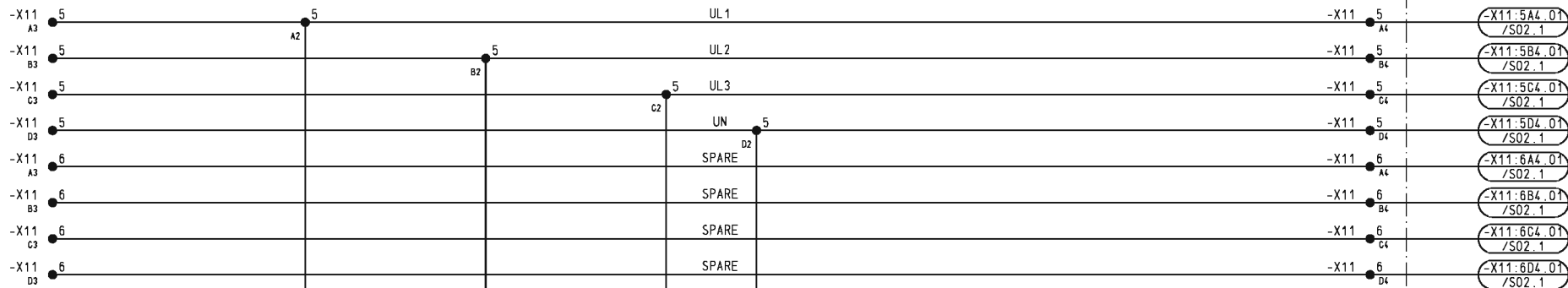
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ABBPLAN 4.61

=AA1 J00
+ J01



UNDERGROUND FEEDER

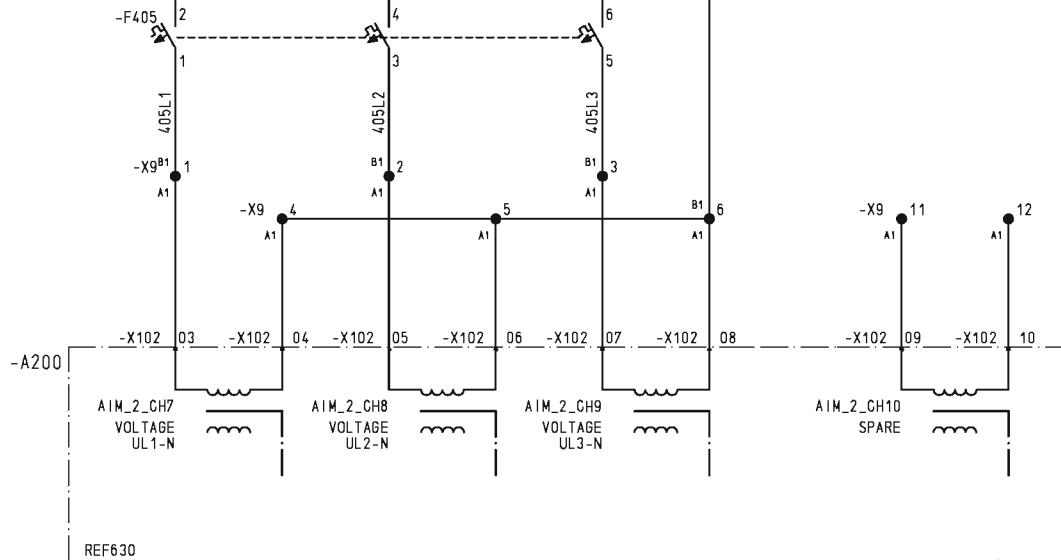
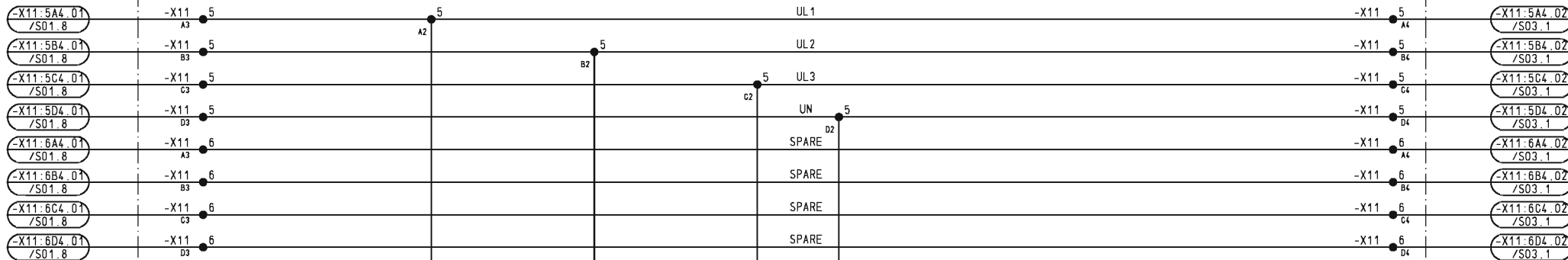
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=AA1 J00
+ J02



UNDERGROUND FEEDER

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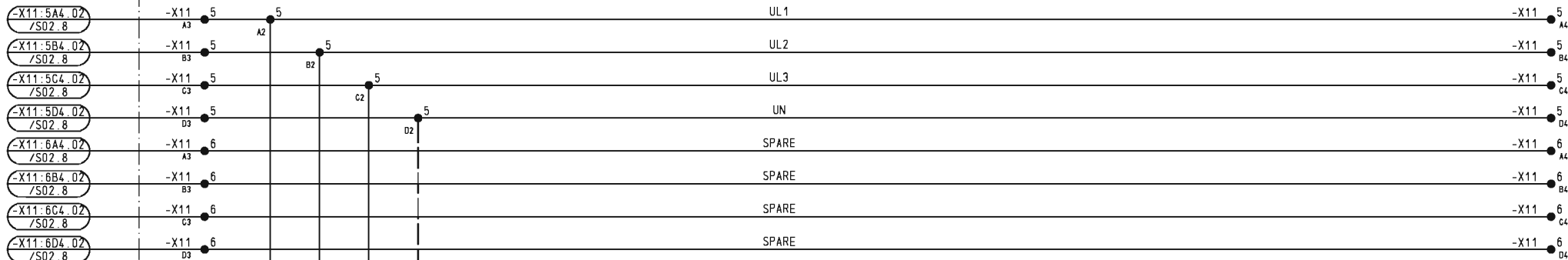
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19.01.2015
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ABBDDN 4.61

=AA1 J00
+ J03



INCOMER

REF630

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19.01.2015

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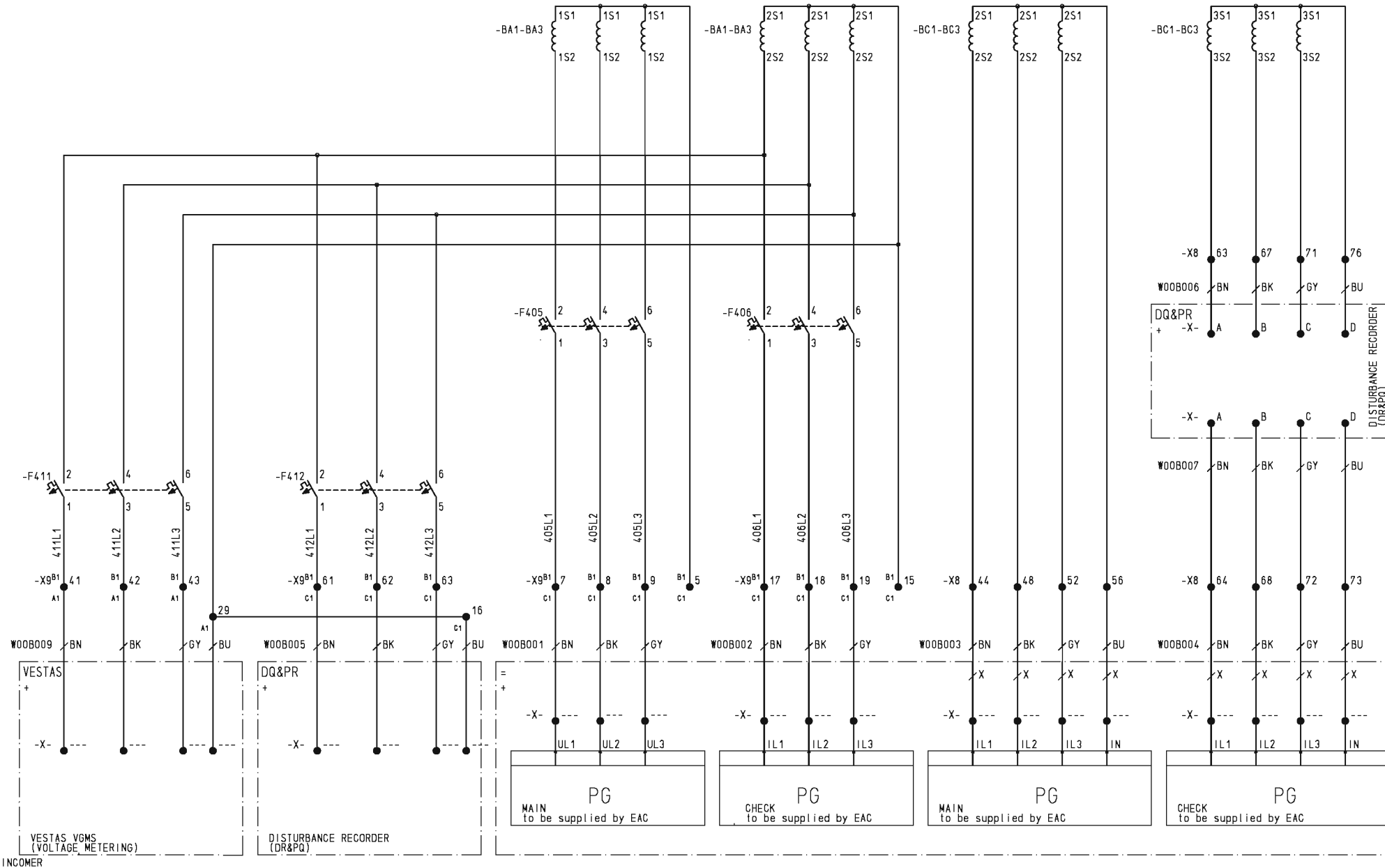
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ABB Color Ring Medium Voltage Products

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Repl.by:



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T.P. AEOLIAN

DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

KEY OF VARIANTS

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19. 1.15
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Fax: (02102) 12-1777

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T.P. AEOLIAN

DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

UNDERGROUND FEEDER

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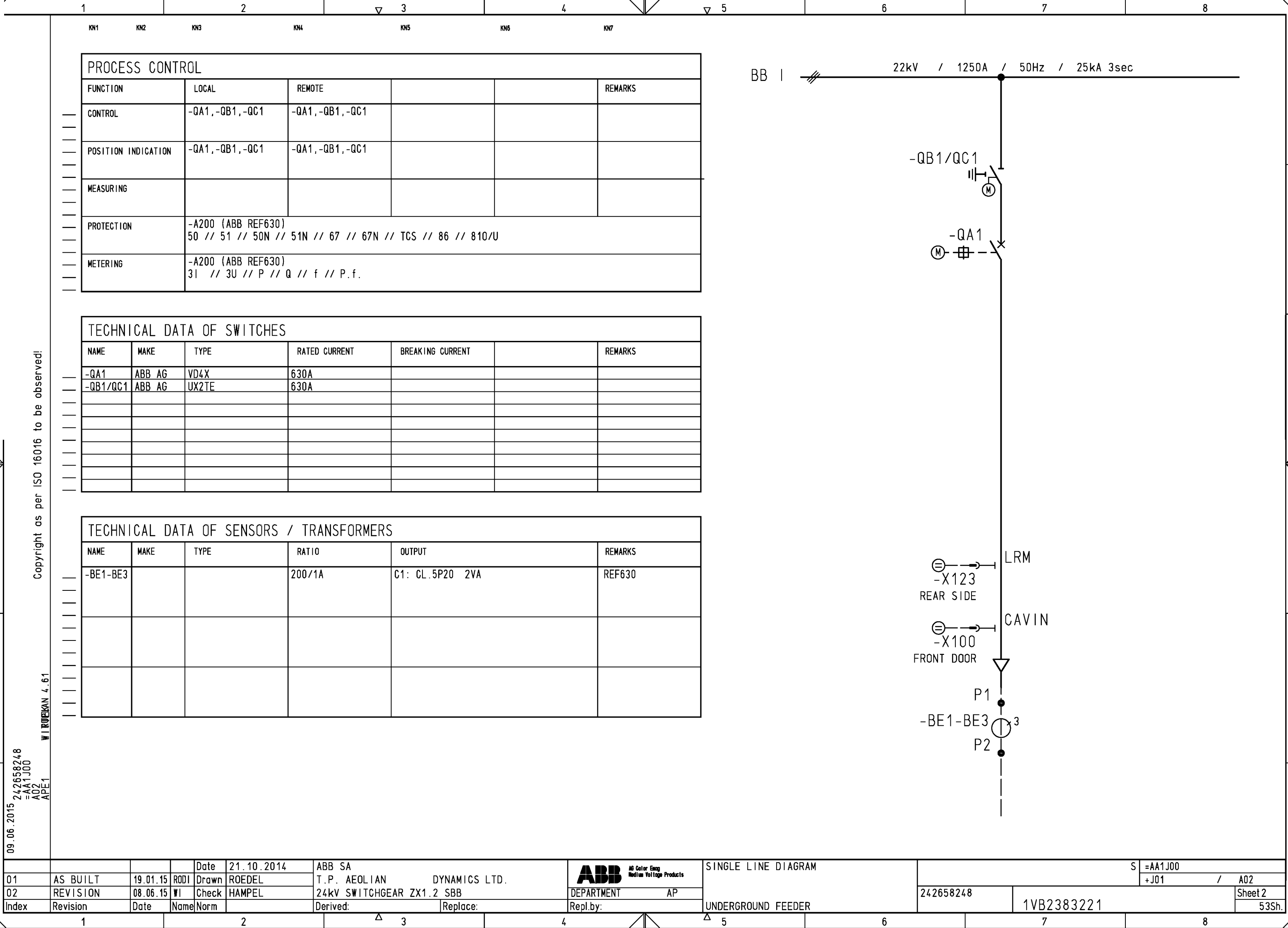
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02 REVISION
No. Revision

19.01.15
08.06.15
Date

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Name

09.06.2015 242658248
=AA1J00
A01
APE1
RUPLAN 4.61



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09.06.2015 242658248 =AA1J00 A02 APE 1

WIRTEK 4.61

01 AS BUILT 19.01.15 RODI Drawn ROEDEL T.P. AEOLIAN DYNAMICS LTD. 24kV SWITCHGEAR ZX1.2 SBB

02 REVISION 08.06.15 WI Check HAMPEL

Index Revision Date Name Norm Derived: Replace: Repl.by:

ABB A6 Color Enag Medium Voltage Products

DEPARTMENT AP

SINGLE LINE DIAGRAM

UNDERGROUND FEEDER

S =AA1J00 +J01 / A02

242658248

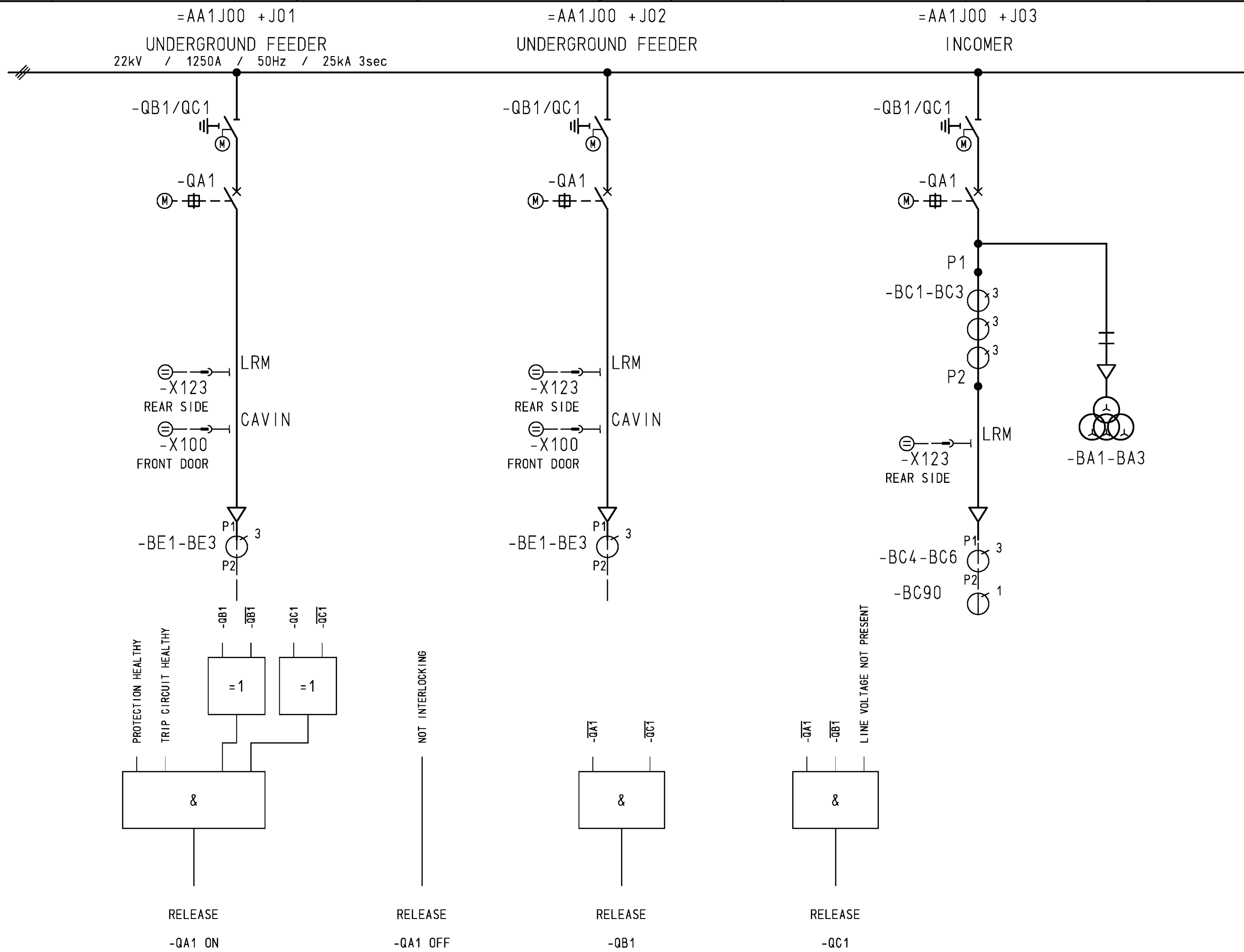
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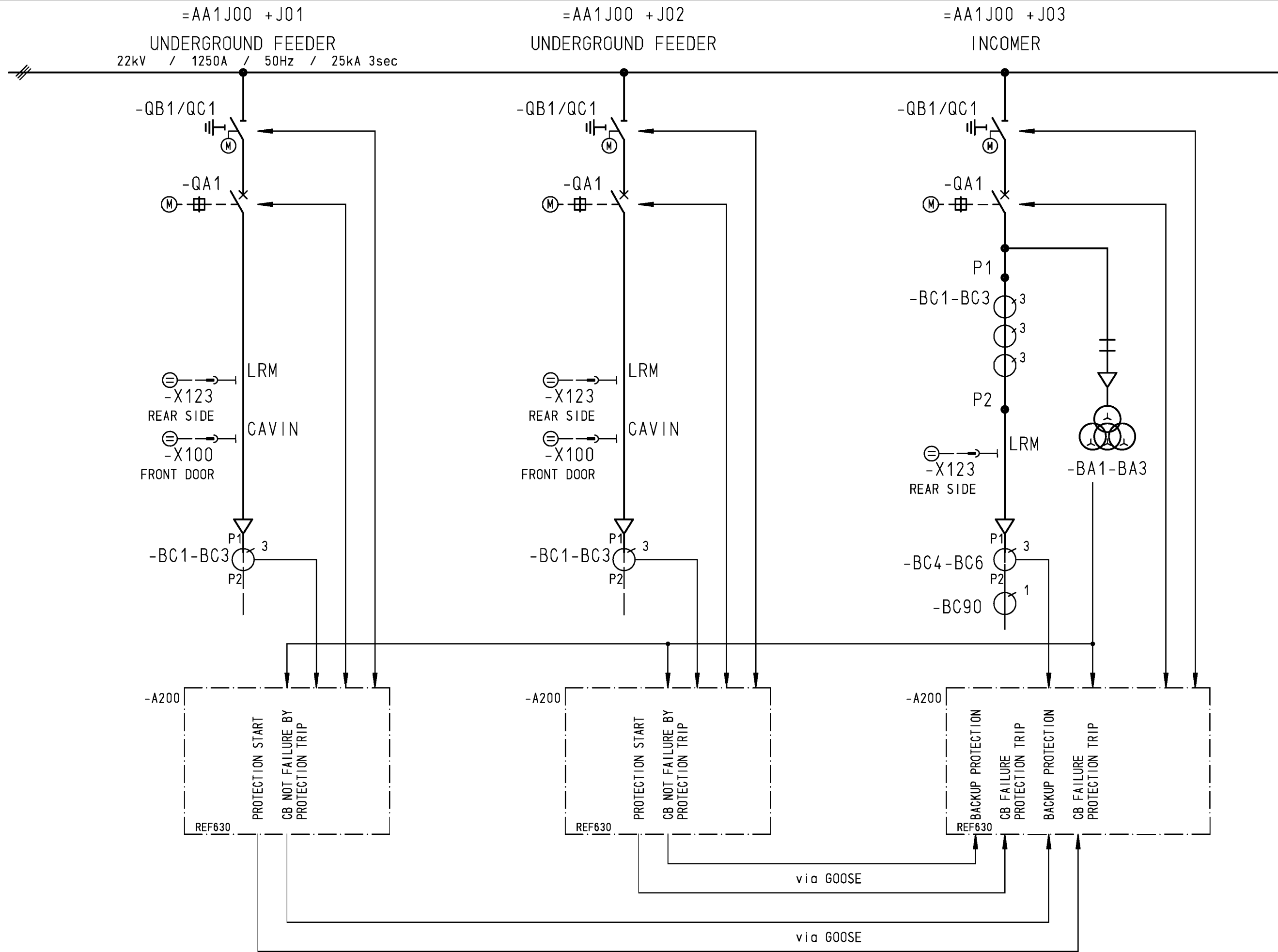
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										Sheet 3		
										53Sh.		

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09.06.2015 242658248
=AA1J00
A04
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		1VB2383221		02 01		S =AA1J00 +J01 A02		2		SINGLE LINE DIAGRAM					
		1VB2383221		02 01		S =AA1J00 +J01 A03		3		INTERLOCKING LOGIC					
		1VB2383221		02 01		S =AA1J00 +J01 A04		4		BACKUP PROTECTION & BFP					
		1VB2383221		02 01		S =AA1J00 +J01 B01		5		TABLE OF CONTENTS					
		1VB2383221		02 01		S =AA1J00 +J01 B02		6		TABLE OF CONTENTS					
		1VB2383221		02 01		S =AA1J00 +J01 D01		7		AC VOLTAGE SUPPLY					
		1VB2383221		02 01		S =AA1J00 +J01 G01		8		DC VOLTAGE SUPPLY					
		1VB2383221		02 01		S =AA1J00 +J01 G02		9		SELF MONITORING					
		1VB2383221		02 01		S =AA1J00 +J01 L01		10		MOTOR DRIVES					
		1VB2383221		02 01		S =AA1J00 +J01 M01		11		CONTROL CIRCUIT BREAKER					
		1VB2383221		02 01		S =AA1J00 +J01 M02		12		2.SHUNT RELEASE OFF					
		1VB2383221		02 01		S =AA1J00 +J01 P01		13		SPARE INTERLOCKING CIRCUITS					
		1VB2383221		02 01		S =AA1J00 +J01 Q01		14		POSITION INDICATIONS		CB/ISOLATOR/EARTHING SWITCH			
		1VB2383221		02 01		S =AA1J00 +J01 Q02		15		ALARMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q03		16		ALARMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q04		17		ALARMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q05		18		OUTPUTS REF		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q06		19		SPARE CONTACTS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q07		20		SPARE-OUTPUTS REF		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Q08		21		SPARE CONTACTS		CIRCUIT BREAKER			
		1VB2383221		02 01		S =AA1J00 +J01 S01		22		CURRENT TRANSFORMER CIRCUIT		CORE 1			
		1VB2383221		02 01		S =AA1J00 +J01 S02		23		BB-VOLTAGE		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 S03		24		CAPACITIVE VOLTAGE INDICATION		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 X01		25		SPARE TERMINALS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Z01		26		LIST OF ITEMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Z02		27		LIST OF ITEMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Z03		28		LIST OF ITEMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Z04		29		LIST OF ITEMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 Z05		30		LIST OF ITEMS		UNDERGROUND FEEDER			
		1VB2383221		02 01		S =AA1J00 +J01 ZZ01		31		LIST OF ITEMS		CIRCUIT BREAKER			
														</	

1		2		3		4		5			6	7	8	
CUSTOMER DOCUMENT NUMBER		MANUFACTURER DOCUMENT NUMBER		REVISION STATE		ORGANIZATIONAL DESIGNATION			SHEET	DESIGNATION				
						TYPE	ASSIGNMENT	NUMBER PART						
		1VB2383221		02	01		S =AA1J00 +J01	ZZ02	32	LIST OF ITEMS	THREE-POSITION SWITCH UNDERGROUND FEEDER			
		1VB2383221		02	01		S =AA1J00 +J01	ZZ04	33	LIST OF ITEMS	REF630 6U UNDERGROUND FEEDER			
		1VB2383221		02	01		S =AA1J00 +J01	ZZ05	34	LIST OF ITEMS	REF630 6U UNDERGROUND FEEDER			
		1VB2383221		02	01		S =AA1J00 +J01	ZZ06	35	LIST OF ITEMS	REF630 6U UNDERGROUND FEEDER			
		1VB2383221		02	01		S =AA1J00 +J01	ZZ07	36	LIST OF ITEMS	REF630 6U UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	001	37	BUS WIRES -X10	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	002	38	BUS WIRES -X11	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	003	39	BUS WIRES -X11	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	004	40	TERMINAL DIAGRAM -X1	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	005	41	X-PLUG -XP1	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	006	42	X-PLUG -XP1	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	007	43	X-PLUG -XP1	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	008	44	X-PLUG -XP1	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	009	45	X-PLUG -XP2	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	010	46	TERMINAL DIAGRAM -X8	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	011	47	TERMINAL DIAGRAM -X9	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	012	48	TERMINAL DIAGRAM -XC	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	013	49	TERMINAL DIAGRAM -XC	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	014	50	TERMINAL DIAGRAM -XC	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	015	51	TERMINAL DIAGRAM -XD	UNDERGROUND FEEDER			
		1VB2383221		02	01		V =AA1J00 +J01	Q01	52	TABLE OF CROSS CONNECTION	UNDERGROUND FEEDER			
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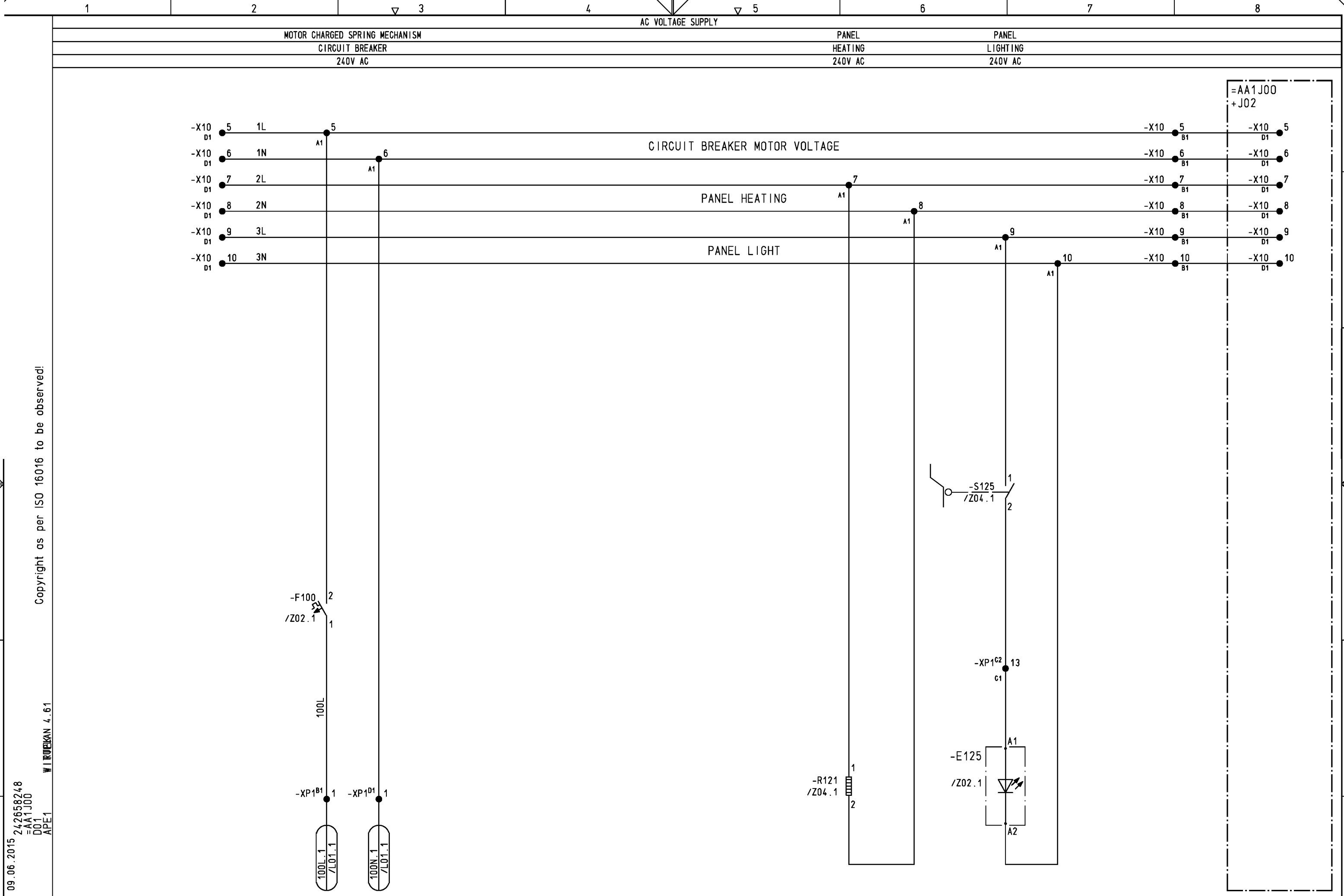
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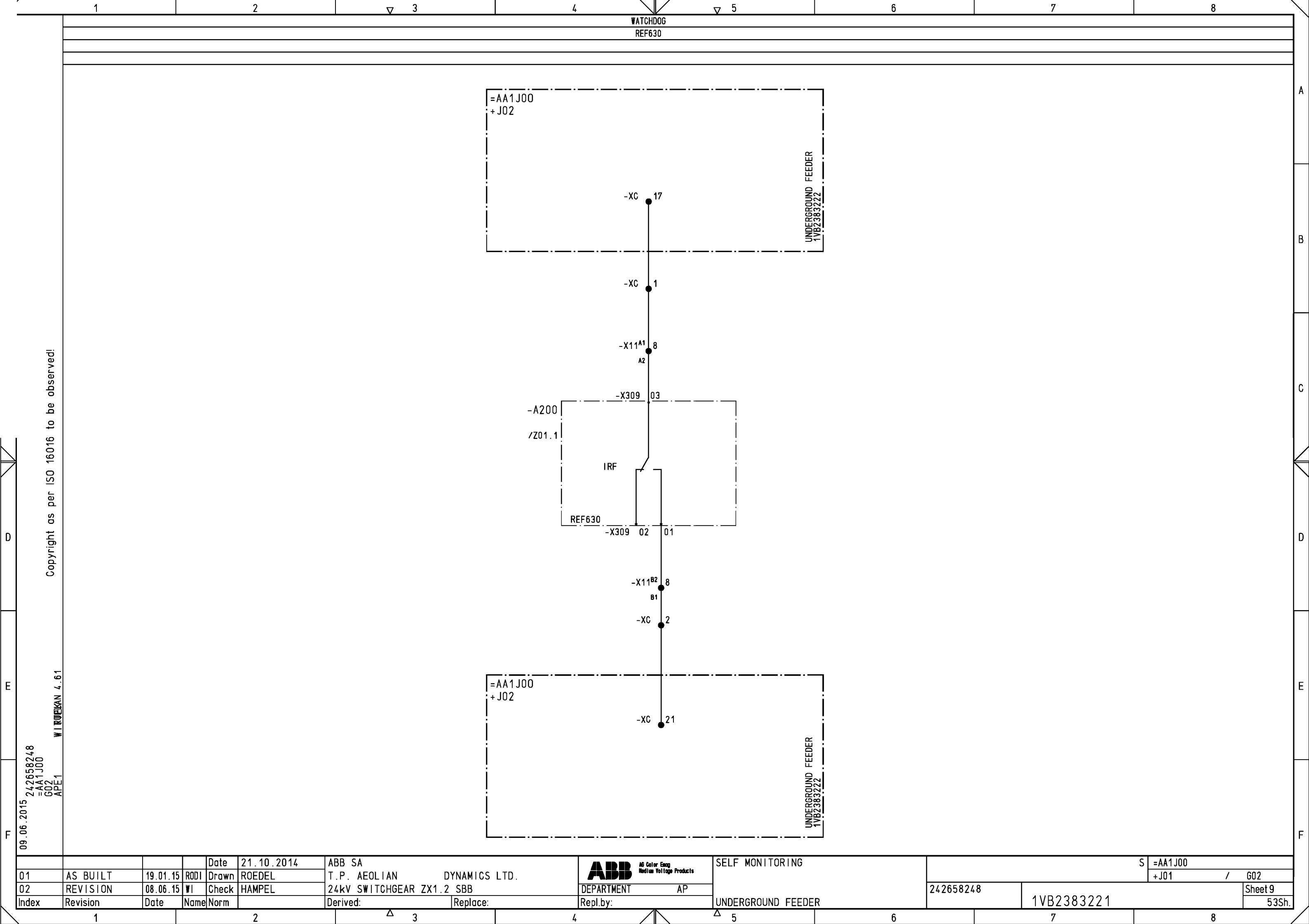
RUPLAN 4.61



AB Color Emag
Medium Voltage Products



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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.		DEPARTMENT AP	242658248	1VB2383221	+J01 /D01				
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					Sheet7				
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	53Sh.						
1		2		3		4		5		6		7		8	



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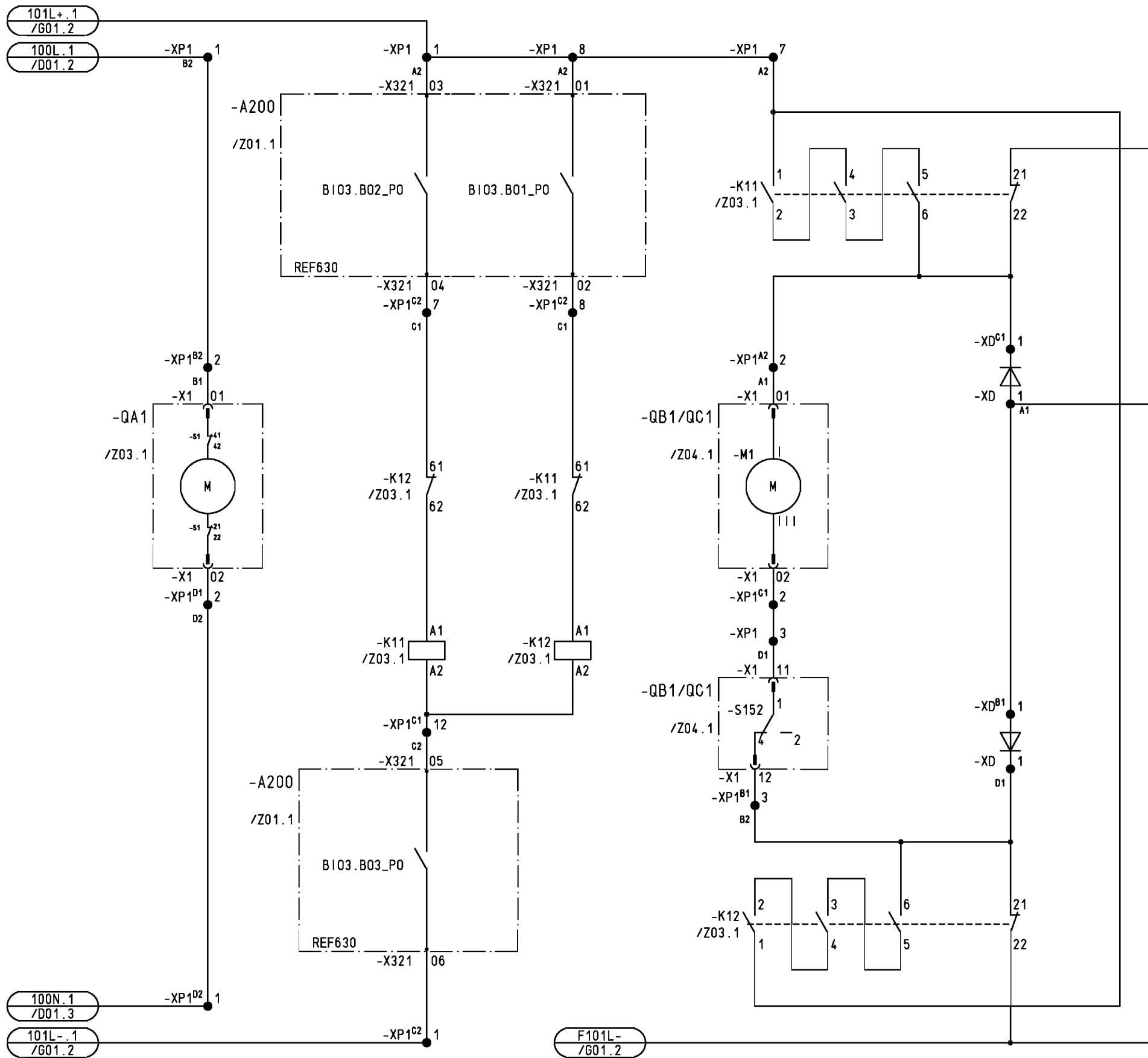
WIRTSCHAFT 4.61

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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 / G02	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
						DEPARTMENT AP				242658248	
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		1VB2383221
											Sheet 9 53Sh.

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				Date	21.10.2014	ABB SA		MOTOR DRIVES		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 / L01	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248	Sheet 10
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383221	53Sh.

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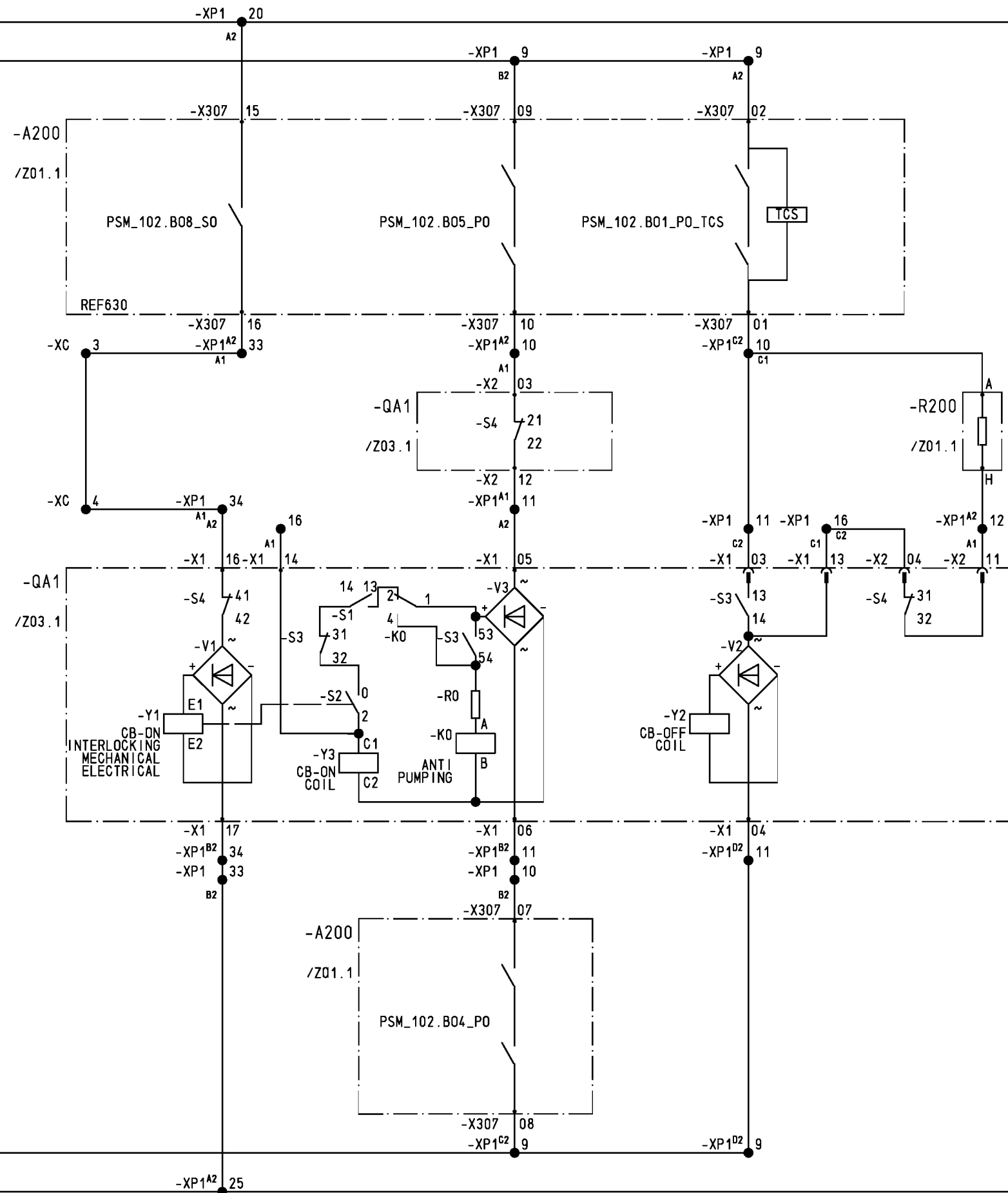
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/G01.3

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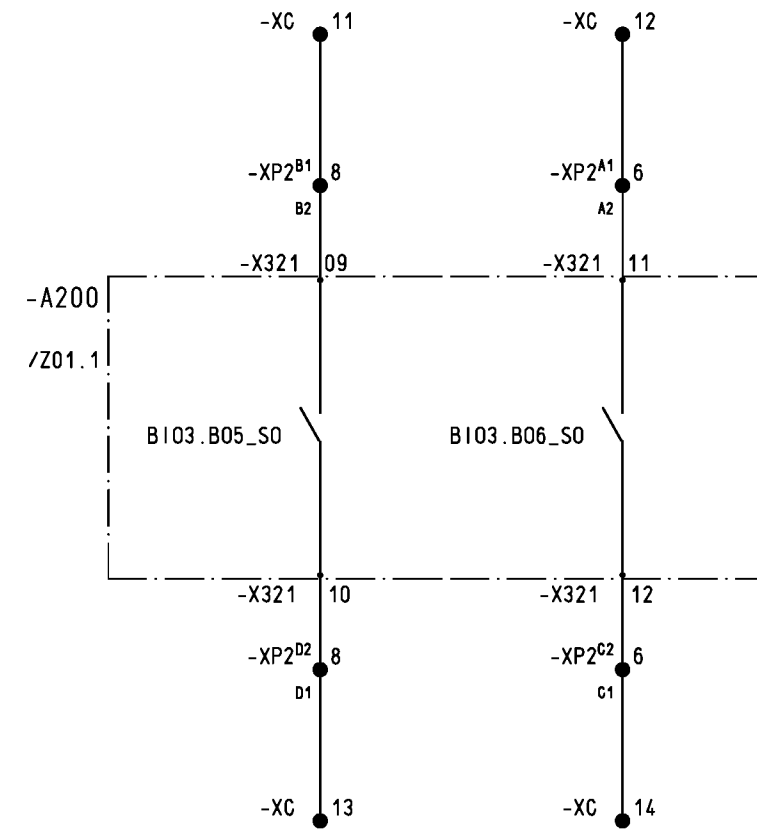
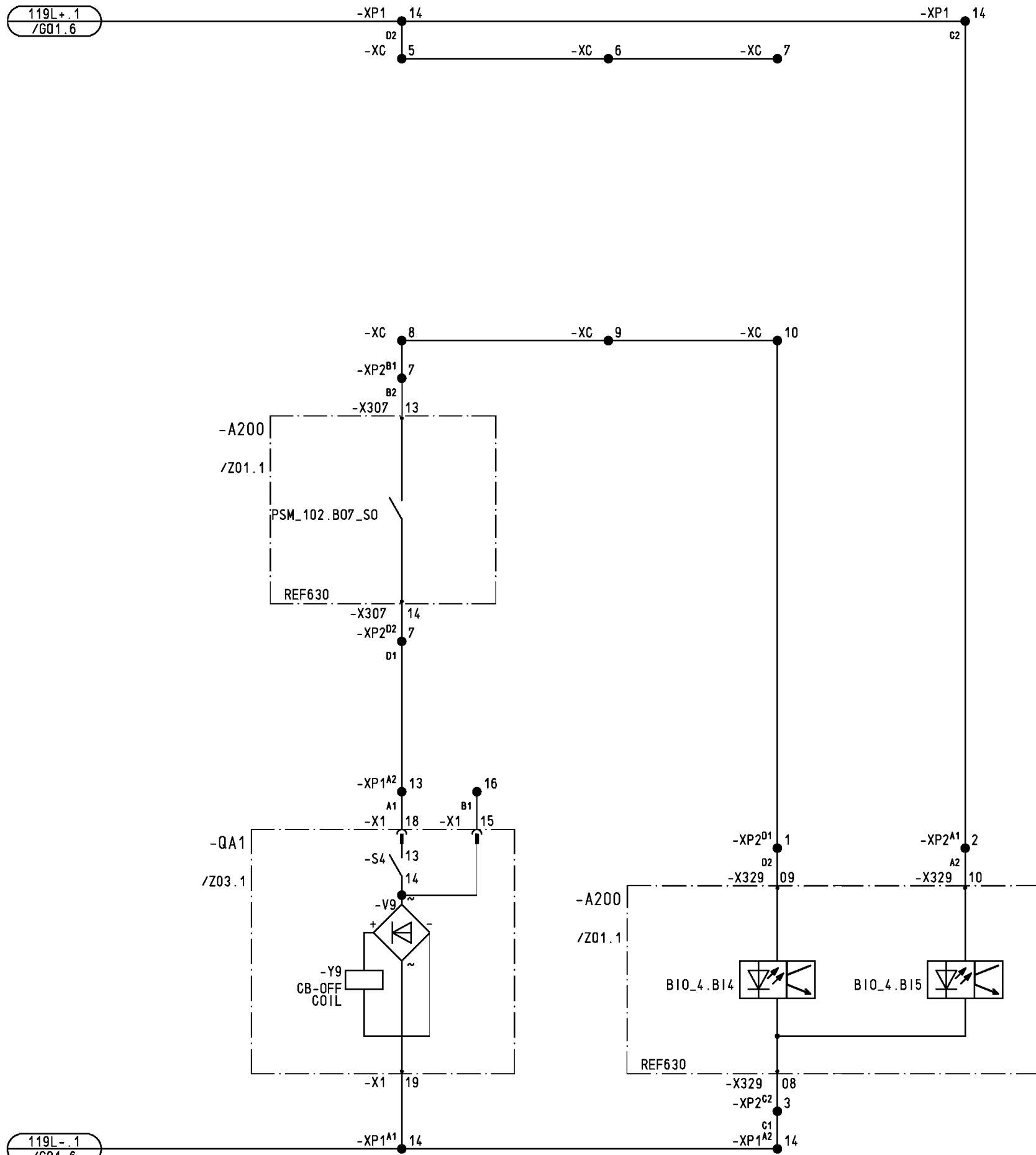
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				Date	21.10.2014	ABB SA	 A6 Color Emag Medium Voltage Products	CONTROL CIRCUIT BREAKER	S =AA1J00			
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			DEPARTMENT	AP	242658248	1VB2383221
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER			53Sh.

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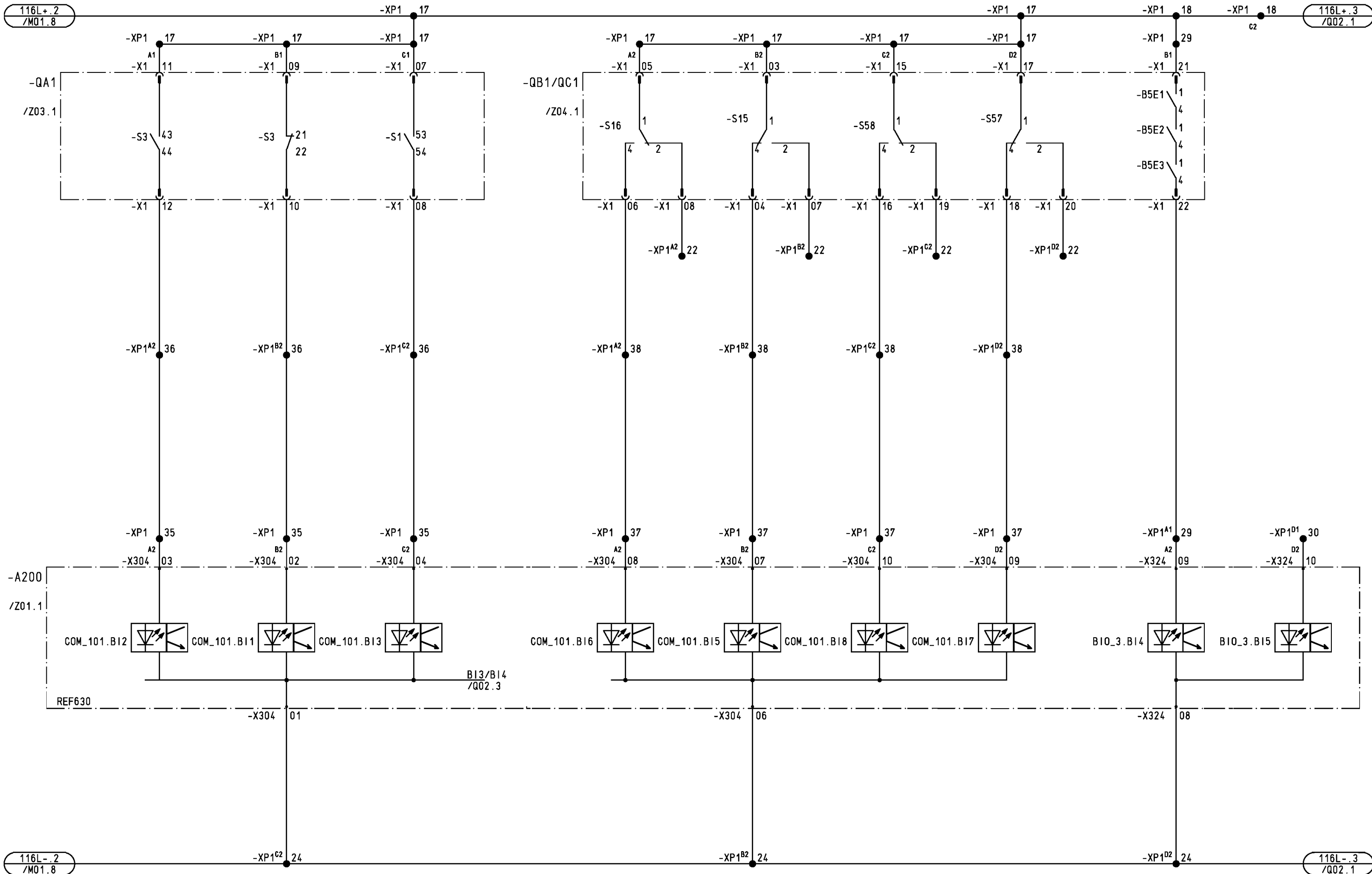


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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		242658248	Sheet 12
Index	Revision	Date	Name	Norm		Derived:	Repl.by:	1VB2383221	53Sh.

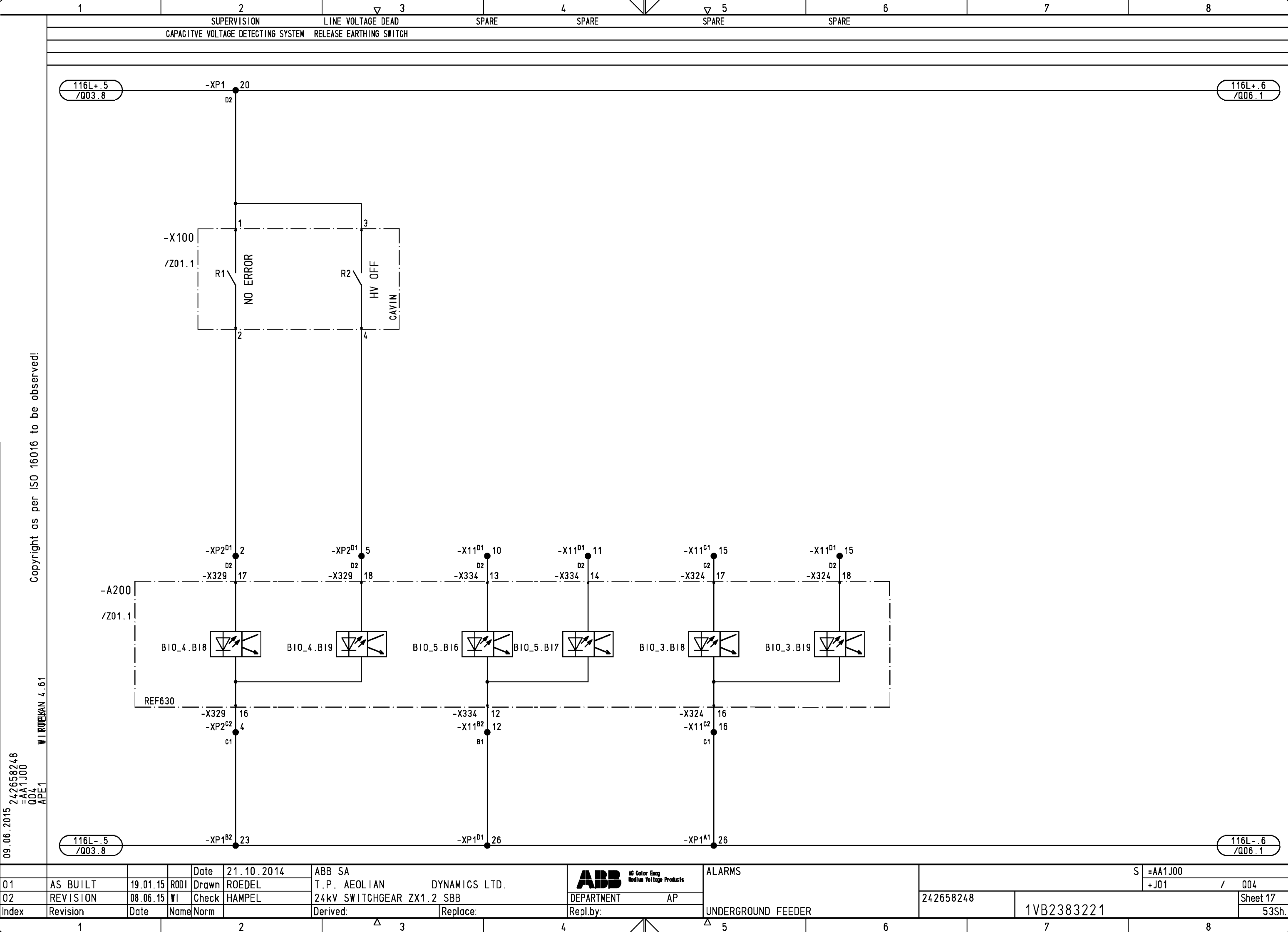
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Date 21.10.2014				ABB SA				POSITION INDICATIONS				S =AA1J00			
01 AS BUILT 19.01.15 RODI Drawn ROEDEL				T.P. AEOLIAN DYNAMICS LTD.				CB/ISOLATOR/EARTHING SWITCH				+J01 / Q01			
02 REVISION 08.06.15 WI Check HAMPEL				24kV SWITCHGEAR ZX1.2 SBB				DEPARTMENT AP				242658248			
Index Revision Date Name Norm				Derived: Replace:				Repl.by:				1VB2383221			
								UNDERGROUND FEEDER				Sheet 14			
												53Sh.			



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Date	21.10.2014	ABB SA	ALARMS		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	+J01 / Q04
02	REVISION	08.06.15	WI	Check	HAMPEL	242658248
Index	Revision	Date	Name	Norm	Derived:	1VB2383221
					Replace:	Sheet 17
					Repl.by:	53Sh.

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ABB

A6 Color Enag

Medium Voltage Products

DEPARTMENT

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UNDERGROUND FEEDER

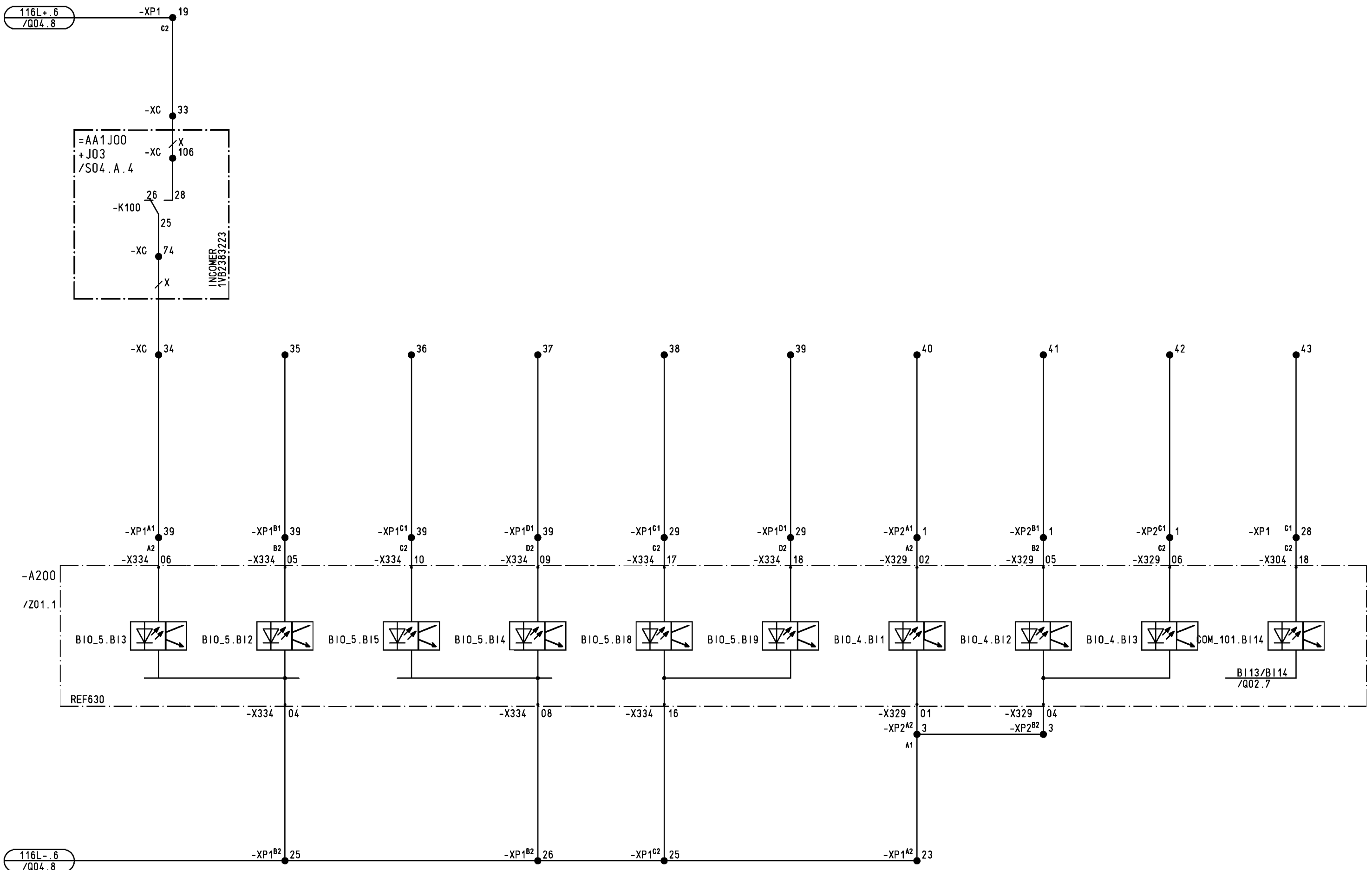


UNDERGROUND FEEDER

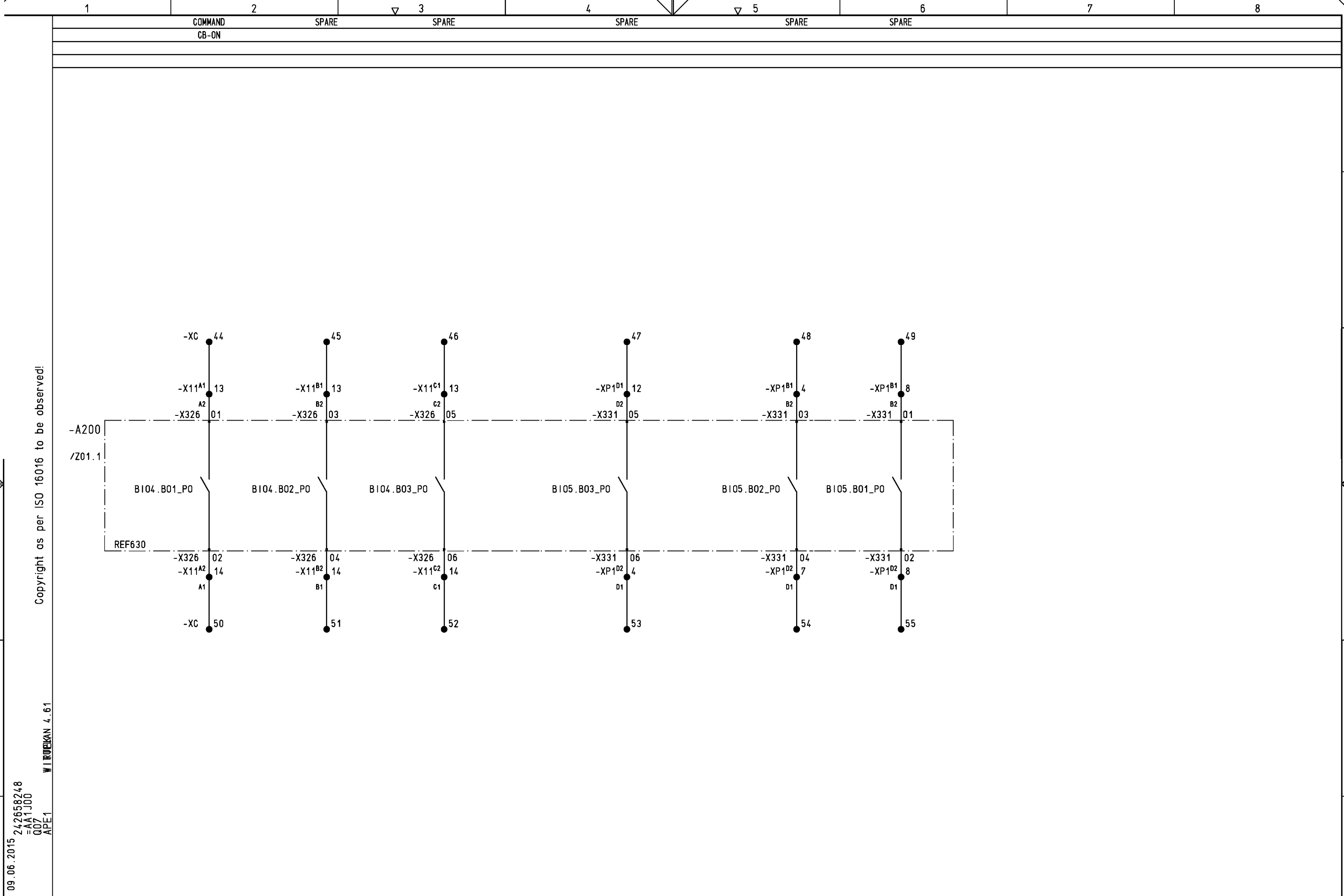
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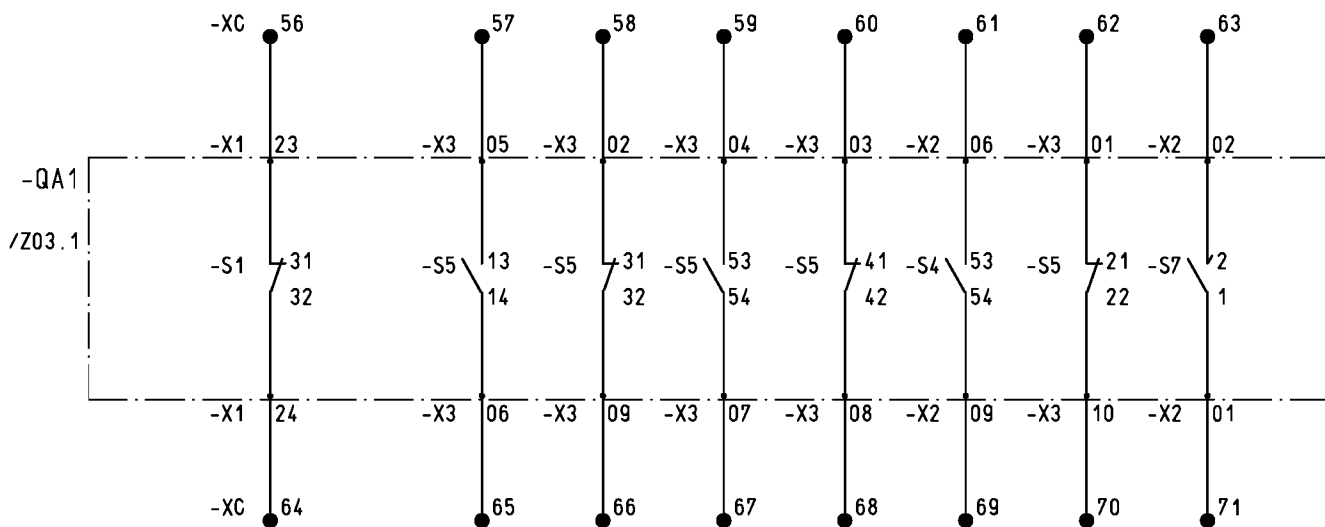
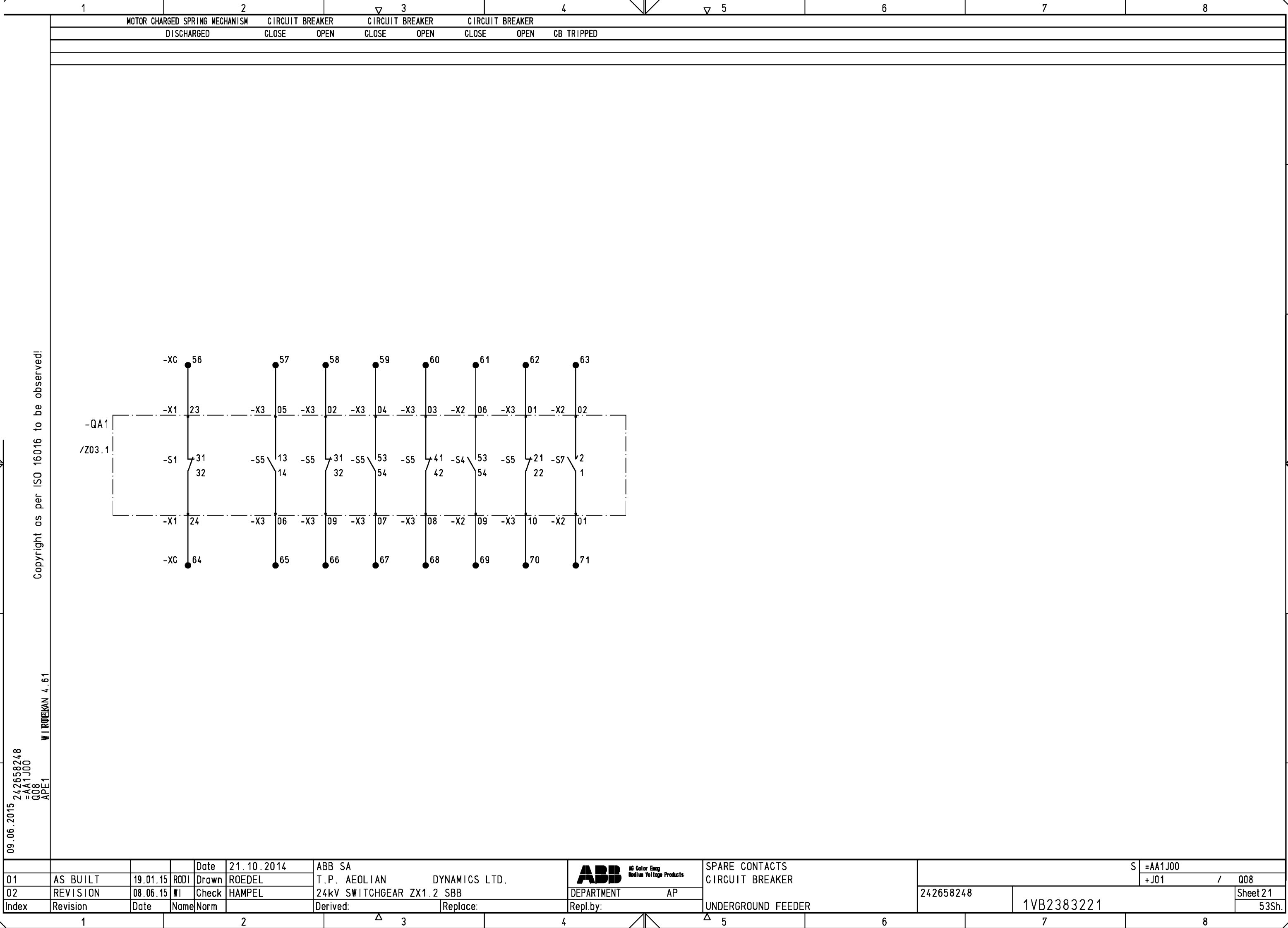


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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 / Q06	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248	Sheet 19
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383221	53Sh.



				Date	21.10.2014	ABB SA	ABB A6 Color Enag Medium Voltage Products	SPARE-OUTPUTS REF	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN			+J01	/	Q07
02	REVISION	08.06.15	WI	Check	HAMPEL	DYNAMICS LTD.					
Index	Revision	Date	Name	Norm		24kV SWITCHGEAR ZX1.2 SBB	DEPARTMENT AP		242658248	1VB2383221	
						Derived:	Repl.by:	UNDERGROUND FEEDER			Sheet 20
						Replace:					53Sh.

1 2 3 4 5 6 7 8

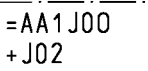


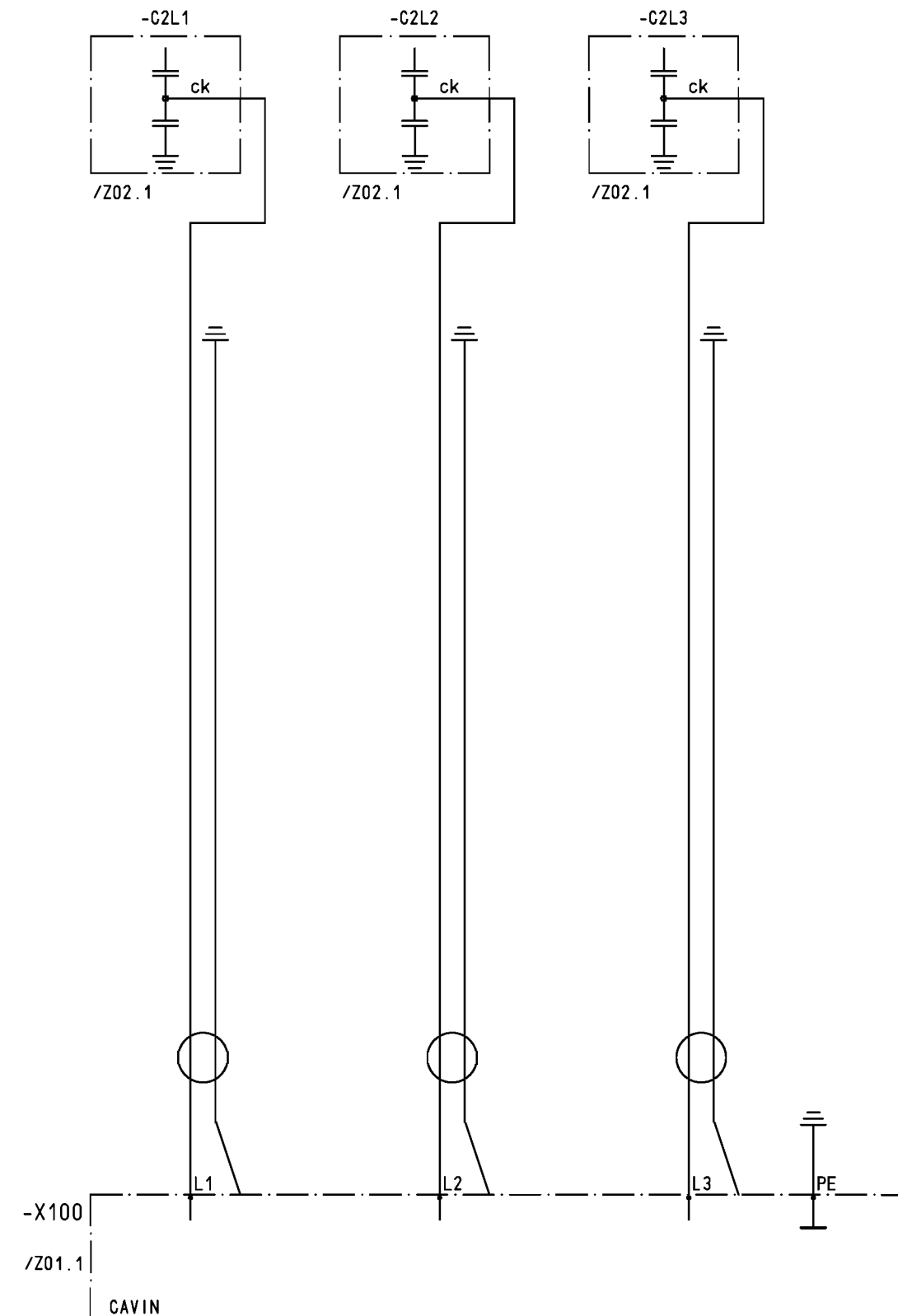


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01
02
Index

				Date	21.10.2014	ABB SA		 ABB AS Color Emag Medium Voltage Products	CURRENT TRANSFORMER CIRCUIT CORE 1	S =AA1J00			
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				DEPARTMENT AP	242658248	+J01 / S01	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						Sheet 22	
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		1VB2383221	53Sh.	





Sheet 24

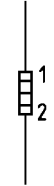
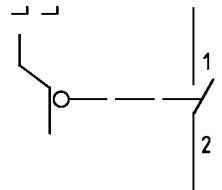
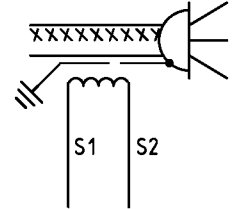
53Sh.

ABB AG Color Emag
Medium Voltage Products

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1	2	3	4	5	6	7	8
1	3-STELLUNGSSCH. ZX THREE-POS. SWITCH ZX TRENNER/ERDER DISCONNECTOR/ESW BEZEICHNUNG DESIGNATION : TECHNICAL COMPONENT TYPE ORDER-NO.	MAKE : ABB Color Emag ORDER-NO: TYPE:UX2TE X MIT HILFSSCHALTERN AUXILIARY SWITCHES	REFERENCE TO GROUP DESCRIPTION IN SCHEMATIC DIAGRAM REFERENCE TO SEPARATE DATA WITH DESCRIPTION DETAILS REFERENCE TO DESCRIPTION FOR APPLIANCE CONNECTIONS				
	APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION			
			+..SS I	-QB1/QC1			
/L01.4	/L01.4	/001.4	/002.5		/ZZ02.1		
1	HEIZUNG HEATER OHNE BEGRENZUNG WITHOUT LIMITATION NENNSPANNUNG RATED VOLTAGE : 240V TECHNICAL COMPONENT TYPE ORDER-NO.	MAKE : RICA ORDER-NO: 1FZEK833B Pos.001 TYPE:1FZEK X NENNLEISTUNG RATED POWER : 50W					
	APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION			
			+..NS	-R121	/D01.6		
1	STEUERSCHALTER CONTROL SWITCH TUERKONTAKT ZELLENTUER DOOR-CONTACT PANEL-DOOR SCHALTELEMENT SWITCHING ELEMENT : TECHNICAL COMPONENT TYPE ORDER-NO.	MAKE : CONRAD ORDER-NO: 707201-77 TYPE:C.3005CB X SCHALTSTUFEN SWITCHING STEPS :					
	APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION			
			+..NS	-S125	/D01.6		
3	KABELUMBAUWANDLER CABLE CURRENT TRANSF IEC/2 IEC/2 PRIMAER PRIMARY : TECHNICAL COMPONENT TYPE ORDER-NO.	MAKE : ORDER-NO: TYPE: X TECHNISCHE DATEN SIEHE UEBERSICHTSSCHALTPLAN TECHNICAL DATA SEE SINGLE LINE SEKUNDAER SECONDARY :					
	APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION			
			+..WR	-BE1	/S01.3		
			+..WR	-BE2	/S01.4		
			+..WR	-BE3	/S01.5		
09.06.2015 242658248 =AA1J00 Z04 APE 1							
	Date	21.10.2014	ABB SA	LIST OF ITEMS		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN	DYNAMICS LTD.
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB	
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:
1	2	3	4	5	6	7	8
242658248 1VB2383221				UNDERGROUND FEEDER		Sheet 29	
						53Sh.	

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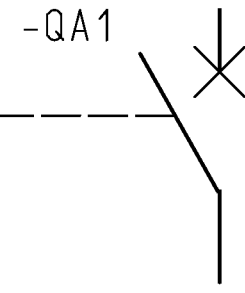
DEPARTMENT AP

The diagram illustrates a test system with three identical units. Each unit has a power source (L1, L2, L3) connected to a load (PE1, PE2, PE3) through a switch. A ground connection (PE) is also shown.

ABB AG Color Engr
Medium Voltage Products

UNDERGROUND FEEDER

Sheet 30
53Sh.



-QA1	01	
-X3	02	
	03	
/Z03.1	04	
	05	
	06	
	07	
	08	
	09	
	10	
	11	
	12	

				Date	21.10.2014	ABB SA		 AB Color Energy Medium Voltage Products	LIST OF ITEMS CIRCUIT BREAKER	S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 / ZZ01	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
						DEPARTMENT		AP		242658248	
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383221	
											Sheet 31 53Sh.

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-QB1/QC1	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
-X1	/L01.4	/L01.4	/L01.5	/L01.5	/L01.4	/L01.4	/L01.5	/L01.5	/L02.6	/L02.6	/L01.4	/L01.4			/L01.6	/L01.6	/L01.6	/L01.6	/L01.6	/L01.7	/L01.7	/L01.7		
/Z04.1																								

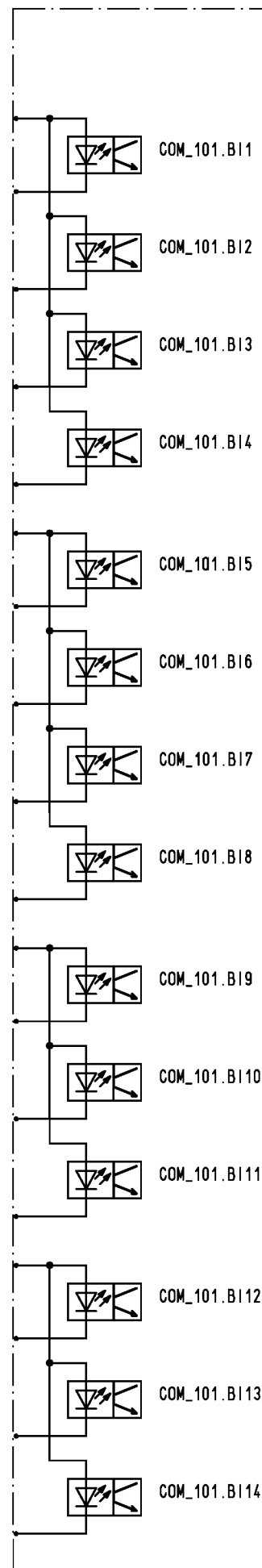
			Date	21.10.2014	ABB SA	 AG Color Enam Medium Voltage Products	LIST OF ITEMS THREE-POSITION SWITCH	S =AA1J00 +J01 / ZZ02			
S BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.						
REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			DEPARTMENT AP			
Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	242658248	1VB2383221	Sheet 32 53Sh.

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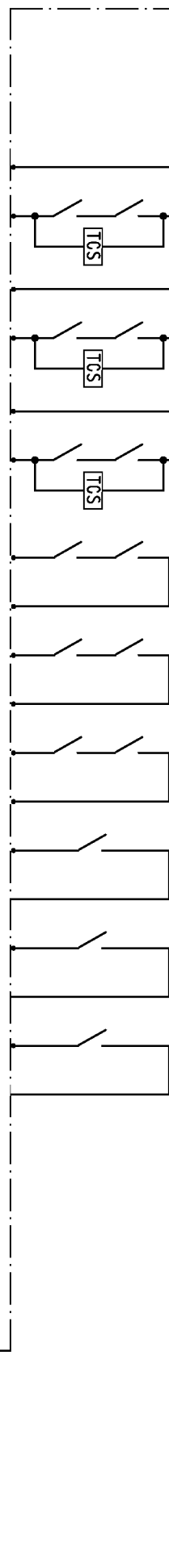
09.06.2015 242658248
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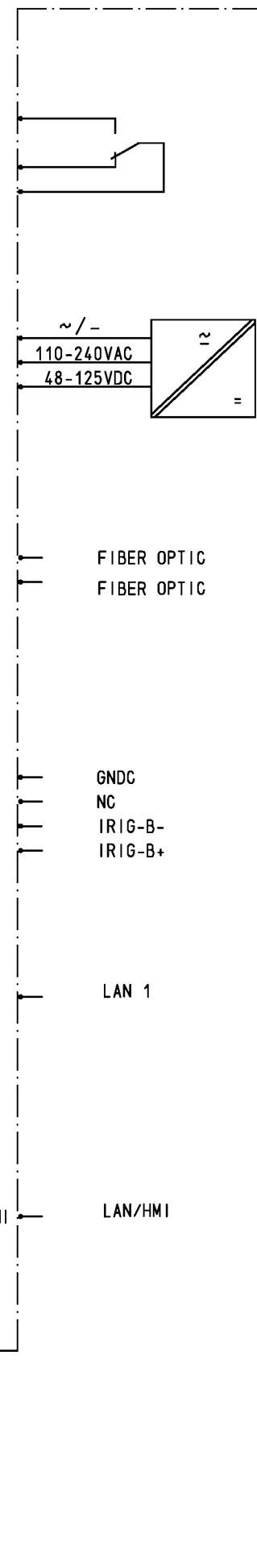
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ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X304	/Q01.2	
MEANING		
POSITION INDICATION CB-OFF	/Q01.2	01
	/Q01.2	02
POSITION INDICATION CB-DN	/Q01.2	03
POSITION INDICATION SPRING MECHANISM CHARGED	/Q01.3	04
MCB TRIPPED CONTROL VOLTAGE	/Q02.3	05
	/Q01.5	06
POSITION INDICATION ISOLATOR OPEN	/Q01.5	07
POSITION INDICATION ISOLATOR CLOSE	/Q01.4	08
POSITION INDICATION EARTHING SWITCH OPEN	/Q01.6	09
POSITION INDICATION EARTHING SWITCH CLOSED	/Q01.6	10
GROUND CLOSED	/Q02.4	11
	/Q02.4	12
MECHANICAL BLOCKING ISOLATOR/EARTHING SWITCH BB I	/Q02.6	13
MCB TRIPPED ALTERNATING VOLTAGE	/Q02.5	14
	/Q02.6	15
GAS PRESSURE OK CB CUBICLE	/Q02.6	16
GAS PRESSURE OK BB COMPARTMENT I	/Q02.7	17
SPARE	/Q06.8	18



LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X307	/M01.6	
MEANING		
COMMAND CB-OFF	/M01.6	01
	/M01.6	02
SPARE	/Q05.3	03
	/Q05.3	04
SPARE	/Q05.5	05
	/Q05.5	06
COMMAND CB-ON	/M01.5	07
COMMAND CB-ON	/M01.5	08
	/M01.5	09
SPARE	/Q05.2	11
	/Q05.2	12
PANEL NOT EARTHED	/M02.3	13
	/M02.3	14
CIRCUIT BREAKER CLOSING BLOCK. MAGNET	/M01.4	15
	/M01.4	16
SPARE	/Q05.3	17
	/Q05.3	18

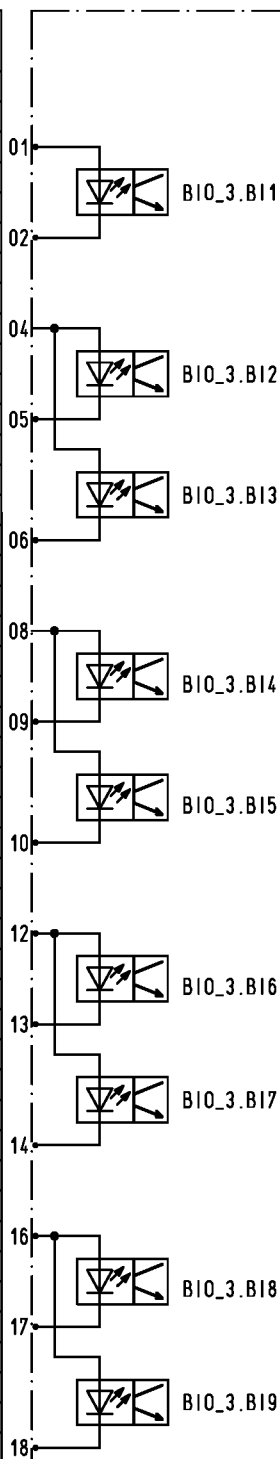
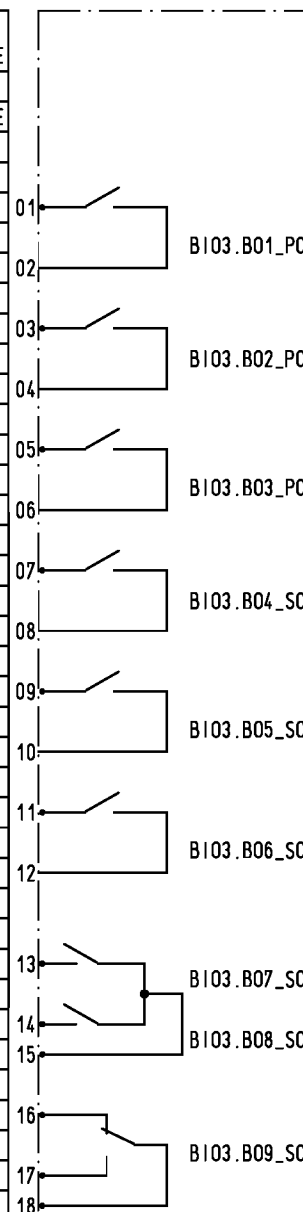


LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X309	/G02.4	
MEANING		
WATCHDOG REF630	/G02.4	01
	/G02.4	02
	/G02.4	03
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X410	/Q02.3	
MEANING		
	/Q02.3	01
	/Q02.3	03
	/Q02.3	02
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X9		
MEANING		
		TX
		RX
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X4		
MEANING		
		01
		02
		03
		04
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X1		
MEANING		
		LAN 1
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X0	/Q02.2	
MEANING		
	/Q02.2	LAN/HMI

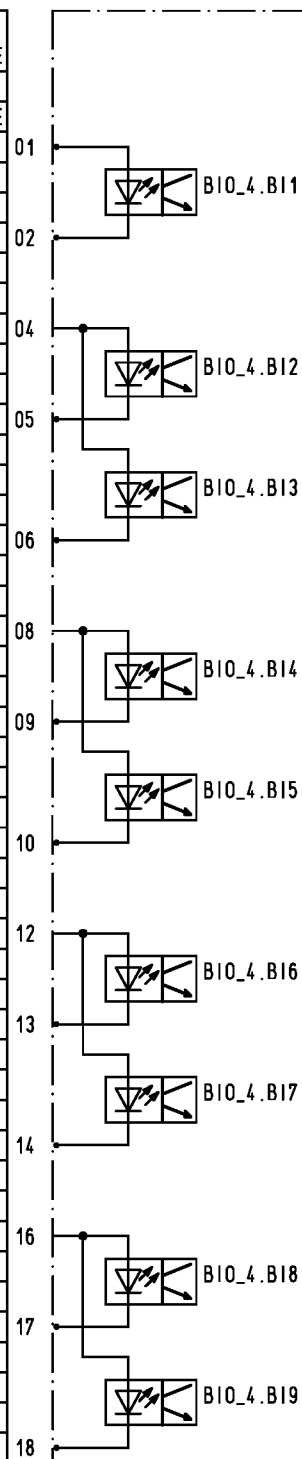
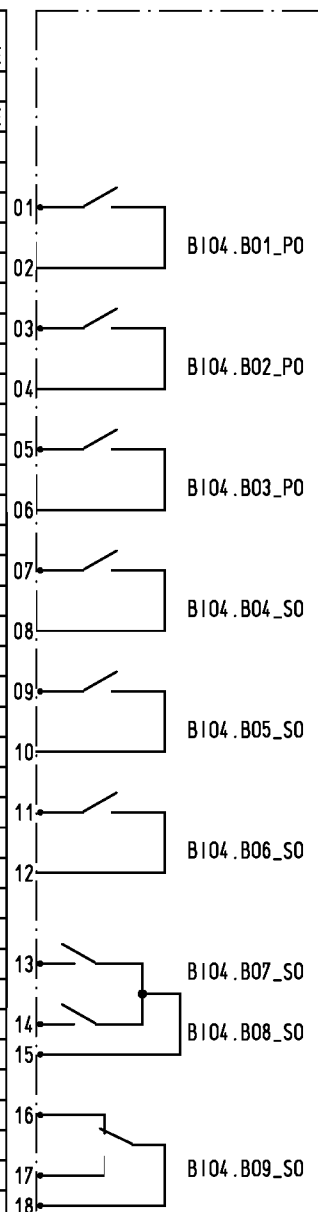


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LOCATION		STATUS	ITEM-
ITEM-DES.	-A200		REFERENCE
TERMINAL	-X324		/Z01.1
MEANING			REFERENCE
MCB TRIPPED			/Q03.2
MOTOR VOLTAGE			
			/Q03.2
			/P01.4
SPARE			
			/P01.4
SPARE			
			/P01.4
			/Q01.7
REEDCONTACTS			
EARTHING SWITCH CLOSED			
			/Q01.7
SPARE			
			/Q01.8
			/Q03.3
SPARE			
			/Q03.3
SPARE			
			/Q03.3
			/Q04.5
SPARE			
			/Q04.5
SPARE			
			/Q04.6

[illegible]

LOCATION	ITEM- REFERENCE	STATUS	ITEM- REFERENCE
ITEM-DES .	-A200		/Z01 . 1
TERMINAL	-X329		REFERENCE
MEANING			/Q06 . 6
SPARE			
			/Q06 . 6
			/Q06 . 7
SPARE			
			/Q06 . 7
SPARE			
			/Q06 . 7
			/M02 . 4
PROTECTION			
TRIPPING			
			/M02 . 4
SPARE			
			/M02 . 5
			/Q03 . 5
SELF MONITORING			
REF630			/Q03 . 5
SPARE			
			/Q03 . 6
			/Q04 . 2
SUPERVISION			
CAPACITIVE VOLTAGE DETECTING	SYSTEM		/Q04 . 2
LINE VOLTAGE DEAD			
RELEASE EARTHING SWITCH			/Q04 . 3

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		S	=AA1J00
		+J01	/ ZZ07
242658248	1VB2383221	Sheet 36	
		53Sh.	

[illegible]

1		2		3		4		5		6		7		8											
NO	CABLE	USED CORES		SHEET\COMPL.		CABLE TYPE Ø		CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE											
												STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WEIDM. ZDU2,5N/4AN PE PHOE. USLK66N											
										-X1															
REFERENCE		NO.												CORES/ LINE	POS.	DESTINATIONS		POT	NO.	DESTINATIONS		POS.	CORES/ LINE	REMARK	
		1	2	3	4	5	6	7	8	9	10	11	12												
/X01.3																									
/X01.3																									
/X01.3																									
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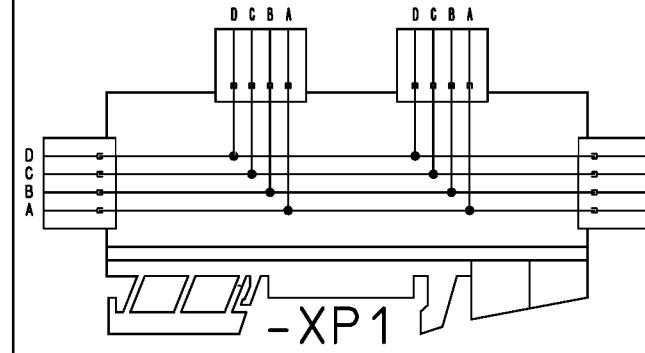
Copyright as per ISO 16016 to be observed!

01		AS BUILT		19.01.15		RODI		Date		21.10.2014		ABB SA		T.P. AEOLIAN DYNAMICS LTD.		ABB AG Color Emag Medium Voltage Products		TERMINAL DIAGRAM -X1		V =AA1J00	
02		REVISION		08.06.15		WI		Check		HAMPEL		24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP				242658248		1VB2383221	
Index		Revision		Date		Name		Norm				Derived:		Replace:		Repl.by:		UNDERGROUND FEEDER		Sheet 40 53Sh.	

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[illegible]



D C B A	Wiring Material				Ø Colour	Terminal Type
	Standard:	H05Z-K	1,0mm ²	BK		Standard: WEIDM. ZRV2,5/4
	POS.:	82	H07Z-K	1,5mm ²	BK	

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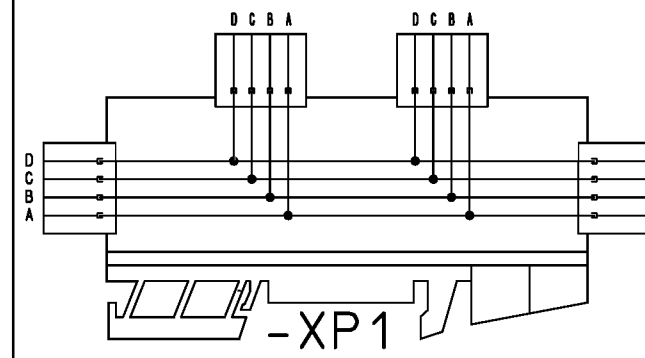
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RUPLAN 4.61

09.06.2015 242658248
=AA1J00+J01
006
APE1

				Date	21.10.2014	ABB SA		 ABB AS Color Emag Medium Voltage Products	X-PLUG -XP1	V =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 /006	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB				242658248	1VB2383221
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		53Sh.

[illegible]



Standard:	H05Z-K	1,0mm2	BK
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K | Standard: WEIDM. ZRV2,5/4

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RUPLAN 4.61

09.06.2015
242658248
=AA1J00+J01
008
APE1

				Date	21.10.2014	ABB SA		 ABB AB Color Engg Medium Voltage Products	X-PLUG -XP1	V =AA1J00		
01	AS_BUILT	19.01.15	R0D1	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J01 /008		
02	REVISION	08.06.15	W1	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB				242658248	1VB2383221	Sheet 44
Index	Revision	Date	Name	Norm		Derived:	Replace:			Repl.by:	UNDERGROUND FEEDER	53Sh.

1		2		3		4		5		6		7		8									
NO	CABLE	USED CORES		SHEET\COMPL.		CABLE TYPE		Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE								
													STANDARD: H07Z-K 1,5mm2 BK		STANDARD: WEIDM. ZDU2,5N/4AN								
-X9																							
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS		POT	NO.	DESTINATIONS		POS.	CORES/ LINE	REMARK
/S02.3																	A 405L1	1 B					
																	C	D					
/S02.4																	A 405L2	2 B					
																	C	D					
/S02.4																	A 405L3	3 B					
																	C	D					
/S02.3																	A UN	4 B					
																	C	D					
/S02.4																	A UN	5 B					
																	C	D					
/S02.5																	A UN	6 B					
																	C	D					
/S02.6																	A	11 B					
																	C	D					
/S02.6																	A	12 B					
																	C	D					
09.06.2015		242658248		=AA1J00+J01		011		APE1		RUPLAN 4.61		Copyright as per ISO 16016 to be observed!											
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	21.10.2014		ABB SA		T.P. AEOLIAN		DYNAMICS LTD.		ABB		A6 Color Emag Medium Voltage Products		TERMINAL DIAGRAM		-X9		V =AA1J00	
02	REVISION	08.06.15	WI	Check	HAMPEL			24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT		AP										+J01 /011	
Index	Revision	Date	Name	Norm		Derived:		Replace:		Repl.by:		UNDERGROUND FEEDER		242658248		1VB2383221		Sheet 47		53Sh.			
1		2		3		4		5		6		7		8									

09.06.2015242658248=AA1J00+J01012APE1

RUPLAN 4.61

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1		2		3		4		5		6		7		8							
NO	CABLE	USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE									
1	X		4					-XC		STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WDU1,5/BLZ/5.08/ZQV									
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK
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/G02.5															=AA1J00 +J02 -XC	21		-X11	8B1		
/M01.3																		-XP1	33A1		
/M01.3																		-XP1	34A1		
/M02.3																		-XP1	14D2		
/M02.3																					
/M02.4																					
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/M02.3																					
/M02.4																		-XP2	1D1		
/M02.7																		-XP2	8B1		
/M02.7																		-XP2	6A1		
/M02.7																		-XP2	8D1		
/M02.7																		-XP2	6C1		
/Q03.3																		-XP1	19D2		
/Q03.4																					
/Q03.5	<-													X	=AA1J00 +J03 -XC	1					
/Q03.6																					
/Q03.3																		-X11	15B1		
/Q03.4																		-XP2	5A1		
/Q03.5	<-													X	=AA1J00 +J03 -XC	2		-XP2	2B1		
/Q03.6																		-XP2	2C1		
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/Q05.6																		-XP2	6B1		
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/Q05.7																		-XP2	5C1		
/Q05.3																		-XP2	8C1		
/Q05.5																		-XP2	4B1		
/Q05.6																		-XP2	6D1		
/Q05.7																		-XP2	7C1		
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/Q06.2	<-													X	=AA1J00 +J03 -XC	74		-XP1	39A1		

01AS BUILT19.01.15RODI

02REVISION08.06.15WI

IndexRevisionDateNameNorm

Date21.10.2014

DrawnROEDEL

CheckHAMPEL

ABB SA

T.P. AEOLIAN

24kV SWITCHGEAR ZX1.2 SBB

Derived:

DYNAMICS LTD.

Replac:

ABB

AG Color Emag

Medium Voltage Products

DEPARTMENT

AP

Repl.by:

TERMINAL DIAGRAM -XC

UNDERGROUND FEEDER

V=AA1J00

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242658248

1VB2383221

Sheet 48

53Sh.

09.06.2015
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APE1

RUPLAN 4.61

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1		2		3		4		5		6		7		8							
NO	CABLE	USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE									
								-XC		STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WDU1,5/BLZ/5.08/ZQV									
										ASSEMBLY											
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK
/Q06.2																	35	-XP1	39B1		
/Q06.3																	36	-XP1	39C1		
/Q06.4																	37	-XP1	39D1		
/Q06.5																	38	-XP1	29C1		
/Q06.5																	39	-XP1	29D1		
/Q06.6																	40	-XP2	1A1		
/Q06.7																	41	-XP2	1B1		
/Q06.7																	42	-XP2	1C1		
/Q06.8																	43	-XP1	28C1		
/Q07.2																	44	-X11	13A1		
/Q07.2																	45	-X11	13B1		
/Q07.3																	46	-X11	13C1		
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/Q08.4																	62	-QA1	-X3	01	
/Q08.4																	63	-QA1	-X2	02	
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/Q08.2																	65	-QA1	-X3	06	
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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA	T.P. AEOLIAN DYNAMICS LTD.		 ABB Color Emag Medium Voltage Products	TERMINAL DIAGRAM -XC	V =AA1J00	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		+J01 /013		
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	242658248	1VB2383221	Sheet 49 53Sh.	

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AG Calor Emag
Medium Voltage Products

ABB AG Power Technologies Division - Calor Emag Medium Voltage Products

Oberhausener Strasse 33

D-40472 Ratingen

D-40472 Ratingen

Telefon: (02102) 12-0

Fax: (02102) 12-1777

ABB SA

T.P. AEOLIAN

DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

UNDERGROUND FEEDER

DEACE CONTRACT NO. : 242658248

DEACE DRAWING NO. : 1VB2383222

S = AA1J00 + J02

/ A01

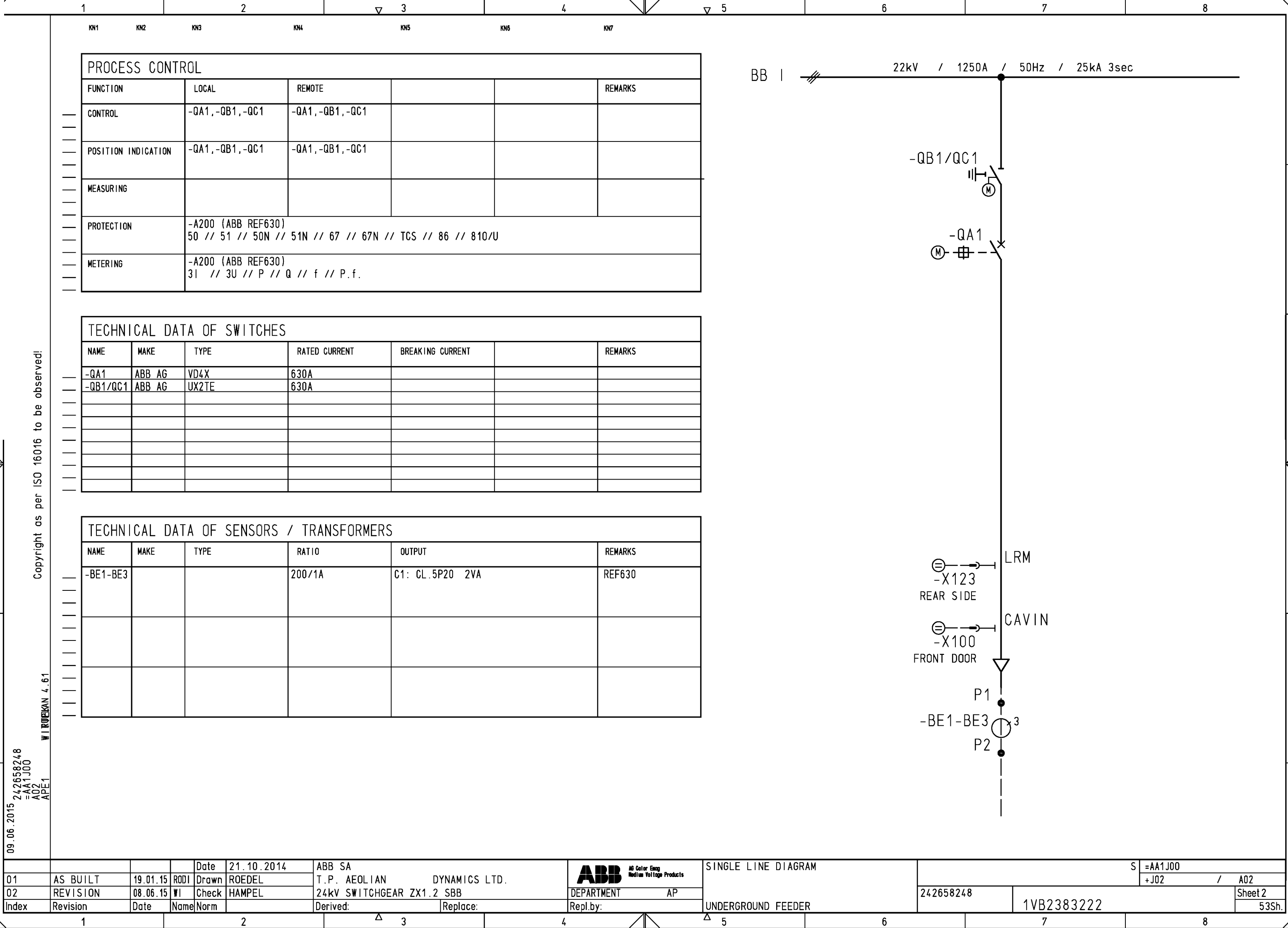
09.06.2015 242658248
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RUPLAN 4.61

01 AS BUILT
02 REVISION
No. Revision

19.01.15
08.06.15
Date

R001
WI
Name

Date :	21.10.2014
Drawn :	ROEDEL
Check :	HAMPEL
Repl.:	



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09.06.2015 242658248 =AA1J00 A02 APE 1

WIRTEK 4.61

01 AS BUILT 19.01.15 RODI Drawn ROEDEL T.P. AEOLIAN DYNAMICS LTD. 24kV SWITCHGEAR ZX1.2 SBB

02 REVISION 08.06.15 WI Check HAMPEL

Index Revision Date Name Norm Derived: Replace: Repl.by:

ABB A6 Color Enag Medium Voltage Products

DEPARTMENT AP

SINGLE LINE DIAGRAM

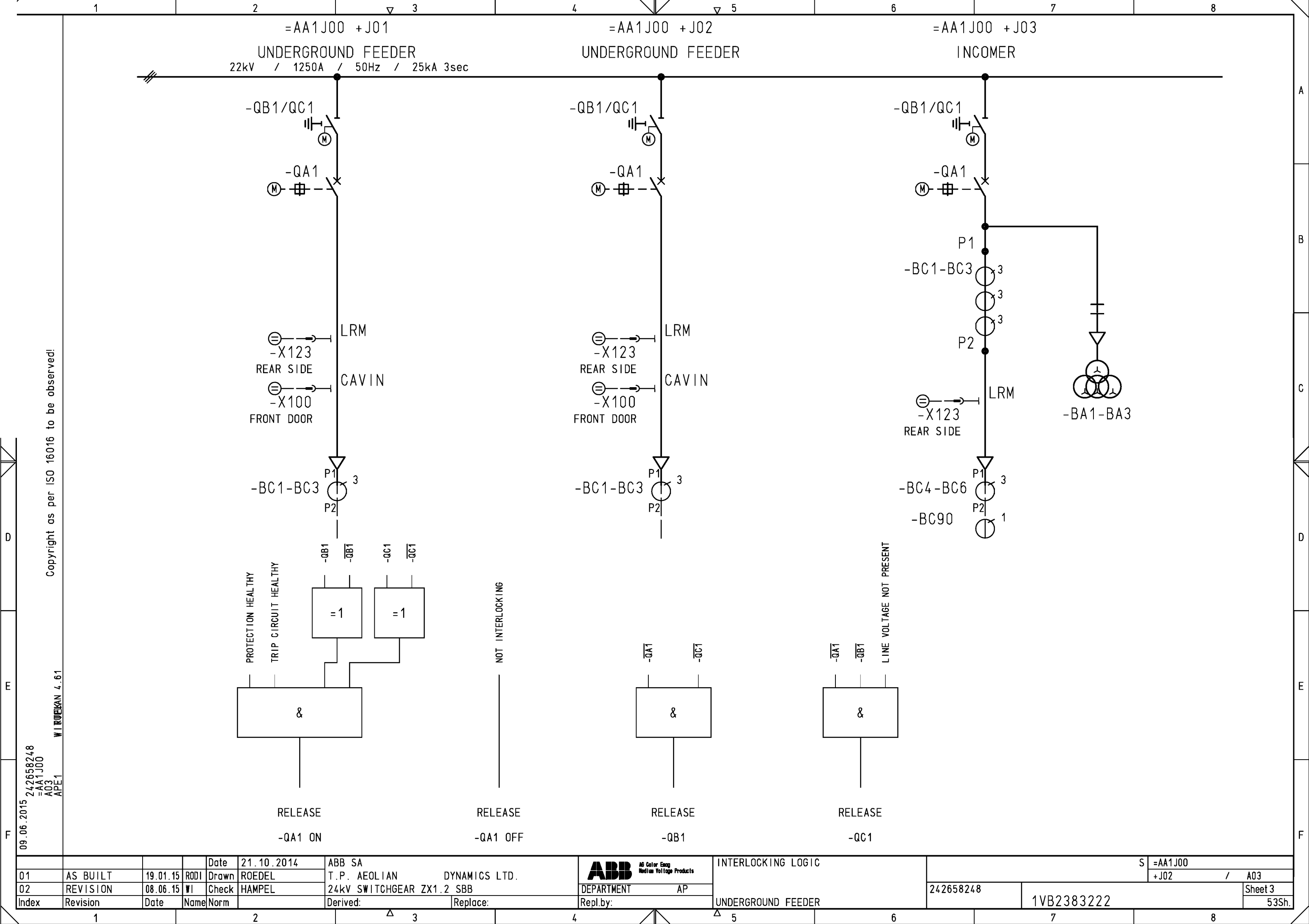
UNDERGROUND FEEDER

S =AA1J00 +J02 / A02

242658248

1VB2383222

Sheet 2 53Sh.



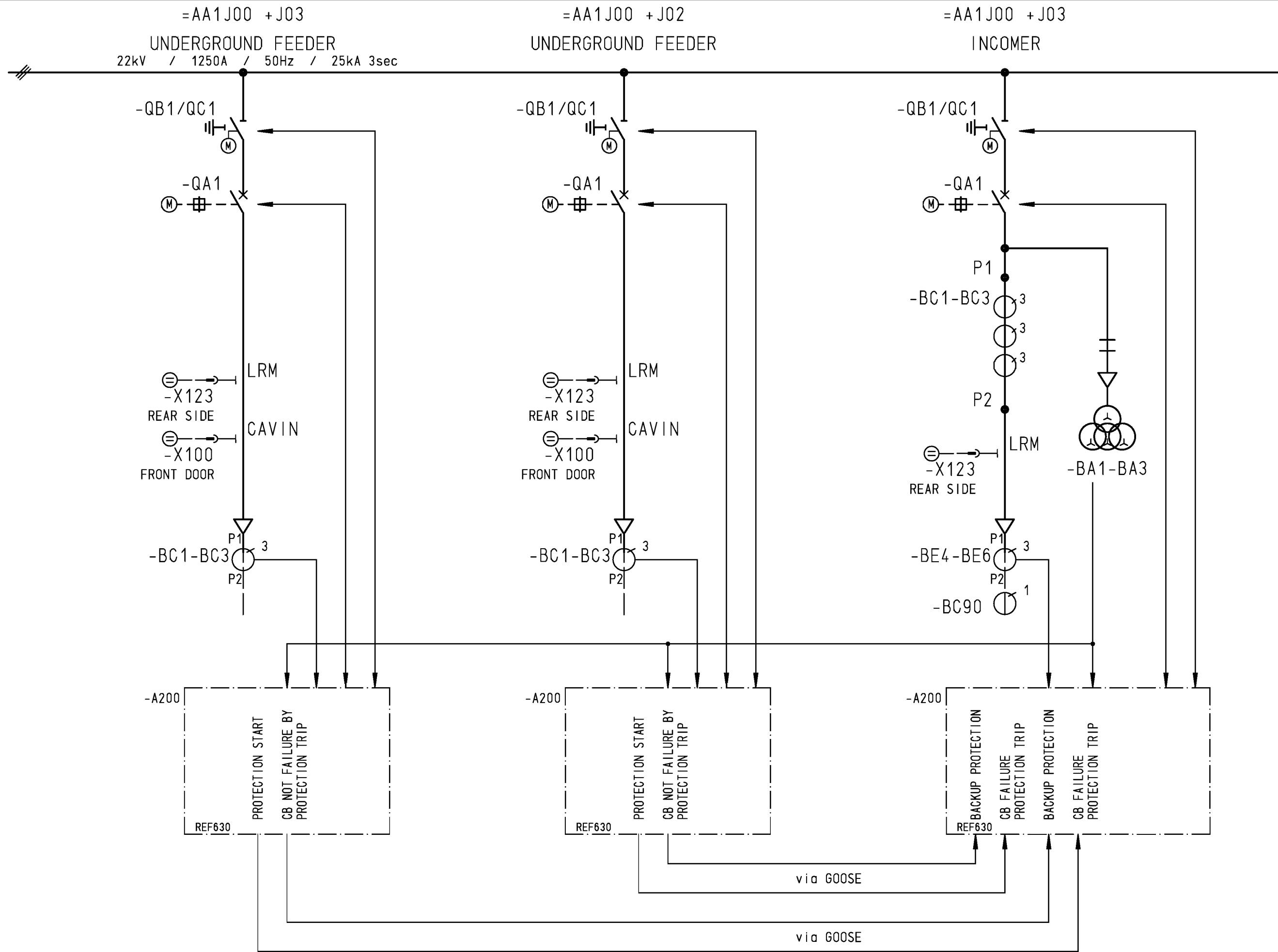
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02		REVISION	08.06.15	WI	Check	HAMPEL				242658248	Sheet 3
Index		Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383222	
										53Sh.	

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09.06.2015 242658248
=AA1J00
A04
APE 1

WIRTEK 4.61

				Date	21.10.2014	ABB SA	BACKUP PROTECTION & BFP		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J02	/ A04
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			242658248	Sheet 4
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	53Sh.



1	2		3		4		5			6	7	8	
CUSTOMER DOCUMENT NUMBER	MANUFACTURER DOCUMENT NUMBER		REVISION STATE		ORGANIZATIONAL DESIGNATION			SHEET	DESIGNATION				
TYPE	ASSIGNMENT		NUMBER PART										
1VB2383222	02		01		S	=AA1J00 +J02		A01	1	COVERSHEET	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		A02	2	SINGLE LINE DIAGRAM	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		A03	3	INTERLOCKING LOGIC	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		A04	4	BACKUP PROTECTION & BFP	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		B01	5	TABLE OF CONTENTS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		B02	6	TABLE OF CONTENTS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		D01	7	AC VOLTAGE SUPPLY	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		G01	8	DC VOLTAGE SUPPLY	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		G02	9	SELF MONITORING	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		L01	10	MOTOR DRIVES	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		M01	11	CONTROL CIRCUIT BREAKER	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		M02	12	2.SHUNT RELEASE OFF	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		P01	13	SPARE INTERLOCKING CIRCUITS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q01	14	POSITION INDICATIONS	CB/ISOLATOR/EARTHING SWITCH UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q02	15	ALARMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q03	16	ALARMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q04	17	ALARMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q05	18	OUTPUTS REF	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q06	19	SPARE CONTACTS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q07	20	SPARE-OUTPUTS REF	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Q08	21	SPARE CONTACTS	CIRCUIT BREAKER UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		S01	22	CURRENT TRANSFORMER CIRCUIT	CORE 1 UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		S02	23	BB-VOLTAGE	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		S03	24	CAPACITIVE VOLTAGE INDICATION	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		X01	25	SPARE TERMINALS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Z01	26	LIST OF ITEMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Z02	27	LIST OF ITEMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Z03	28	LIST OF ITEMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Z04	29	LIST OF ITEMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		Z05	30	LIST OF ITEMS	UNDERGROUND FEEDER		
1VB2383222	02		01		S	=AA1J00 +J02		ZZ01	31	LIST OF ITEMS	CIRCUIT BREAKER UNDERGROUND FEEDER		

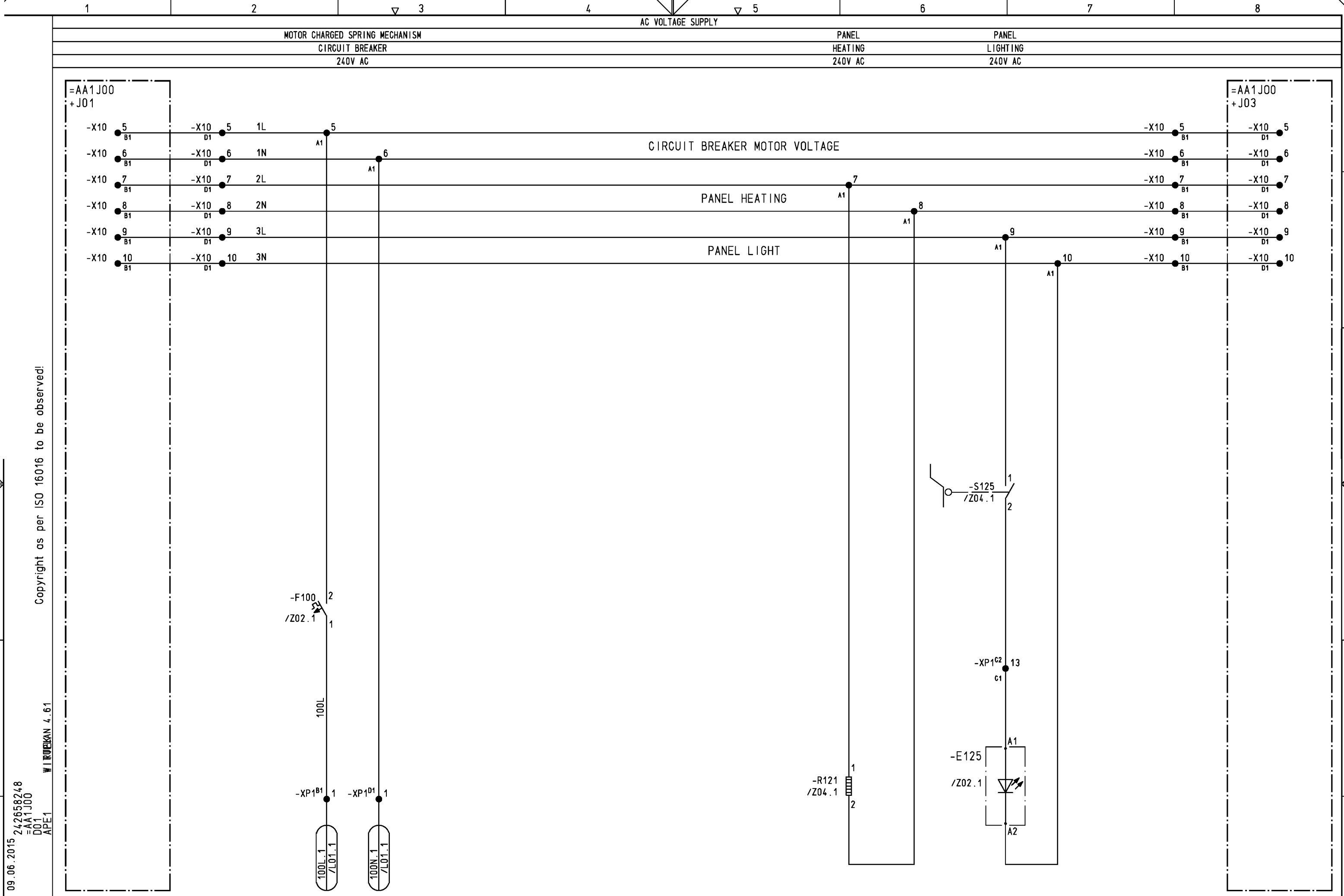
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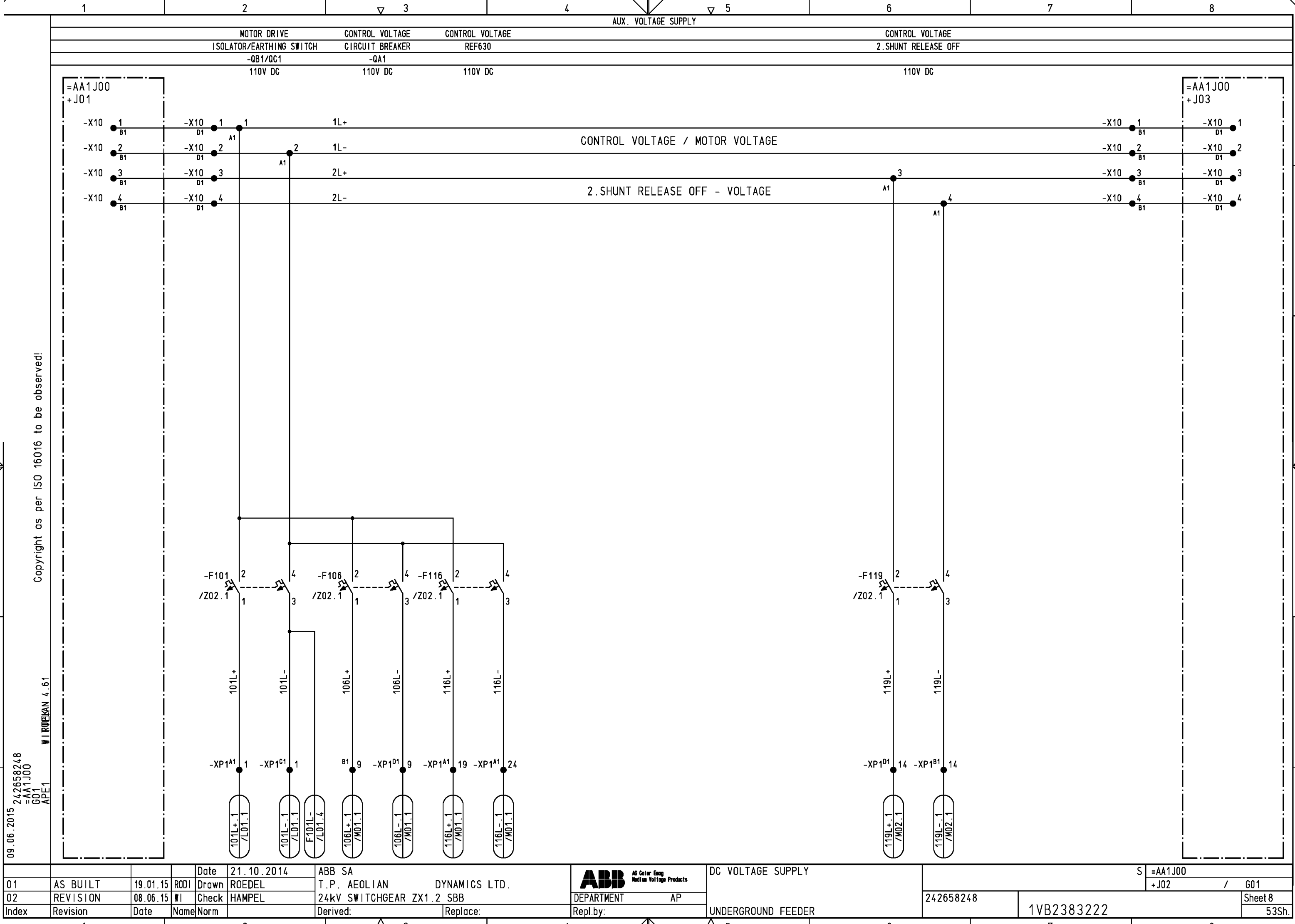
Copyright as per ISO 16016 to be observed!

01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	21.10.2014	ABB SA	T.P. AEOLIAN	DYNAMICS LTD.	ABB	AB Color Emag Medium Voltage Products	TABLE OF CONTENTS	S	=AA1J00																																																																																																																																																																																																																																											
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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.		DEPARTMENT AP	242658248	1VB2383222	+J02 /D01
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		Sheet7 53Sh.	
1		2		3		4		5		6	
7		8									

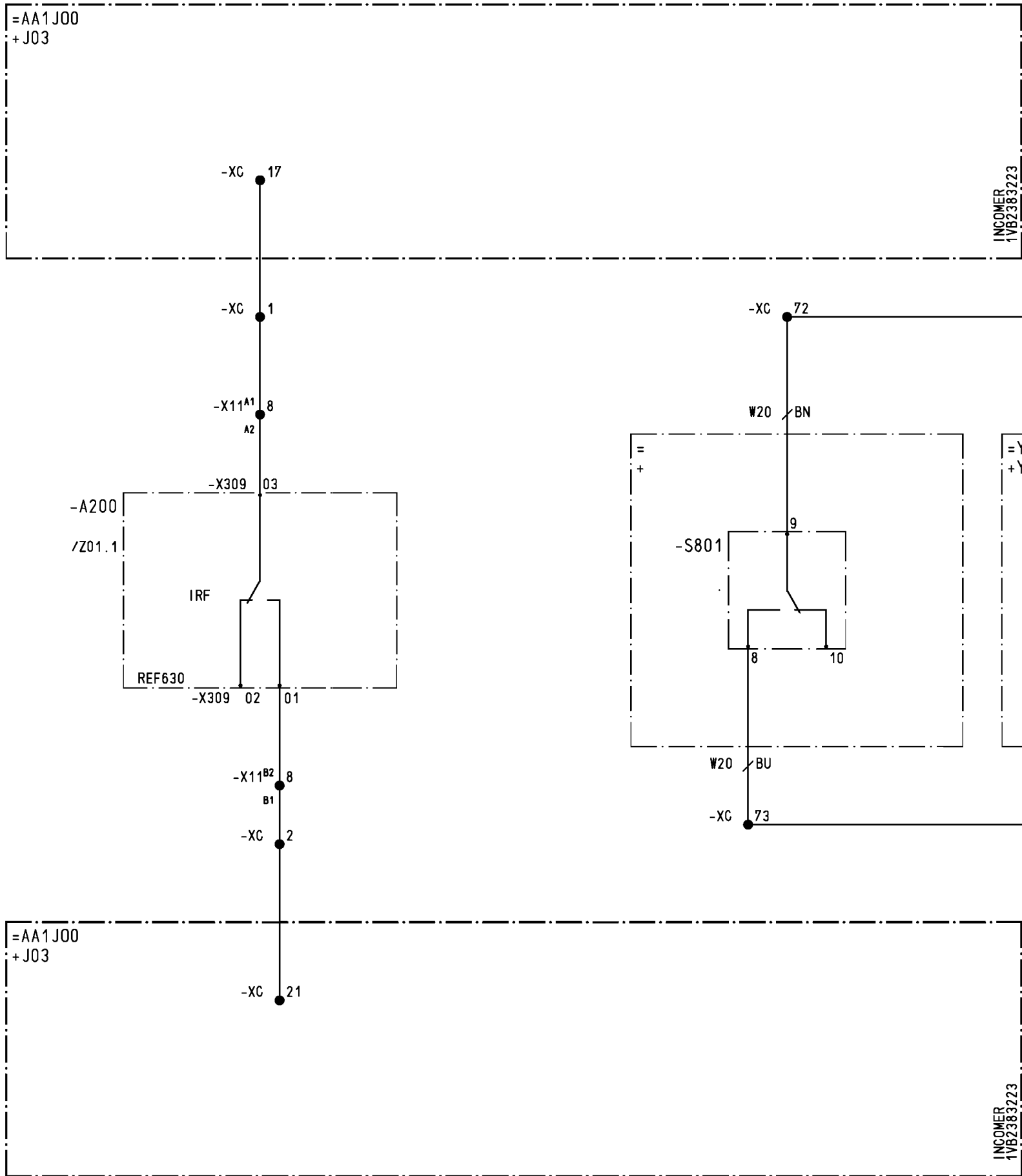


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Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	242658248	1VB2383222
1	2	3	4	5	6	7	8			

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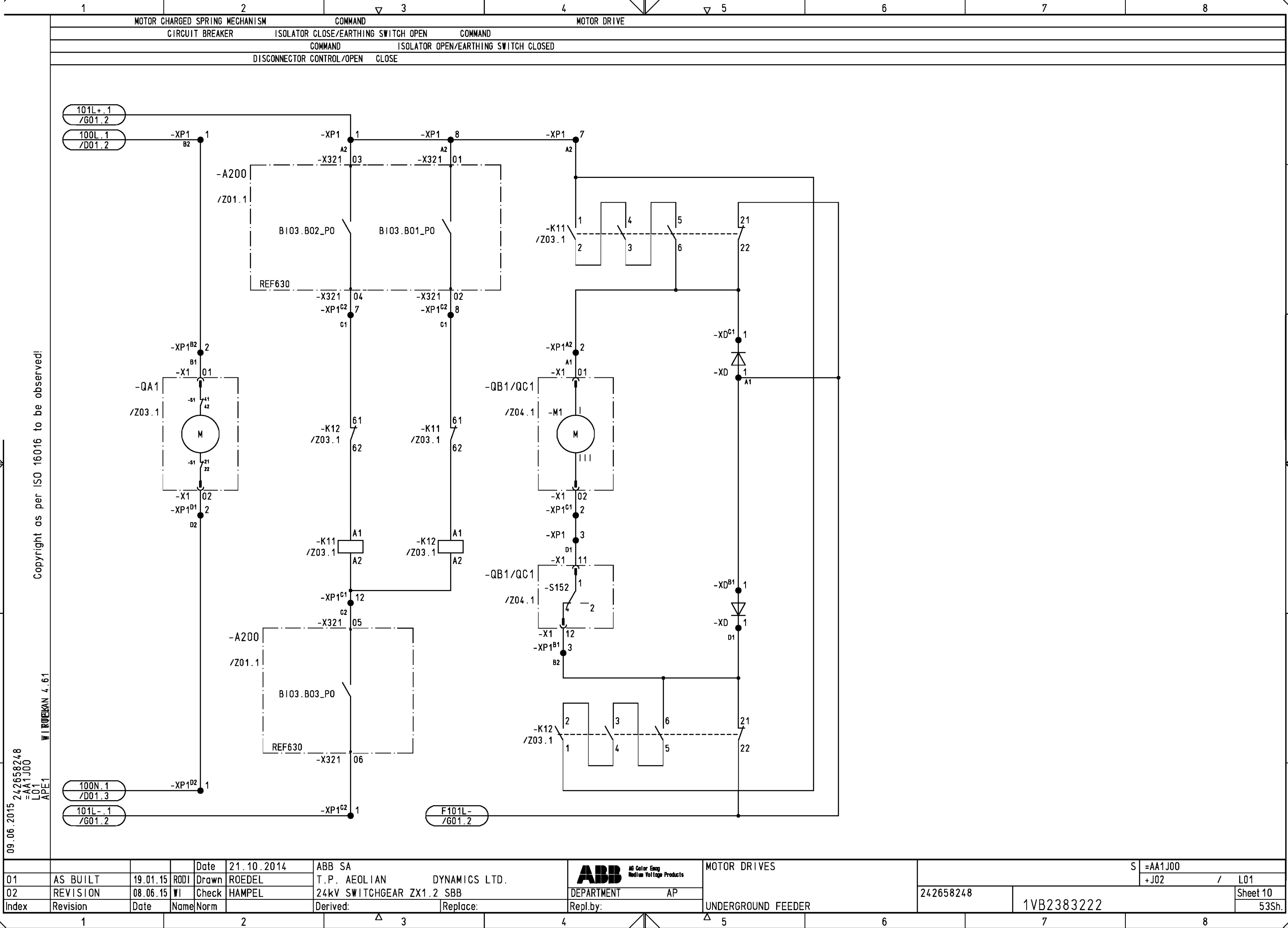
				Date	21.10.2014	ABB SA		S	=AA1J00
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN	DYNAMICS LTD.		+J02 / G02
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		242658248	Sheet 9
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	53Sh.



SELF MONITORING

UNDERGROUND FEEDER

1VB2383222



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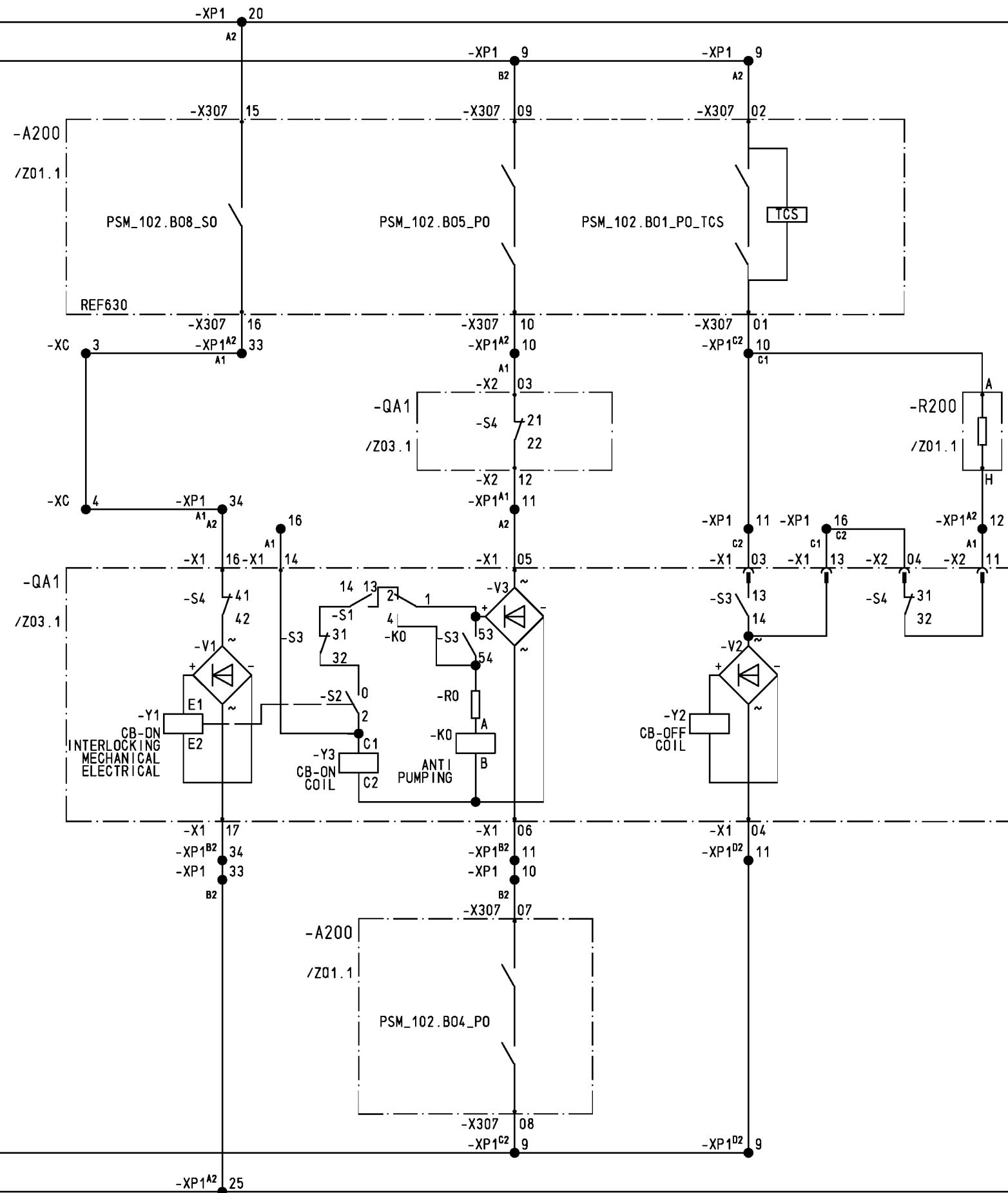
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/Q01.1

106L-.1
/G01.3

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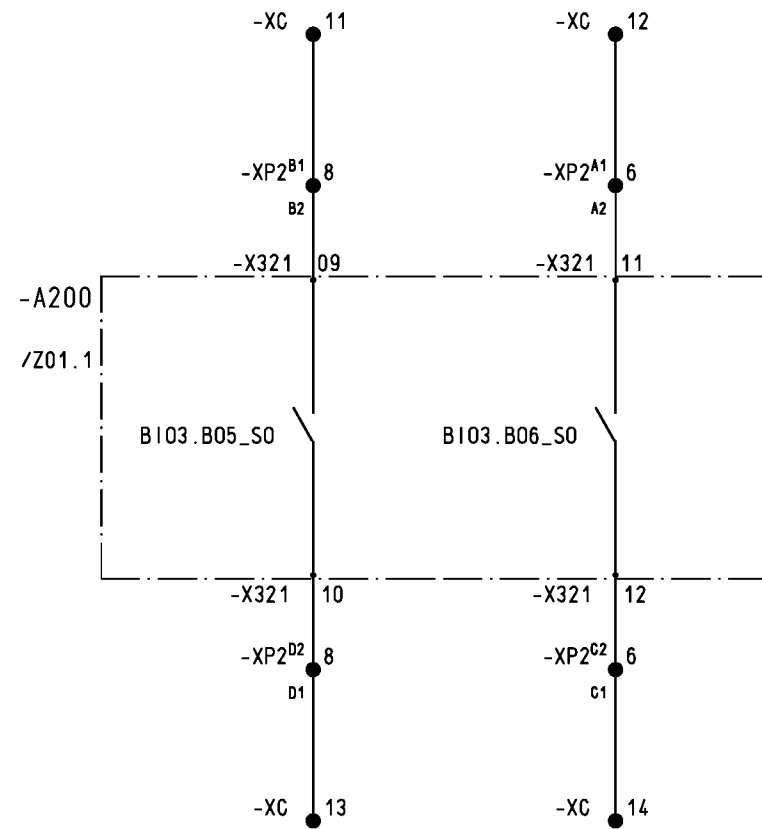
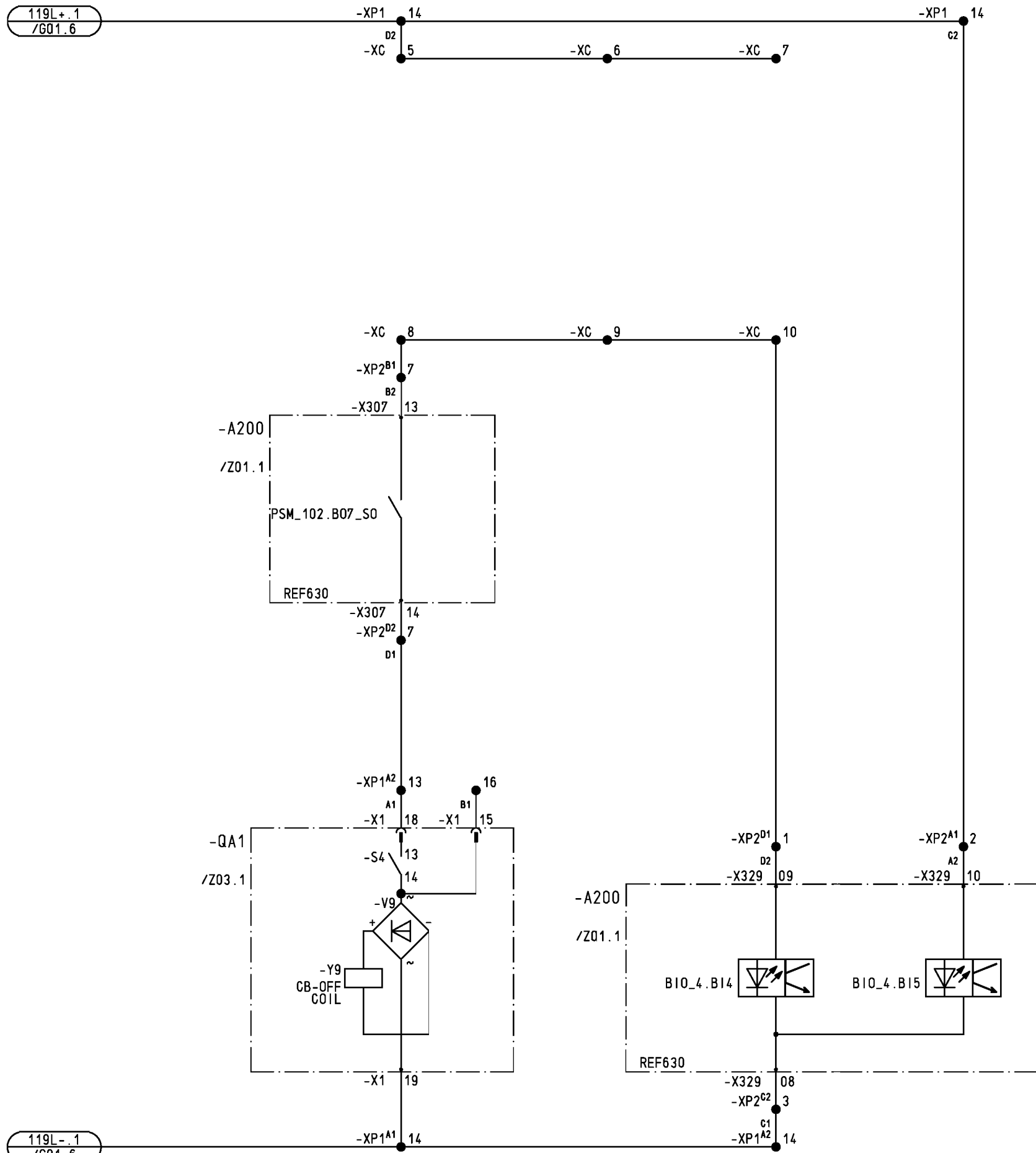
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248	
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		1VB2383222	
										Sheet 11	
										53Sh.	

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APE 1 WIRPLAN 4.61

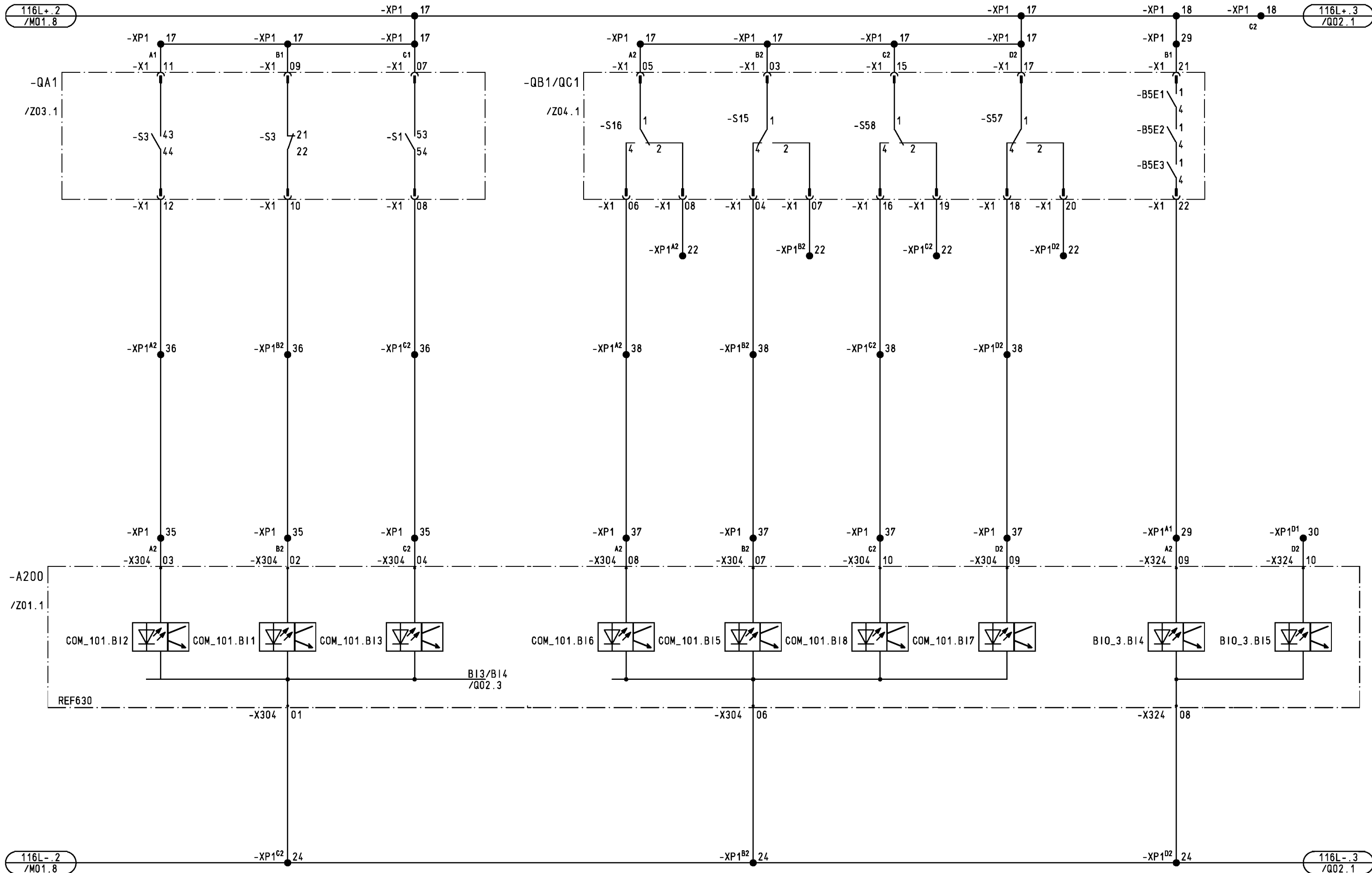


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02	REVISION	08.06.15	WI	Check	HAMPEL	DYNAMICS LTD.			
Index	Revision	Date	Name	Norm		24kV SWITCHGEAR ZX1.2 SBB			
					Derived:	Replace:	Repl.by:	242658248	Sheet 12
							UNDERGROUND FEEDER	1VB2383222	53Sh.

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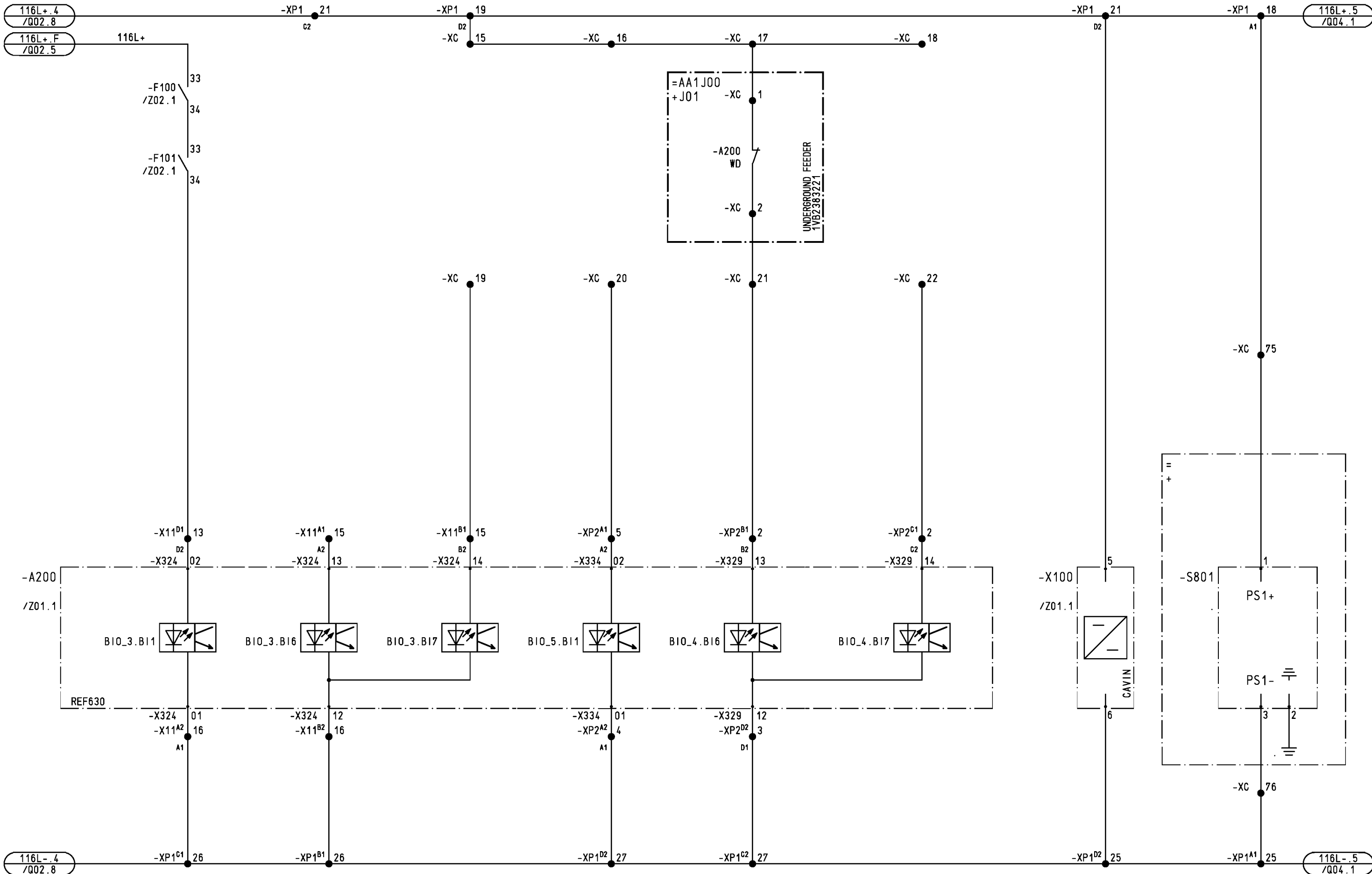


Date 21.10.2014				ABB SA				POSITION INDICATIONS				S =AA1J00			
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02 REVISION 08.06.15 WI Check HAMPEL				24kV SWITCHGEAR ZX1.2 SBB				DEPARTMENT AP				242658248			
Index Revision Date Name Norm				Derived: Replace:				Repl.by:				1VB2383222			
								UNDERGROUND FEEDER				Sheet 14			
												53Sh.			

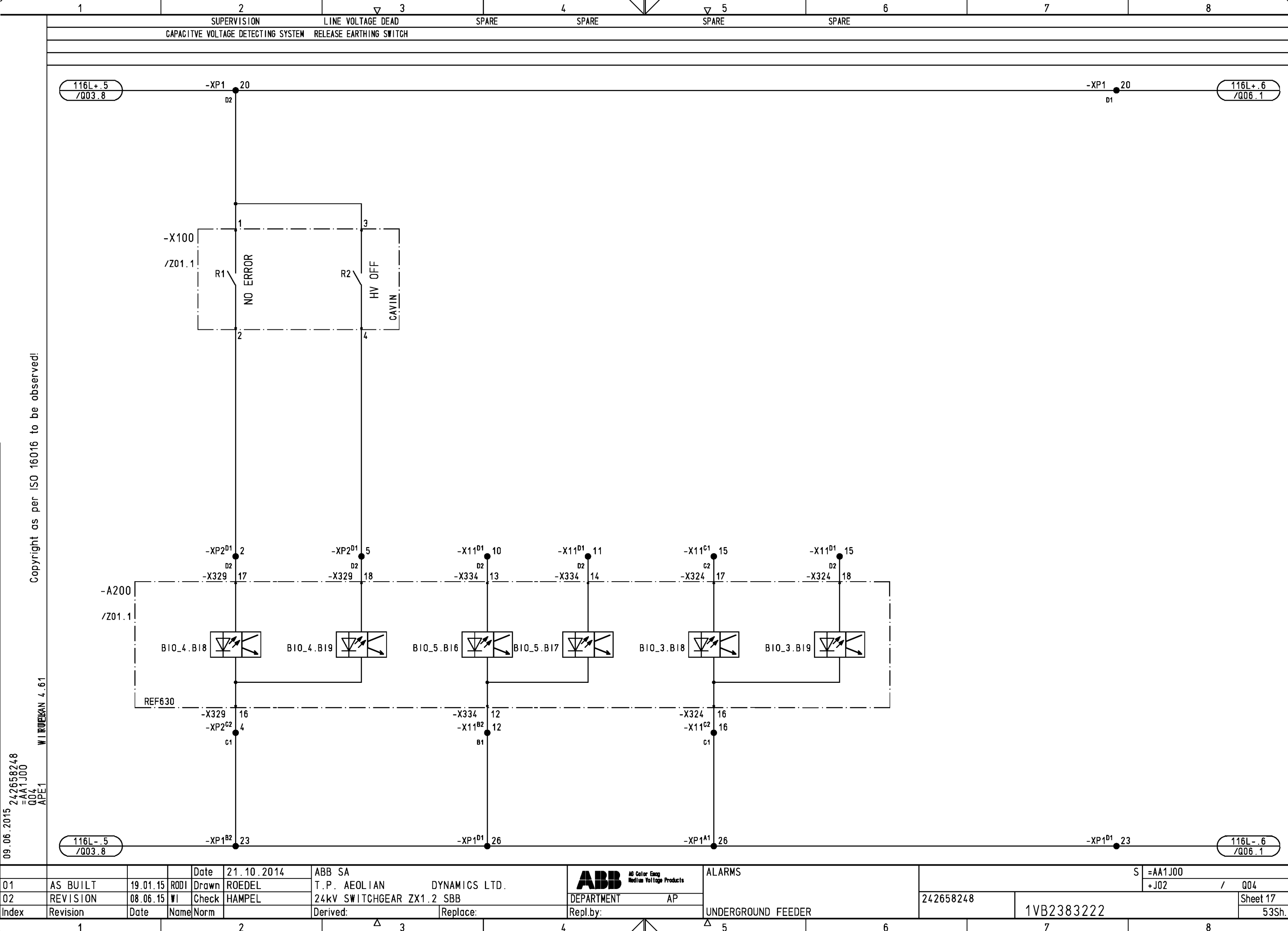
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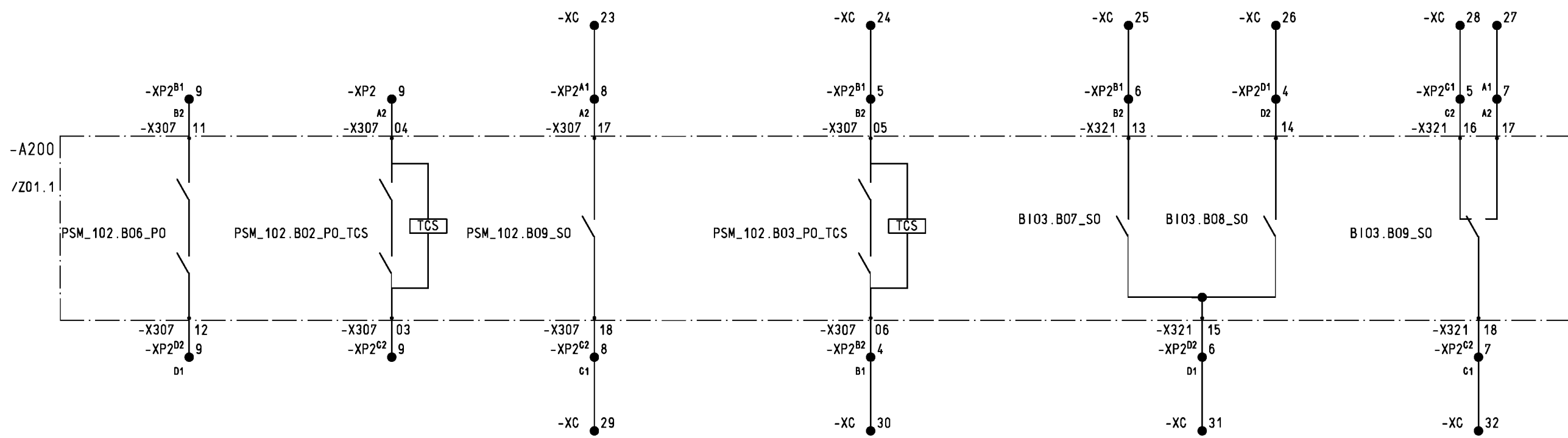
				Date	21.10.2014	ABB SA	 A6 Color Engr Medium Voltage Products	ALARMS	S =AA1J00	
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			242658248	1VB2383222
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	53Sh.



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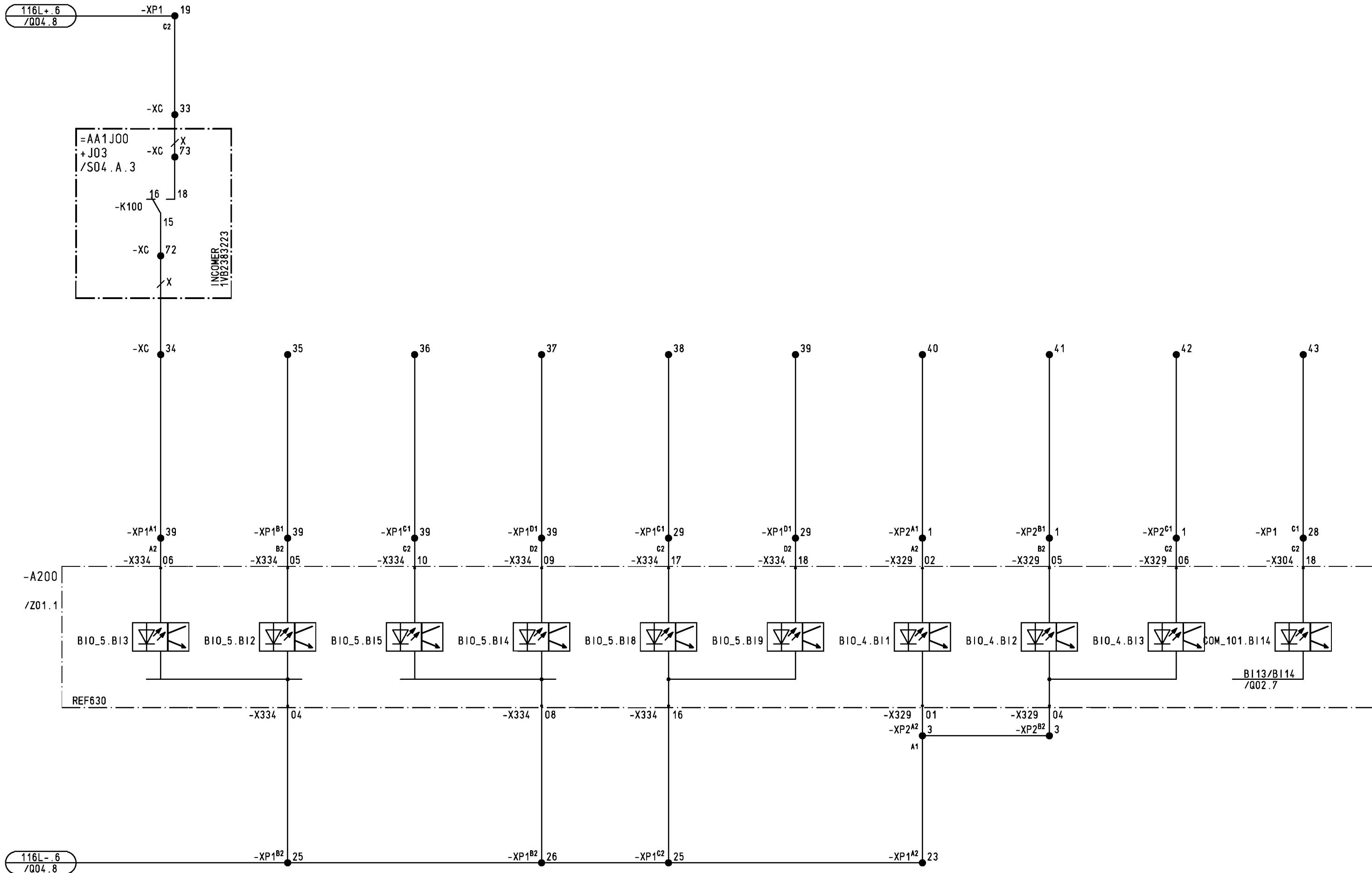


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02		REVISION	08.06.15	WI	Check	HAMPEL		T.P. AEOLIAN DYNAMICS LTD.				
Index		Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		242658248	Sheet 18
											1VB2383222	53Sh.

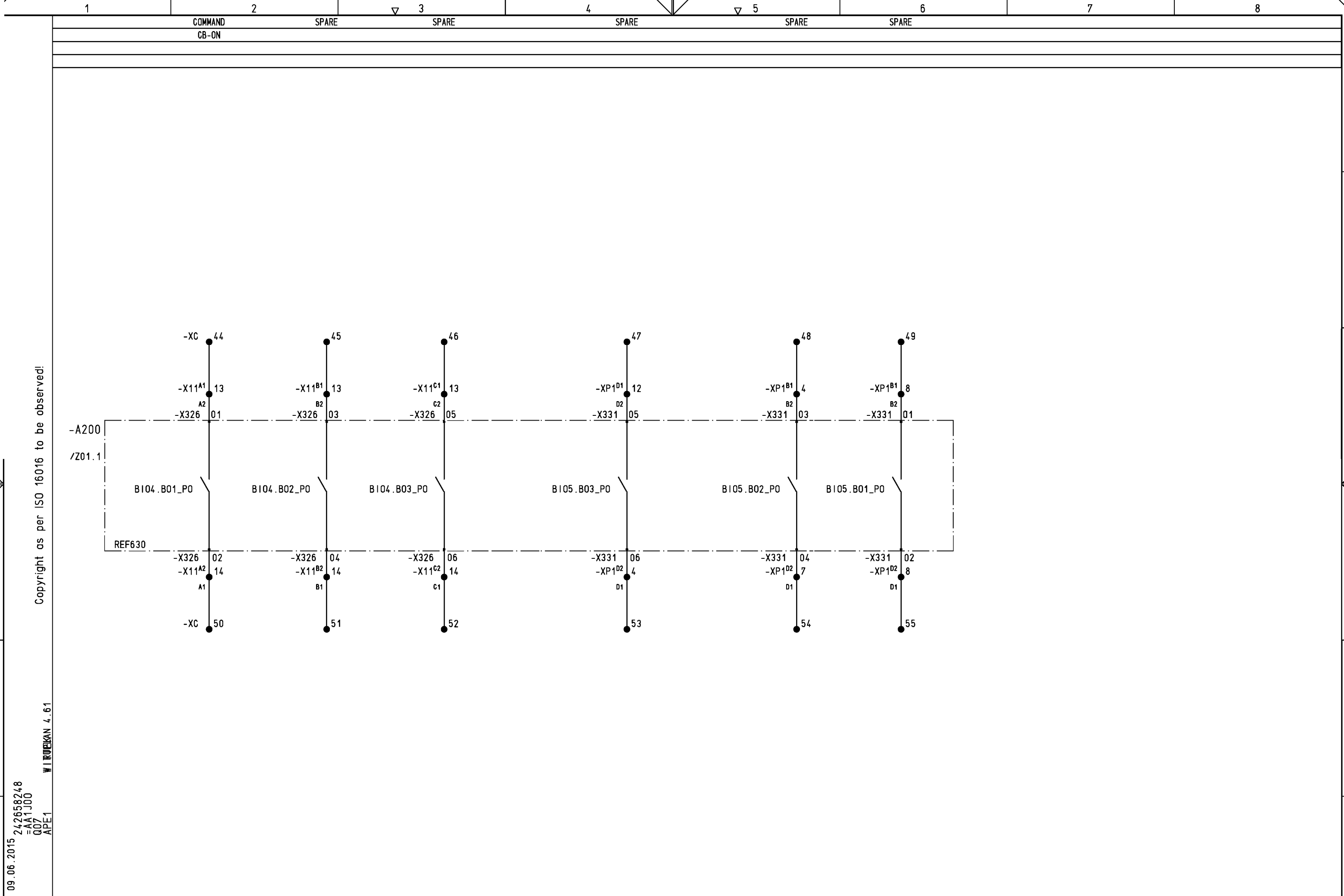
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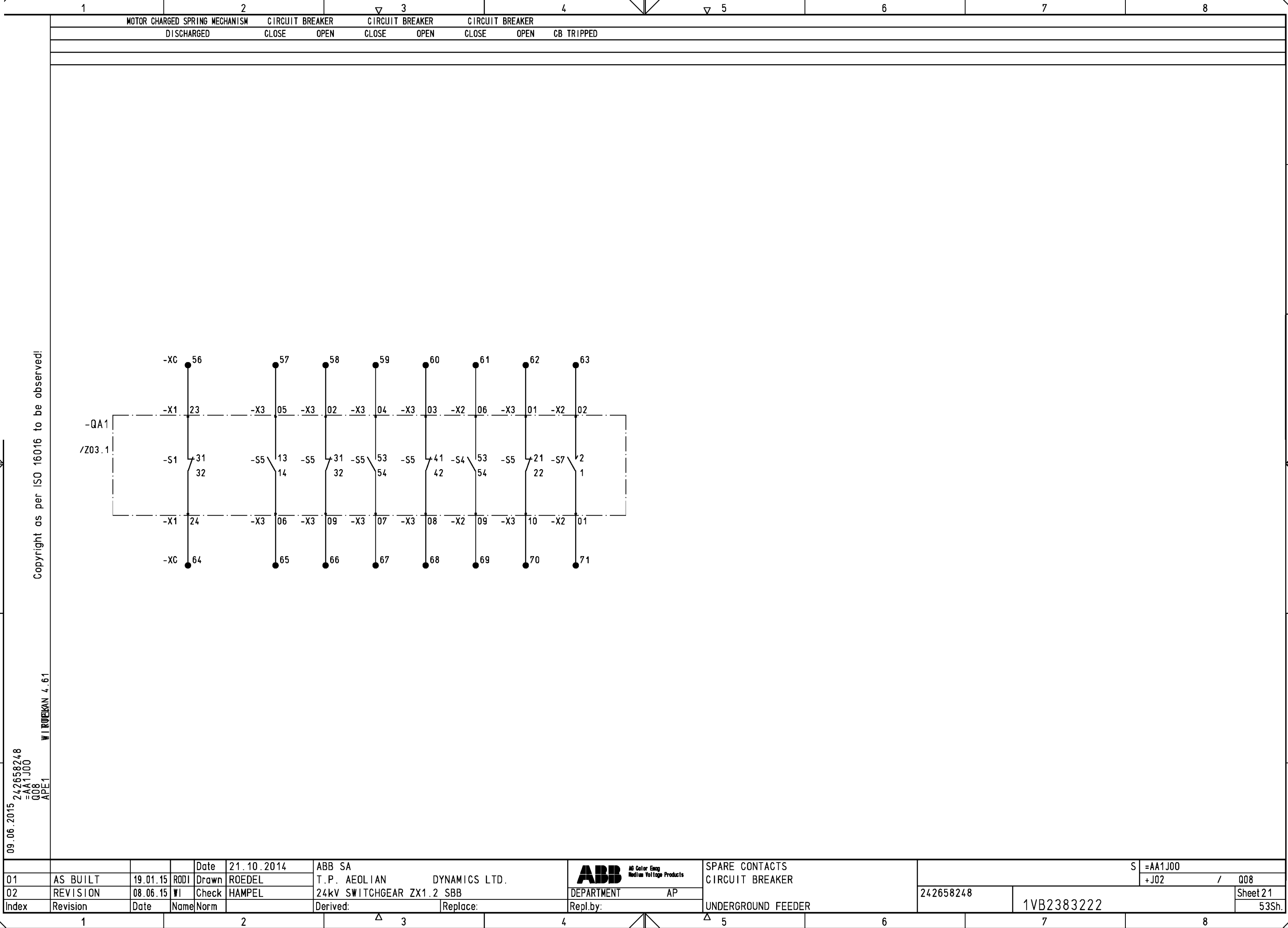
WIRUPEKAN 4.61



				Date	21.10.2014	ABB SA		SPARE CONTACTS		S =AA1J00	
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248	Sheet 19
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		1VB2383222	53Sh.



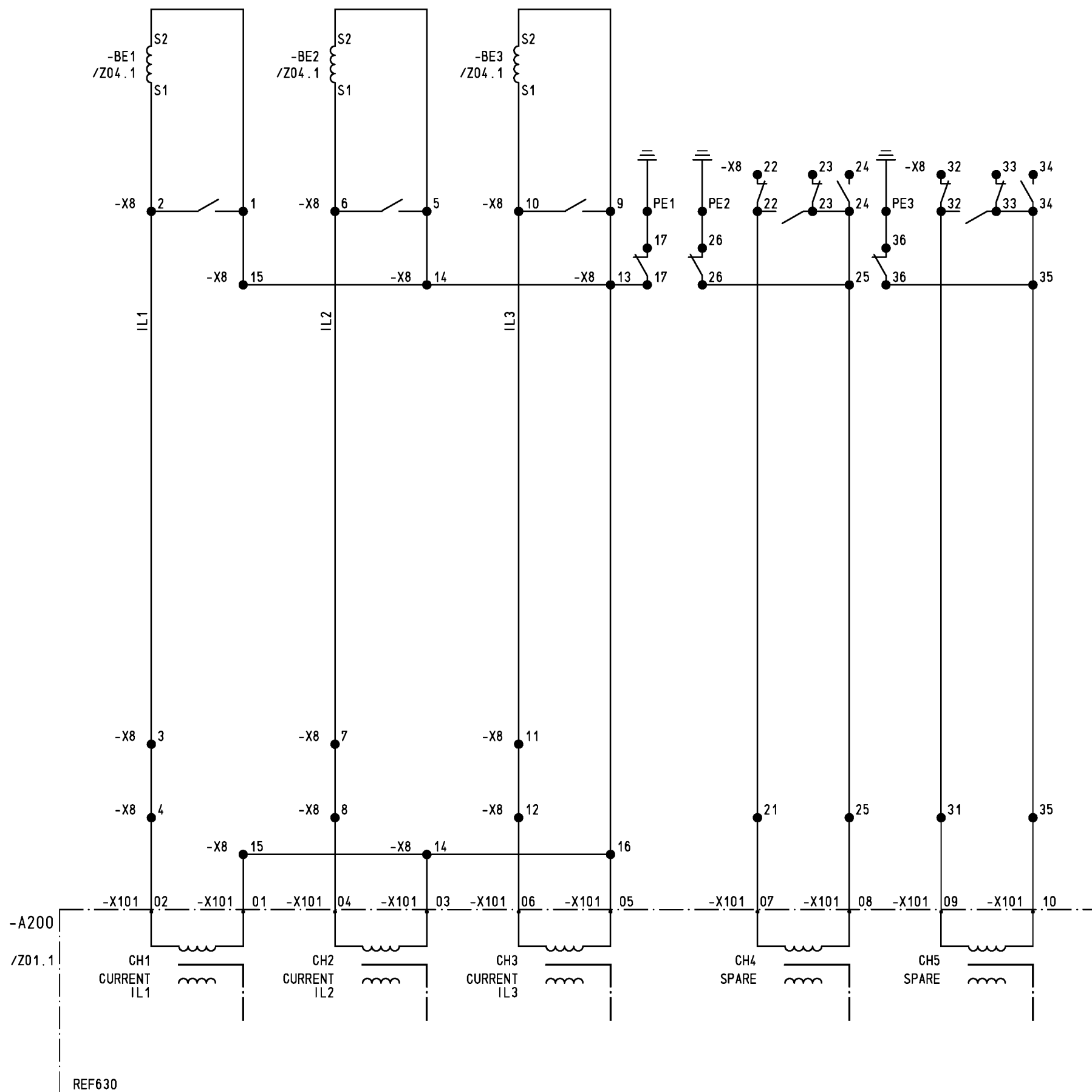
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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J02 / Q07		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
Index	Revision	Date	Name	Norm		Derived:	Repl.by:	UNDERGROUND FEEDER	242658248	1VB2383222	Sheet 20 53Sh.



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APE1

WIRTEKAN 4.61



				Date	21.10.2014	ABB SA		CURRENT TRANSFORMER CIRCUIT		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.		CORE 1		+J02 / S01	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248	Sheet 22
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383222	53Sh.

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09.06.2015 242658248
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S02
APE1

WIRTEKAN 4.61

=AA1J00
+J01

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5 A2

5 B2

5 C2

5 D2

UL1

UL2

UL3

UN

SPARE

SPARE

SPARE

SPARE

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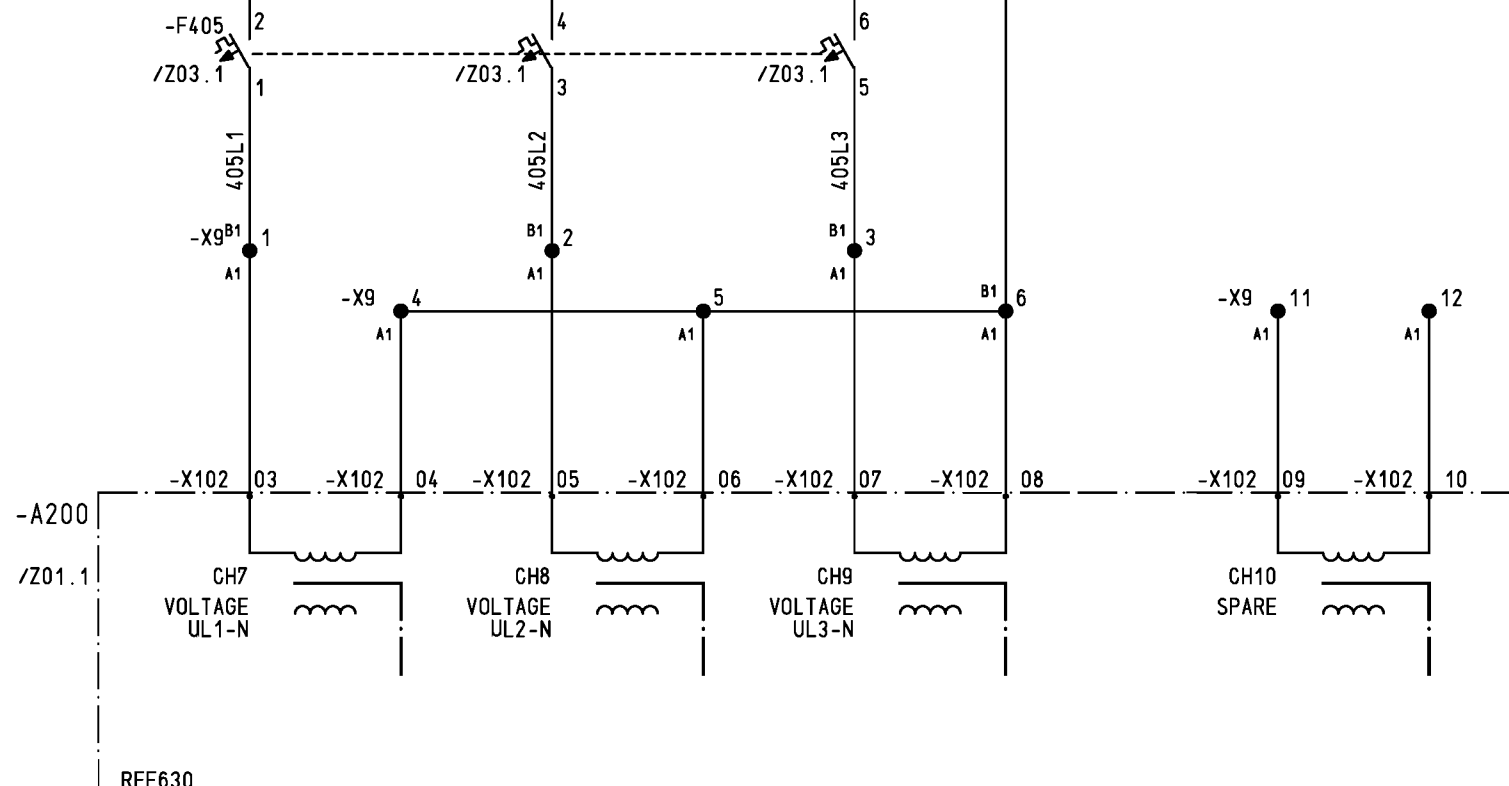
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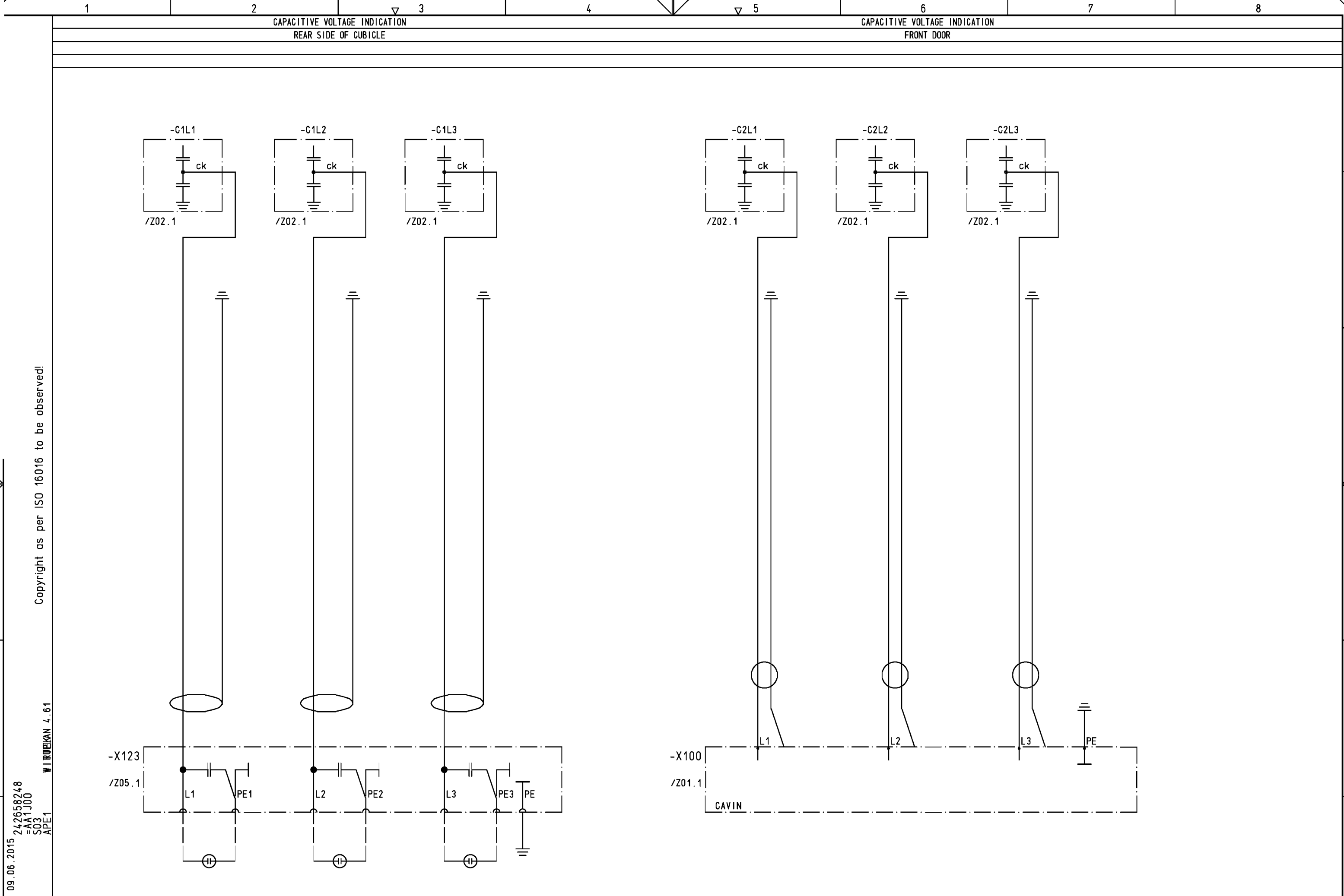
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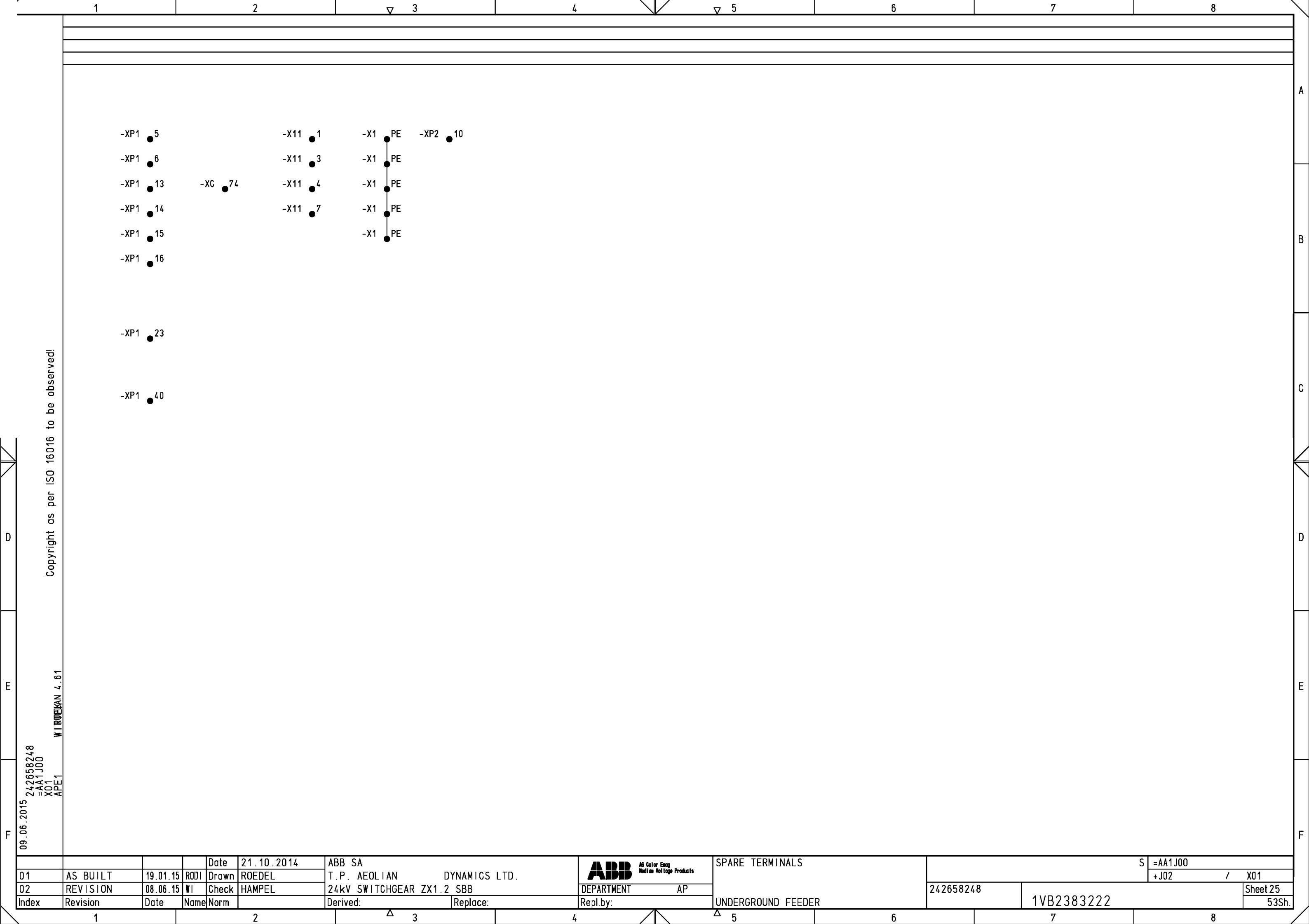


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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB	DEPARTMENT AP		242658248	1VB2383222
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:		



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APE1
WIRUNG 4.61
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				Date	21.10.2014	ABB SA	 46 Color Emag Medium Voltage Products	CAPACITIVE VOLTAGE INDICATION	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J02 / S03		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
							DEPARTMENT	AP	242658248	Sheet 24	
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383222	53Sh.



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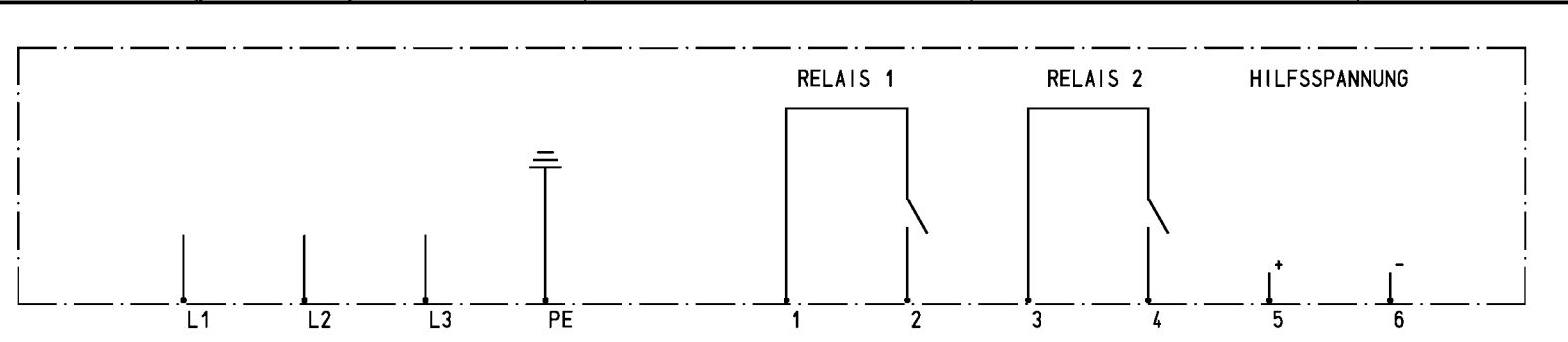
				Date	21.10.2014	ABB SA	 A6 Color Enag Medium Voltage Products	SPARE TERMINALS	S =AA1J00	
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			242658248	Sheet 25
Index	Revision	Date	Name	Norm		Derived:	Repl.by:	UNDERGROUND FEEDER	1VB2383222	53Sh.

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09.06.2015 242658248
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Z01
APE1

1	KAP.SPG.ANZEIGE CAP.VOLT.INDICATION UND MELDESYSTEM AND INDIC. SYSTEM SYSTEM SYSTEM : LRM	MAKE : JORDAN ORDER-NO: 2039113 MIT DISPLAYTEST WITH DISPLAY TEST	TYPE:CAVIN X TUEREINBAU BUILT-IN : UH= 24-230V AC/DC
TECHNICAL COMPONENT		TYPE	ORDER-NO.



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NT	-X100

/S03.5	/S03.6	/S03.7	/S03.7	/Q04.2	/Q04.2	/Q04.3	/Q04.3	/Q03.7	/Q03.7
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1	BAUGRUPPE SUBASSEMBLY	MAKE : ABB-STOTZ ORDER-NO: 1SNA 114 533 R0000	TYPE:EBS/A
BESTUECKUNG COMPONENT PARTS : WIDERSTAND		ANZAHL PIECES : 1	
TECHNICAL COMPONENT		TYPE	ORDER-NO.
2	1	ABSCHLUSSPLATTE WIDERSTAND 33 KOHM / 4W	VITROHM 591-0 1SNA113095R0400 29 E 768



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NS	-R200

/M01.7	/M01.7								
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1	FELBLEITGERAET BAY CONTROL UNIT	MAKE : ABB OY ORDER-NO: TBFNABADAAAZECNBXC	TYPE:REF630 1.2
41BI 36B0		4CT 1CT0,1/0,5A 4VT ETH.(LC) + SGF(ST)	AUFBAU 6U,DISPL. EXT
41BI 36B0		4CT 1CT0,1/0,5A 4VT ETH.(LC) + SGF(ST)	BUILT-ON 6U, D. EXT
GRUNDEINHEIT BASIC UNIT : PS 110-250=		1/0 KARTE 1/0 BOARD	: BIO 15-221V DC
TECHNICAL COMPONENT		TYPE	ORDER-NO.
1	STECKERSATZ F. 4I+5U/5I+4U (6U)		2RCA021736

REFERENCE TO GROUP DESCRIPTION IN SCHEMATIC DIAGRAM

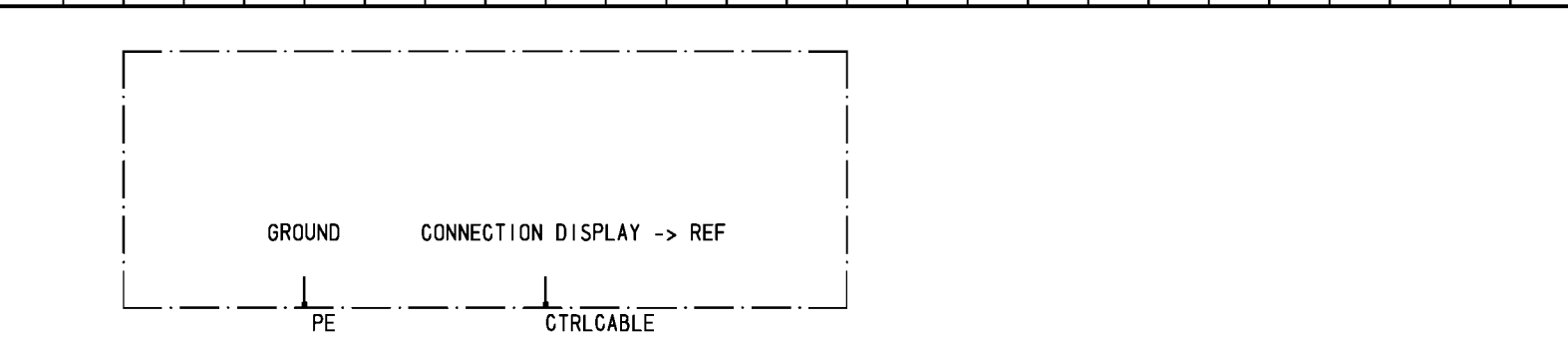
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REFERENCE TO DESCRIPTION FOR APPLIANCE CONNECTIONS

APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NS	-A200

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1	UEBERWACHUNG SUPERVISION	MAKE : ABB OY ORDER-NO:	TYPE:HMI X
BEDIENEINHEIT FUER DISPLAY FOR REF 630		REF630	
HILFSSPANNUNG AUX. VOLTAGE :		:	
TECHNICAL COMPONENT		TYPE	ORDER-NO.



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NT	-A210

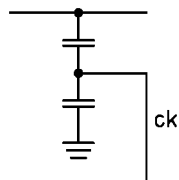
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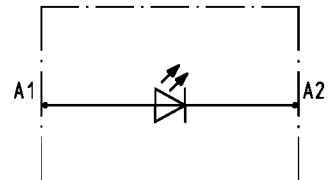
6	KAP.SPG.ANZEIGE CAP.VOLT.INDICATION	MAKE : ABB Color Emag ORDER-NO:	TYPE: X
	KAP. ABGRIFF CAP. TAP	KABELSTECKBUCHSE CABLE PLUG-IN	
	SYSTEM	:	
	TECHNICAL COMPONENT	TYPE	ORDER-NO.



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.WR	-C1L1
		+.WR	-C1L2
		+.WR	-C1L3
		+.WR	-C2L1
		+.WR	-C2L2
		+.WR	-C2L3

/S03.2																	
/S03.2																	
/S03.3																	
/S03.5																	
/S03.6																	
/S03.6																	

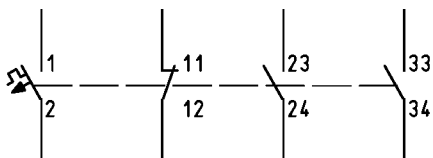
1	BELEUCHTUNG LIGHTING	MAKE : PEAKS ORDER-NO:	TYPE:LL10-W-3 X
	LED-LICHTLEISTE LED LIGHT	FARBE SCHWARZ COLOR BLACK	
	NENNSPANNUNG RATED VOLTAGE	: 110-230V ~/= 50-60HzLEISTUNG : 3W	
	TECHNICAL COMPONENT	TYPE	ORDER-NO.



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NS	-E125

/D01.6		/D01.6															
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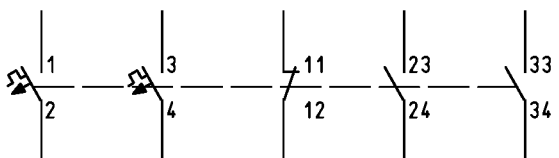
1	SICHERUNGSAUTOMAT MINIATURE C.B.	MAKE : ABB-STOTZ ORDER-NO: GHS 281 0164 R0187	TYPE:S281 UC-K0,75	
	1-POLIG SINGLE-POLE			
	NENNSTROM RATED CURRENT	: 0,75A	HILFSSCHALTER AUXILIARY SWITCH	: 2S/10E
	TECHNICAL COMPONENT	TYPE	ORDER-NO.	
1	HILFSSCHALTER	2S/10E	S2-H21	GHS 270 1936 R0001



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NS	-F100

/D01.2		/Q03.2															
--------	--	--------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

4	SICHERUNGSAUTOMAT MINIATURE C.B.	MAKE : ABB-STOTZ ORDER-NO: GHS 282 0164 R0217	TYPE:S282 UC-K1	
	2-POLIG TWO-POLE			
	NENNSTROM RATED CURRENT	: 1A	HILFSSCHALTER AUXILIARY SWITCH	: 2S/10E
	TECHNICAL COMPONENT	TYPE	ORDER-NO.	
1	HILFSSCHALTER	2S/10E	S2-H21	GHS 270 1936 R0001



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.NS	-F101
		+.NS	-F106
		+.NS	-F116
		+.NS	-F119

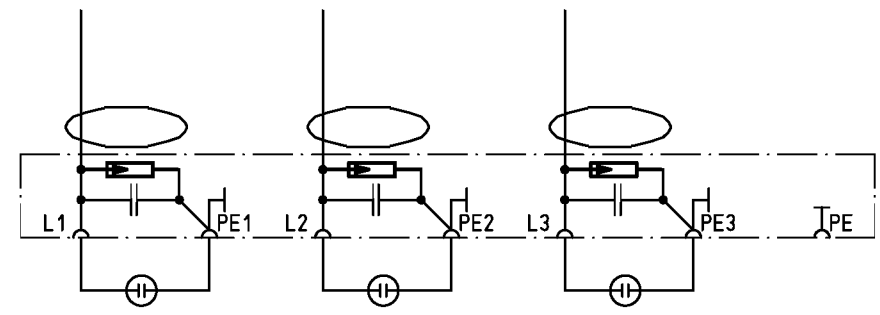
/G01.2	/G01.2	/Q02.4	/Q02.3	/Q03.2													
/G01.3	/G01.3	/Q02.4	/Q02.3														
/G01.3	/G01.4																
/G01.6	/G01.6	/Q02.4	/Q02.3														

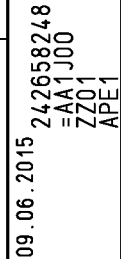
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA T.P. AEOLIAN DYNAMICS LTD.	ABB AS Color Emag Medium Voltage Products	LIST OF ITEMS	S =AA1J00 +J02 / Z02	242658248	1VB2383222	Sheet 27 53Sh.
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER				

1 SICHERUNGSAUTOMAT MINIATURE C.B. 3-POLIG THREE-POLE NENNSTROM RATED CURRENT : 0,5A HILFSSCHALTER AUXILIARY SWITCH : 2S/10E TECHNICAL COMPONENT HILFSSCHALTER 2S/10E TYPE S2-H21 ORDER-NO. GHS 270 1936 R0001																							
APPLICATION SETTING RANGE LOCATION ITEM-DESIGNATION				/S02.3 /S02.4 /S02.4 /Q02.5																			
2 UEBERWACHUNG SUPERVISION DRUCKWAECHTER FUER ZX PRESSURE MONITOR FORZX HILFSSPANNUNG AUX.VOLTAGE : TECHNICAL COMPONENT TYPE ORDER-NO.																							
APPLICATION SETTING RANGE LOCATION ITEM-DESIGNATION				/Q02.6 /Q02.7 /Q02.7 /Q02.7																			
2 SCHUETZ RELAY ABB-STOTZ 010136 DILEM-01-G NENNSPANNUNG RATED VOLTAGE 110V DC KONTAKTE CONTACTS 3S + 10E 1 1S/10E 1S/10E 11 DILE 0102240																							
APPLICATION SETTING RANGE LOCATION ITEM-DESIGNATION				+ NS -K11 + NS -K12 /L01.3 /L01.4 /L01.4 /L01.5 /L01.5 /L01.3																			
1 LEISTUNGSSCHALTER CIRCUIT BREAKER MAKE : ABB Calor Emag ORDER-NO: TYPE:VD4X X MIT HILFSSCHALTERN AUXILIARY SWITCH NENNSPANNUNG RATED VOLTAGE : SIEHE SINGLE LINE NENNSTROM RATED CURRENT : SIEHE SINGLE LINE TECHNICAL COMPONENT TYPE ORDER-NO. 1 -M0 MOTORAUFZUG 220-240V AC/DC 1 -Y1 SPERRMAGNET 110V AC/DC 1 -Y2;-Y3 A/E-AUSLOES. 110V AC/DC 1 -Y9 2.ARB-STR-AUSL. 110-125V AC/DC APPLICATION SETTING RANGE LOCATION ITEM-DESIGNATION				REFERENCE TO GROUP DESCRIPTION IN SCHEMATIC DIAGRAM REFERENCE TO SEPARATE DATA WITH DESCRIPTION DETAILS REFERENCE TO DESCRIPTION FOR APPLIANCE CONNECTIONS																			
APPLICATION SETTING RANGE LOCATION ITEM-DESIGNATION				+ LS -QA1 /L01.1 /M01.5 /M01.3 /M02.2 /Q01.1 /Q08.1 /ZZ01.1 /ZZ01.4 /ZZ01.6																			
AS BUILT 19.01.15 RODI Drawn ROEDEL REVISION 08.06.15 WI Check HAMPEL Revision Date Name Norm				21.10.2014 ABB SA T.P. AEOLIAN DYNAMICS LTD. 24kV SWITCHGEAR ZX1.2 SBB Derived: Replace:				ABB AG Calor Emag Medium Voltage Products DEPARTMENT AP Repl.by:				LIST OF ITEMS UNDERGROUND FEEDER				S =AA1J00 +J02 /Z03 242658248 1VB2383222				Sheet 28 53Sh			

[illegible]

1	2	Δ 3	4	Δ 5	6	7	8
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-QA1	01	
-X3	02	
	03	
/Z03.1	04	
	05	
	06	
	07	
	08	
	09	
	10	
	11	
	12	

				Date	21.10.2014	ABB SA	 ABB AB Color Energy Medium Voltage Products	LIST OF ITEMS CIRCUIT BREAKER	S =AA1J00		
01	AS BUILT	19.01.15	R0DI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J02 / ZZ01		
02	REVISION	08.06.15	W1	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	242658248	1VB2383222	Sheet 31 53Sh.

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WIRTEK 4.61

09.06.2015 242658248
=AA1J00
ZZ02
APE1

-QB1/QC1	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
-X1	/L01.4	/L01.4	/L01.5	/L01.5	/L01.4	/L01.4	/L01.5	/L01.5	/L02.6	/L02.6	/L01.4	/L01.4			/L01.6	/L01.6	/L01.6	/L01.6	/L01.6	/L01.7	/L01.7	/L01.7		
/Z04.1																								

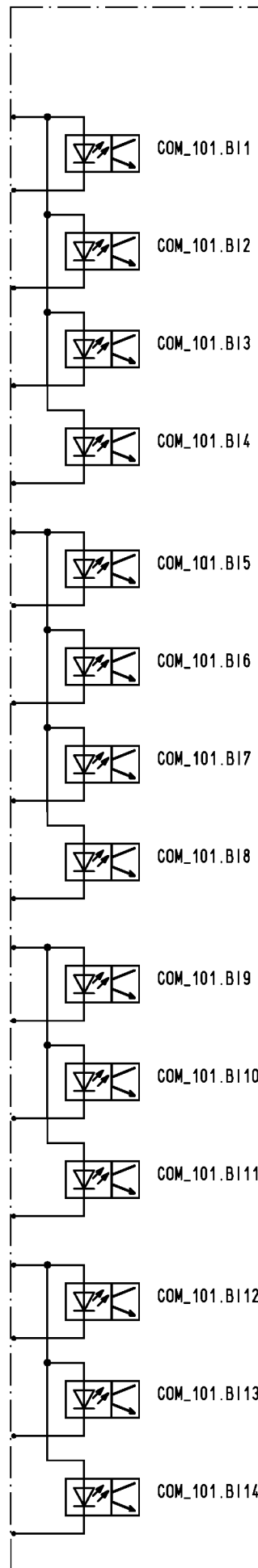
01		AS BUILT	19.01.15	RODI	Date	21.10.2014	ABB SA	T.P. AEOLIAN DYNAMICS LTD.		DEPARTMENT AP		LIST OF ITEMS THREE-POSITION SWITCH		S =AA1J00			
02		REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB							+J02 / ZZ02			
Index		Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		242658248		1VB2383222		Sheet 32	
																53Sh.	

Copyright as per ISO 16016 to be observed!

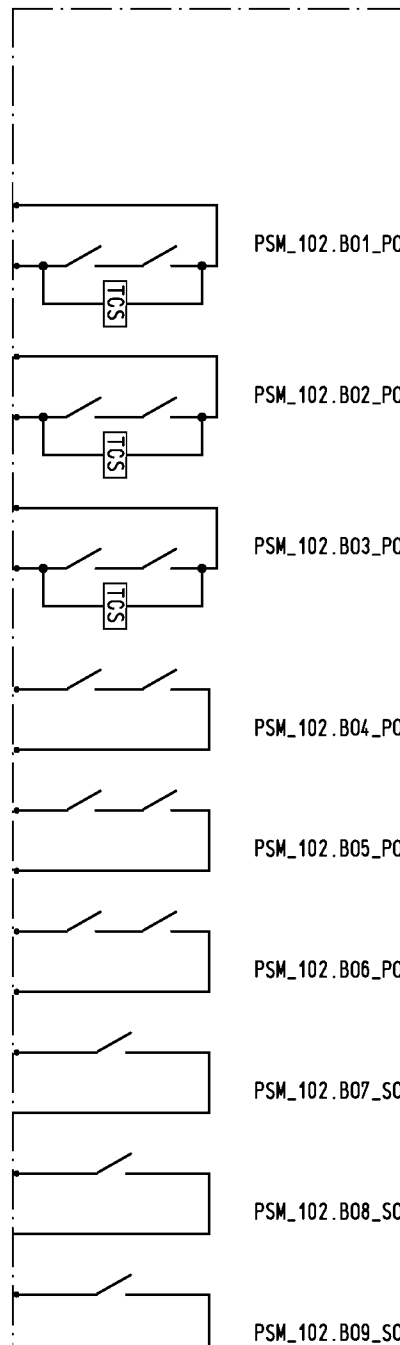
09.06.2015 242658248
=AA1J00
ZZ04
APE1

WIRING 4.61

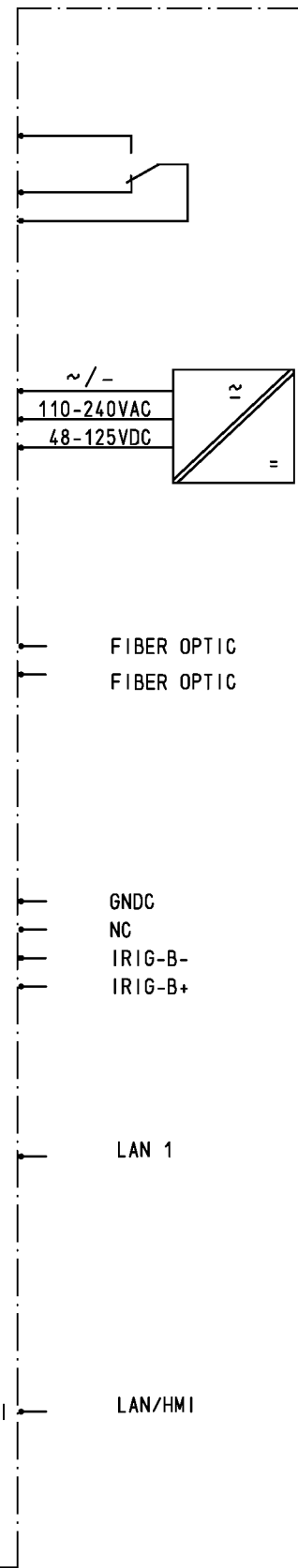
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X304	/Q01.2	01
MEANING	/Q01.2	02
POSITION INDICATION CB-OFF	/Q01.2	03
POSITION INDICATION CB-DN	/Q01.2	04
POSITION INDICATION SPRING MECHANISM CHARGED	/Q01.3	05
MCB TRIPPED CONTROL VOLTAGE	/Q02.3	06
POSITION INDICATION ISOLATOR OPEN	/Q01.5	07
POSITION INDICATION ISOLATOR CLOSE	/Q01.4	08
POSITION INDICATION EARTHING SWITCH OPEN	/Q01.6	09
POSITION INDICATION EARTHING SWITCH CLOSED	/Q01.6	10
GROUND CLOSED	/Q02.4	11
MECHANICAL BLOCKING ISOLATOR/EARTHING SWITCH BB I	/Q02.4	12
MCB TRIPPED ALTERNATING VOLTAGE	/Q02.5	13
GAS PRESSURE OK CB CUBICLE	/Q02.6	14
GAS PRESSURE OK BB COMPARTMENT I	/Q02.6	15
SPARE	/Q02.7	16
	/Q06.8	17
		18



LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X307	/M01.6	01
MEANING	/M01.6	02
COMMAND CB-OFF	/Q05.3	03
SPARE	/Q05.3	04
SPARE	/Q05.5	05
COMMAND CB-ON	/M01.5	06
COMMAND CB-ON	/M01.5	07
SPARE	/M01.5	08
PANEL NOT EARTHED	/Q05.2	09
CIRCUIT BREAKER CLOSING BLOCK. MAGNET	/M02.3	10
SPARE	/M02.3	11
	/Q05.3	12
	/Q05.3	13
	/Q05.3	14
	/Q05.3	15
	/Q05.3	16
	/Q05.3	17
	/Q05.3	18



LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X309	/G02.4	01
MEANING	/G02.4	02
WATCHDOG REF630	/G02.4	03
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X410	/Q02.3	01
MEANING	/Q02.3	03
	/Q02.3	02
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X9		TX
MEANING		RX
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X4		01
MEANING		02
		03
		04
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X1		LAN 1
MEANING		
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X0	/Q02.2	LAN/HMI
MEANING		



01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA	LIST OF ITEMS	S	=AA1J00
02	REVISION	08.06.15	WI	Check	HAMPEL	T.P. AEOLIAN	REF630 6U	+J02	/ ZZ04
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	242658248	Sheet 33
								1VB2383222	53Sh.



ABB Color Emag
Medium Voltage Products

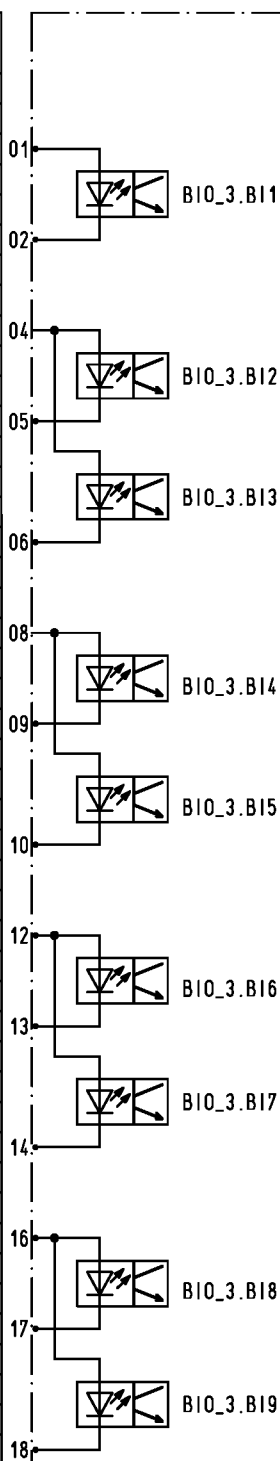
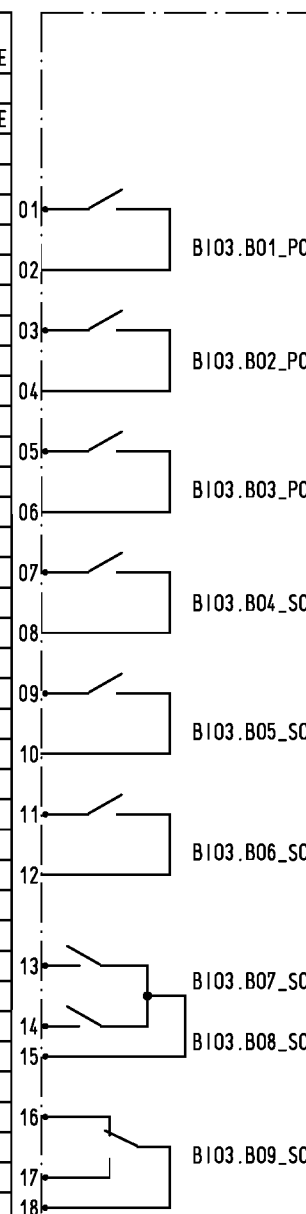
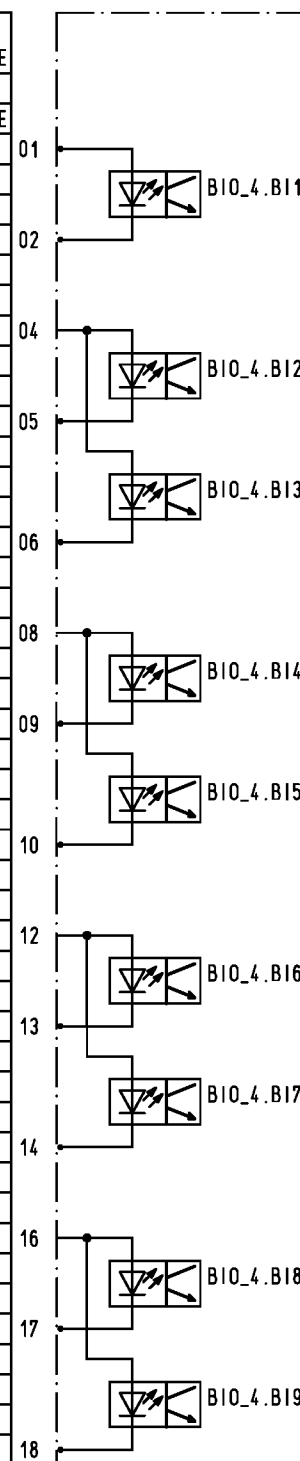
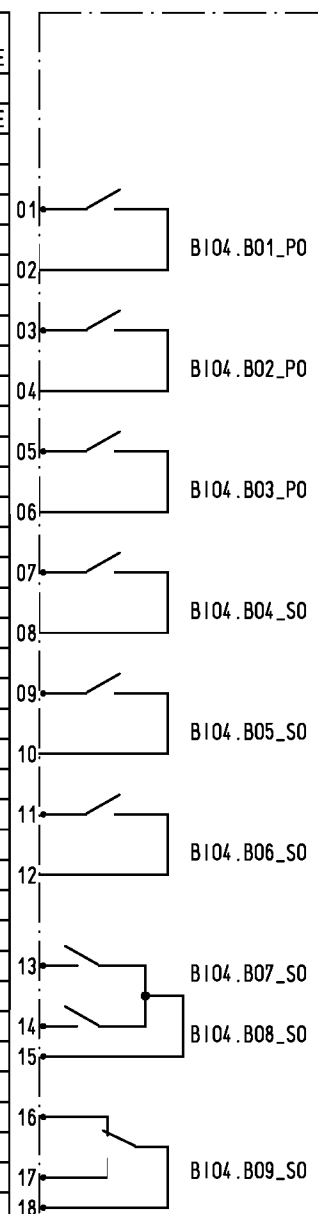
DEPARTMENT AP

UNDERGROUND FEEDER

WIRPEKAN 4.61

		S	=AA1J00	
			+J02	/ ZZ05
242658248	1VB2383222			Sheet 34
				53Sh.

LOCATION		STATUS	ITEM-
ITEM-DES.	-A200		REFERENCE
TERMINAL	-X324		/Z01.1
MEANING			REFERENCE
MCB TRIPPED			/Q03.2
MOTOR VOLTAGE			
			/Q03.2
			/P01.4
SPARE			
			/P01.4
SPARE			
			/P01.4
			/Q01.7
REEDCONTACTS			
EARTHING SWITCH CLOSED			
			/Q01.7
SPARE			
			/Q01.8
			/Q03.3
SPARE			
			/Q03.3
SPARE			
			/Q03.3
			/Q04.5
SPARE			
			/Q04.5
SPARE			
			/Q04.6

[illegible][illegible][illegible]

				Date	21.10.2014	ABB SA		 ABB AS Color Emag Medium Voltage Products	LIST OF ITEMS REF630 6U	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J02 / ZZ06		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
						DEPARTMENT		AP		242658248		Sheet 35
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER		1VB2383222	53Sh.

WIRUPEKAN 4.61

The timing diagram for the B105 module shows the following signals and their transitions:

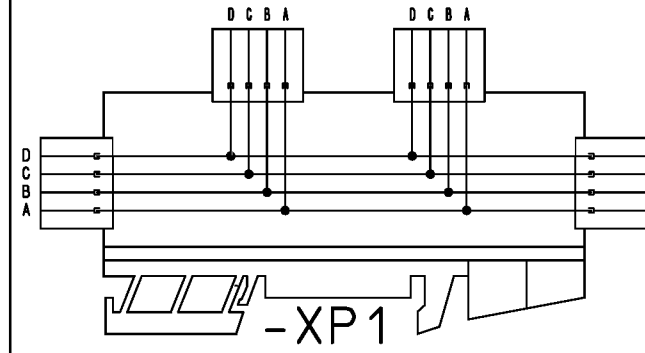
- B01_PO:** Signal 01 (input) transitions from low to high, then back to low. Signal 02 (output) transitions from low to high when 01 is high, and back to low when 01 returns to low.
- B02_PO:** Signal 03 (input) transitions from low to high, then back to low. Signal 04 (output) transitions from low to high when 03 is high, and back to low when 03 returns to low.
- B03_PO:** Signal 05 (input) transitions from low to high, then back to low. Signal 06 (output) transitions from low to high when 05 is high, and back to low when 05 returns to low.
- B04_SO:** Signal 07 (input) transitions from low to high, then back to low. Signal 08 (output) transitions from low to high when 07 is high, and back to low when 07 returns to low.
- B05_SO:** Signal 09 (input) transitions from low to high, then back to low. Signal 10 (output) transitions from low to high when 09 is high, and back to low when 09 returns to low.
- B06_SO:** Signal 11 (input) transitions from low to high, then back to low. Signal 12 (output) transitions from low to high when 11 is high, and back to low when 11 returns to low.
- B07_SO:** Signal 13 (input) transitions from high to low. Signal 14 (output) transitions from high to low when 13 is high, and back to high when 13 returns to low. Signal 15 (output) transitions from high to low when 13 is high, and back to high when 13 returns to low.
- B08_SO:** Signal 14 (input) transitions from high to low. Signal 15 (output) transitions from high to low when 14 is high, and back to high when 14 returns to low.
- B09_SO:** Signal 16 (input) transitions from high to low. Signal 17 (output) transitions from high to low when 16 is high, and back to high when 16 returns to low. Signal 18 (output) transitions from high to low when 16 is high, and back to high when 16 returns to low.

LIST OF ITEMS REF630 6U	S =AA1J00 +J02 / ZZ07		
UNDERGROUND FEEDER	242658248	1VB2383222	Sheet 36 53Sh.

[illegible]

1				2				3				4				5				6				7				8								
SCHEMATIC DIAGRAM				ASSEMBLY												WIRING MATERIAL				Ø COLOUR				TERMINAL TYPE												
																STANDARD: H07Z-K 2,5mm2 BK				STANDARD: WEIDM. ZRV2,5/4																
																POS.: 0 H05Z-K 1,0mm2 BK				82 H07Z-K 1,5mm2 BK																
REFERENCE	POS.	POT	CORES	A	POS.	POT	CORES	B	POS.	POT	CORES	C	POS.	POT	CORES	D	POS.	POT	CORES	D	CORES	POT	POS.	C	CORES	POT	POS.	B	CORES	POT	POS.	A	CORES	POT	POS.	
/X01.2																																				
				=AA1J00J01 -X11 2A4				=AA1J00J01 -X11 2B4				=AA1J00J01 -X11 2C4				=AA1J00J01 -X11 2D4								=AA1J0003 -X11 2D3			=AA1J0003 -X11 2C3			=AA1J0003 -X11 2B3			=AA1J00J03 -X11 2A3			
/P01.2																																				
/X01.2																																				
/X01.2																																				
/X01.2																																				
/S02.3		UL1		=AA1J00J01 -X11 5A4		UL2		=AA1J00J01 -X11 5B4		UL3		=AA1J00J01 -X11 5C4		UN		=AA1J00J01 -X11 5D4								=AA1J0003 -X11 5D3		UN	=AA1J0003 -X11 5C3		UL3	=AA1J0003 -X11 5B3		UL2	=AA1J00J03 -X11 5A3		UL1	
/S02.2		SPARE		=AA1J00J01 -X11 6A4		SPARE		=AA1J00J01 -X11 6B4		SPARE		=AA1J00J01 -X11 6C4		SPARE		=AA1J00J01 -X11 6D4								=AA1J0003 -X11 6D3		SPARE	=AA1J0003 -X11 6C3		SPARE	=AA1J0003 -X11 6B3		SPARE	=AA1J00J03 -X11 6A3		SPARE	
/X01.2																																				
/G02.4				-XC 1				-XC 2																												

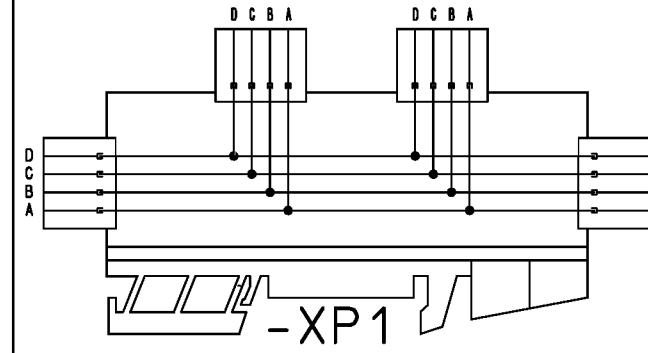
[illegible]



Standard:	H05Z-K	1,0mm2	BK
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Standard: WEIDM. ZRV2,5/4

Reference	Line	Pot	q	A1	Line	Pot	q	B1	Line	Pot	q	C1	Line	Pot	q	D1	D2	C2	B2	A2	D1	C1	B1	A1	Nr	D2	q	Pot	Line	C2	q	Pot	Line	B2	q	Pot	Line	A2	q	Pot	Line
/G01.2		101L+		-F101 1		100L		-F100 1		101L-	q	-F101 3		1N		-X10 6 A1								1				-A200 -X321 06		101L-			-A200 -X321 03		101L+						
/L01.4				-QB1/QC1 -X1 01		100L		-QA1 -X1 01				-QB1/QC1 -X1 02		100N		-QA1 -X1 02								2							-K11 6	q									
/L01.4								-QB1/QC1 -X1 12								-QB1/QC1 -X1 11								3						-K12 6	q										
/Q07.5								-XC 48								-XC 53								4	-A200 -X331 06				-A200 -X331 03												
/X01.1																																									



D C B A	Wiring Material			Ø Colour	Terminal Type
	Standard:	H05Z-K	1,0mm ²	BK	Standard: WEIDM. ZRV2,5/4
	POS.:	82	H07Z-K	1,5mm ²	BK

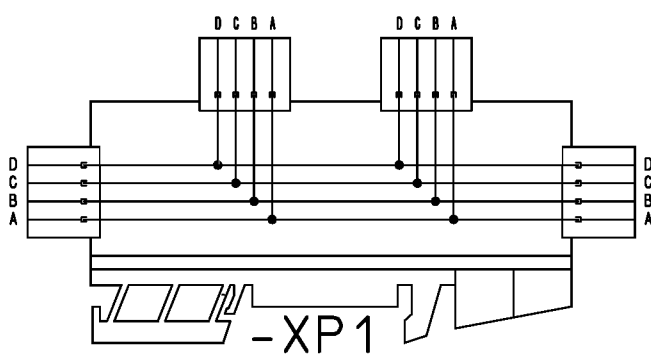
Refe- rence	Line Pot q	A1	Line Pot q	B1	Line Pot q	C1	Line Pot q	D1	D2	C2	B2	A2	D1	C1	B1	A1	Nr	D2	q Pot Line	C2	q Pot Line	B2	q Pot Line	A2	q Pot Line	
/M01.6			106L+	-F106 1			106L-	-F106 3	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	● — ● ○ ○ ○ ○	○ — ● ○ ○ ○ ○			9			-A200 -X307 08	106L-	-A200 -X307 09	106L+	-A200 -X307 02	106L+	
/M01.5		-QA1 -X2 03				-R200 A			○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ — ● ● ○ ○ ○	● — ● ○ ○ ○ ○	○ ○ ○ ○ ○ ○		10			-A200 -X307 01		-A200 -X307 07		-A200 -X307 10		
/M01.5		-QA1 -X2 12							○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	● — ● ○ ○ ○ ○	● — ● ○ ○ ○ ○	○ ○ ○ ○ ○ ○		11	-QA1 -X1 04	106L-	-QA1 -X1 03		-QA1 -X1 06		-QA1 -X1 05		
/M01.7		-QA1 -X2 11			q	-K11 A2		-XC 47	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○		12	-A200 -X331 05		-A200 -X321 05				-R200 H		
/M02.3		-QA1 -X1 18			82	-E125 A1			○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○		13			-S125 2		82		-XP2 7D1		
/M02.3	119L-	-QA1 -X1 19	119L-	-F119 3			119L+	-F119 1	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ — ● ○ ○ ○ ○	○ — ● ○ ○ ○ ○			14	-XC 5	119L+	-XP2 2A1	119L+			-XP2 3C1	119L-	
/X01.1									○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○		15									
/M01.4		-QA1 -X1 14		-QA1 -X1 15		-QA1 -X1 13			○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○	○ ○ ○ ○ ○ ○		16			-QA1 -X2 04						

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RUPLAN 4.61

09.06.2015
242658248
=AA1J00+J02
006
APE1

				Date	21.10.2014	ABB SA		 ABB AB Color Comp Medium Voltage Products	X-PLUG -XP1	V =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J02 /006		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
						DEPARTMENT AP			242658248	Sheet 42		
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	UNDERGROUND FEEDER	1VB2383222	53Sh.	

1		2		3		4		5		6		7		8																			
Circuit Diagram										Wiring Material		Ø Colour		Terminal Type																			
										Standard: H05Z-K		1,0mm2 BK		Standard: WEIDM. ZRV2,5/4																			
										POS.: 0 H05Z-K		1,0mm2 BK																					
Reference	Line	Pot	q	A1	Line	Pot	q	B1	Line	Pot	q	C1	Line	Pot	q	D1	Line	Pot	q	A2	Line	Pot	q	B2	Line	Pot	q	A2	Line	Pot	q		
/Q01.2		116L+		-QA1 -X1 11		116L+		-QA1 -X1 09		116L+		-QA1 -X1 07																					
/Q03.8	0	116L+		-XC 75																													
/G01.3		116L+		-F116 1																													
/M01.4																																	
/Q02.3																																	
/Q01.5																																	
/Q06.6																																	
/G01.4		116L-		-F116 3																													
/Q03.8	0	116L-		-XC 76																													
/Q04.5		116L-		-X11 16C1	0	116L-		-X11 16	0	116L-		-X11 16 A1	116L-																				
/Q02.7		116L+		-B0G 4		116L+		-B1G 4																									
/Q02.6				-B0G 6				-B1G 6				-XC 43																					
/Q01.7				-QB1/QC1 -X1 22		116L+		-QB1/QC1 -X1 21				-XC 38				-XC 39																	
/Q02.6		116L+		-QB1/QC1 -X1 09																													
/Q02.6				-QB1/QC1 -X1 10																													
/Q02.4				-F119 12				-F119 23				-F405 24																					

[illegible]

09.06.2015
242658248
=AA1J00+J02
010
APE1

RUPLAN 4.61

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1		2		3		4		5		6		7		8									
NO	CABLE	USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE											
										STANDARD: H07Z-K 2,5mm2 BK POS.: 88 H07Z-K 4,0mm2 GN/YE		STANDARD: WEIDM. WTL6/1NEU/ABB PE1-PE3 PHOE. USLK66N											
								ASSEMBLY															
														-X8									
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK		
/S01.3																-X8	15	N		-BE1	S2		
/S01.3																		IL1		-BE1	S1		
/S01.3																		IL1					
/S01.3																-A200	-X101	02	IL1				
/S01.4																-X8	14	N		-BE2	S2		
/S01.4																		IL2		-BE2	S1		
/S01.4																		IL2					
/S01.4																-A200	-X101	04	IL2				
/S01.5																-X8	13	N		-BE3	S2		
/S01.5																		IL3		-BE3	S1		
/S01.5																		IL3					
/S01.5																-A200	-X101	06	IL3				
/S01.5																		N		-X8	9		
/S01.4																-A200	-X101	03	N		-X8	5	
/S01.3																-A200	-X101	01	N		-X8	1	
/S01.5																-A200	-X101	05	N				
/S01.5																88	-X8	PE1	N				
/S01.6																-A200	-X101	07					
/S01.6																							
/S01.6																							
/S01.6																-X8	25						
/S01.6																-A200	-X101	08			-X8	24	
/S01.5																88	-X8	PE2	N				
/S01.6																-A200	-X101	09					
/S01.6																							
/S01.7																							
/S01.7																-X8	35						
/S01.7																-A200	-X101	10			-X8	34	
/S01.6																88	-X8	PE3	N				
/S01.5																			N		-X8	17	88
/S01.5																			N		-X8	26	88
/S01.6																			N		-X8	36	88

ABB
AG Color Emag
Medium Voltage Products

DEPARTMENT AP

Repl.by:

TERMINAL DIAGRAM -X8

UNDERGROUND FEEDER

V =AA1J00
+J02 /010

242658248

1VB2383222

Sheet 46
53Sh.

1		2		3		4		5		6		7		8	
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	TERMINAL DIAGRAM -X8		UNDERGROUND FEEDER		V =AA1J00 +J02 /010		Sheet 46 53Sh.	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.									
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB									

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09.06.2015
242658248
=AA1J00+J02
012
APE1

RUPLAN 4.61

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1		2		3		4		5		6		7		8							
NO	CABLE	USED CORES	SHEET/COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE									
1	X		2					-XC		STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WDU1,5/BLZ/5.08/ZQV									
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK
/G02.4															=AA1J00 +J03 -XC	17		-X11	8A1		
/G02.5															=AA1J00 +J03 -XC	21		-X11	8B1		
/M01.3																		-XP1	33A1		
/M01.3																		-XP1	34A1		
/M02.3																		-XP1	14D2		
/M02.3																					
/M02.4																					
/M02.3																		-XP2	7B1		
/M02.3																					
/M02.4																		-XP2	1D1		
/M02.7																		-XP2	8B1		
/M02.7																		-XP2	6A1		
/M02.7																		-XP2	8D1		
/M02.7																		-XP2	6C1		
/Q03.3																		-XP1	19D2		
/Q03.4																					
/Q03.5															=AA1J00 +J01 -XC	1					
/Q03.6																					
/Q03.3																		-X11	15B1		
/Q03.4																		-XP2	5A1		
/Q03.5															=AA1J00 +J01 -XC	2		-XP2	2B1		
/Q03.6																		-XP2	2C1		
/Q05.3																		-XP2	8A1		
/Q05.5																		-XP2	5B1		
/Q05.6																		-XP2	6B1		
/Q05.6																		-XP2	4D1		
/Q05.7																		-XP2	7A1		
/Q05.7																		-XP2	5C1		
/Q05.3																		-XP2	8C1		
/Q05.5																		-XP2	4B1		
/Q05.6																		-XP2	6D1		
/Q05.7																		-XP2	7C1		
/Q06.2	<-													X	=AA1J00 +J03 -XC	73		-XP1	19C2		
/Q06.2	<-													X	=AA1J00 +J03 -XC	72		-XP1	39A1		

01AS BUILT19.01.15RODI

02REVISION08.06.15WI

IndexRevisionDateNameNorm

Date21.10.2014

DrawnROEDEL

CheckHAMPEL

ABB SA

T.P. AEOLIAN

24kV SWITCHGEAR ZX1.2 SBB

Derived:

DYNAMICS LTD.

REPLACE:

ABB

AG Color Emag

Medium Voltage Products

DEPARTMENT

AP

Repl.by:

TERMINAL DIAGRAM -XC

UNDERGROUND FEEDER

V=AA1J00

+J02

242658248

1VB2383222

Sheet 48

53Sh.

09.06.2015
242658248
=AA1J00+J02
013
APE1

RUPLAN 4.61

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1		2		3		4		5		6		7		8							
NO	CABLE	USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE									
								-XC		STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WDU1,5/BLZ/5.08/ZQV									
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK
	1	2	3	4	5	6	7	8	9	10	11	12									
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/Q06.3																	36		-XP1	39C1	
/Q06.4																	37		-XP1	39D1	
/Q06.5																	38		-XP1	29C1	
/Q06.5																	39		-XP1	29D1	
/Q06.6																	40		-XP2	1A1	
/Q06.7																	41		-XP2	1B1	
/Q06.7																	42		-XP2	1C1	
/Q06.8																	43		-XP1	28C1	
/Q07.2																	44		-X11	13A1	
/Q07.2																	45		-X11	13B1	
/Q07.3																	46		-X11	13C1	
/Q07.4																	47		-XP1	12D1	
/Q07.5																	48		-XP1	4B1	
/Q07.6																	49		-XP1	8B1	
/Q07.2																	50		-X11	14A1	
/Q07.2																	51		-X11	14B1	
/Q07.3																	52		-X11	14C1	
/Q07.4																	53		-XP1	4D1	
/Q07.5																	54		-XP1	7D1	
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/Q08.2																	56		-QA1	-X1	23
/Q08.2																	57		-QA1	-X3	05
/Q08.3																	58		-QA1	-X3	02
/Q08.3																	59		-QA1	-X3	04
/Q08.3																	60		-QA1	-X3	03
/Q08.3																	61		-QA1	-X2	06
/Q08.4																	62		-QA1	-X3	01
/Q08.4																	63		-QA1	-X2	02
/Q08.2																	64		-QA1	-X1	24
/Q08.2																	65		-QA1	-X3	06
/Q08.3																	66		-QA1	-X3	09
/Q08.3																	67		-QA1	-X3	07
/Q08.3																	68		-QA1	-X3	08

01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA		 ABB Color Emag Medium Voltage Products	TERMINAL DIAGRAM	-XC	V	=AA1J00	
02	REVISION	08.06.15	WI	Check	HAMPEL	T.P. AEOLIAN	DYNAMICS LTD.		DEPARTMENT	AP		+J02	/013
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:		UNDERGROUND FEEDER		242658248	1VB2383222	Sheet 49
													53Sh.

1												2												3												4												5												6												7												8											
NO		CABLE				USED CORES				SHEET\COMPL.				CABLE TYPE				Ø				CORES				CAB. ROUTE				SCHEMATIC DIAGRAM				-XC												WIRING MATERIAL												TERMINAL TYPE																																					
1		W20								2				2				2x2,5mm²												STANDARD: H05Z-K 1,0mm2 BK																STANDARD: WDU1,5/BLZ/5.08/ZQV																																																	
REFERENCE		NO.												CORES/ LINE		POS.		DESTINATIONS												POT		NO.		DESTINATIONS												POS.		CORES/ LINE		REMARK																																													
		1	2	3	4	5	6	7	8	9	10	11	12																																																																																		
/Q08.3																																69		-QA1 -X2 09																																																													
/Q08.4																																70		-QA1 -X3 10																																																													
/Q08.4																																71		-QA1 -X2 01																																																													
/G02.7		<-												BN		= +		-S801												9				=Y01 +Y01 -X41 5																																																													
/G02.6		<-												BU		= +		-S801												8				=Y01 +Y01 -X41 6																																																													
/X01.2																																74																																																															
/Q03.8																= +		-S801												1		116L+		-XP1 18A1																																																													
/Q03.8																= +		-S801												3		116L-		-XP1 25A1																																																													

09.06.2015 242658248
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014
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RUPLAN 4.61

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				Date		21.10.2014		ABB SA		T.P. AEOLIAN DYNAMICS LTD.		24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		TERMINAL DIAGRAM -XC		V =AA1J00 +J02 /014		242658248		1VB2383222		Sheet 50 53Sh.	
01		AS BUILT		19.01.15		RODI		Drawn ROEDEL																	
02		REVISION		08.06.15		WI		Check HAMPEL																	
Index		Revision		Date		Name		Norm		Derived:		Replace:		Repl.by:		UNDERGROUND FEEDER									

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1		2		3		4		5		6		7		8					
CROSS CONNECTION																			
BACK REFERENCE		POTENTIAL POSITION CROSS SECTION		WIRING MATERIAL COLOR		START TERMINAL		DESTINATION 1		DESTINATION 2		DESTINATION 3		DESTINATION 4		DESTINATION 5		DESTINATION 6	
G01		1L+ 83 2,5mm2		H07Z-K BK		-X10 1 A1		-F101 2		-F106 2		-F116 2							
G01		1L- 83 2,5mm2		H07Z-K BK		-X10 2 A1		-F101 4		-F106 4		-F116 4							
L01		0 1,0mm2		H05Z-K BK				-K11 2		-K11 4									
L01		0 1,0mm2		H05Z-K BK				-K11 3		-K11 5									
L01		0 1,0mm2		H05Z-K BK				-K11 62		-K12 A1									
L01		0 1,0mm2		H05Z-K BK				-K11 A1		-K12 62									
L01		0 1,0mm2		H05Z-K BK				-K12 2		-K12 4									
L01		0 1,0mm2		H05Z-K BK				-K12 3		-K12 5									
L01		101L- 0 1,0mm2		H05Z-K BK		-XD 1 A1		-XP1 1 C1		-F101 3		-K11 21		-K12 22					
L01		0 1,0mm2		H05Z-K BK		-XD 1 C1		-XP1 2 A2		-K11 6		-K11 22							
L01		0 1,0mm2		H05Z-K BK		-XD 1 D1		-XP1 3 B2		-K12 6		-K12 21							
L01		0 1,0mm2		H05Z-K BK		-XP1 7 A2		-K11 1		-K12 1									
L01		0 1,0mm2		H05Z-K BK		-XP1 12 C1		-K11 A2		-K12 A2									
Q02		PE 88 4,0mm2		H07Z-K GN/YE				-A210 PE		PE PE									
Q02		0 1,0mm2		H05Z-K BK				-F101 12		-F106 11									
Q02		0 1,0mm2		H05Z-K BK				-F101 23		-F106 24									
Q02		0 1,0mm2		H05Z-K BK				-F106 12		-F119 11									
Q02		0 1,0mm2		H05Z-K BK				-F106 23		-F119 24									
Q02		116L+ 0 1,0mm2		H05Z-K BK		-XP1 21 A2		-F101 11		-F405 23		-F100 33		-F101 24					
Q03		0 1,0mm2		H05Z-K BK				-F100 34		-F101 33									

[illegible]



AG Calor Emag
Medium Voltage Products

ABB AG Power Technologies Division - Calor Emag Medium Voltage Products

Oberhausener Strasse 33

D-40472 Ratingen

D-40472 Ratingen

Telefon: (02102) 12-0

Fax: (02102) 12-1777

ABB SA

T.P. AEOLIAN

DYNAMICS LTD.

24kV SWITCHGEAR ZX1.2 SBB

INCOMER

DEACE CONTRACT NO. : 242658248

DEACE DRAWING NO. : 1VB2383223

S = AA1J00 + J03

/ A01

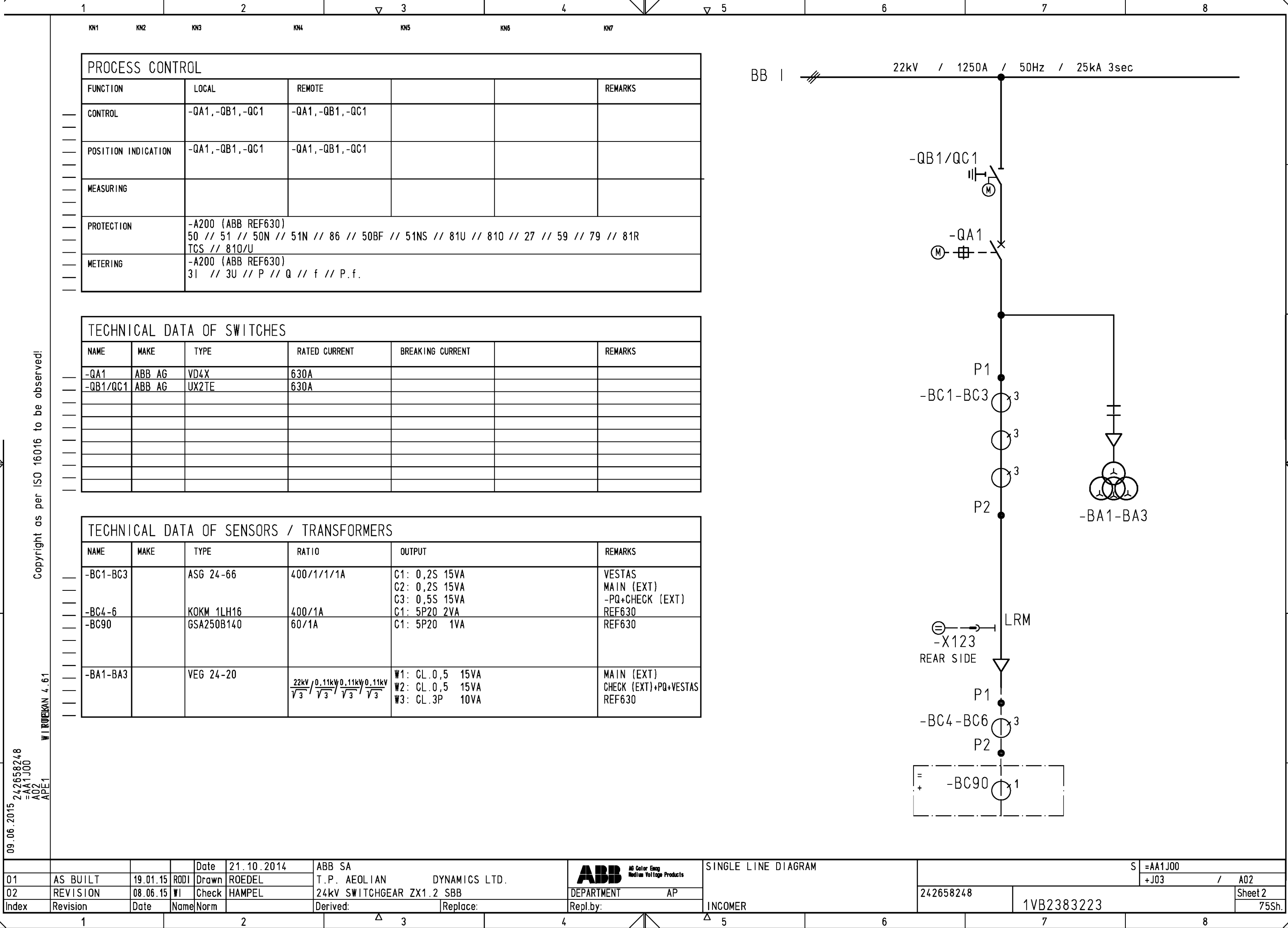
09.06.2015 242658248
=AA1J00
A01
APE1
RUPLAN 4.61

01 AS BUILT
02 REVISION
No. Revision

19.01.15
08.06.15
Date

R001
WI
Name

Date :	21.10.2014
Drawn :	ROEDEL
Check :	HAMPEL
Repl.:	



$$= AA1J00 + J03$$

INCOMER



01
02
Index

				Date	21.10.2014	ABB SA	 A6 Color Easy Medium Voltage Products	INTERLOCKING LOGIC	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J03 / A03		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					
Index	Revision	Date	Name	Norm		Derived:	Replace:	Real by:	242658248	1VB2383223	Sheet 3
								INCOMER			75Sh

1		2		3		4		5		6	7	8
CUSTOMER DOCUMENT NUMBER		MANUFACTURER DOCUMENT NUMBER		REVISION STATE		ORGANIZATIONAL DESIGNATION		SHEET	DESIGNATION			
						TYPE	ASSIGNMENT	NUMBER PART				
			1VB2383223	02	01	S	=AA1J00 +J03	A01	1	COVERSHEET	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	A02	2	SINGLE LINE DIAGRAM	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	A03	3	INTERLOCKING LOGIC	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	A04	4	BACKUP PROTECTION & BFP	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	B01	5	TABLE OF CONTENTS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	B02	6	TABLE OF CONTENTS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	B03	7	TABLE OF CONTENTS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	D01	8	AC VOLTAGE SUPPLY	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	D02	9	AC VOLTAGE SUPPLY	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	G01	10	DC VOLTAGE SUPPLY	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	G02	11	DC VOLTAGE SUPPLY 1L+/1L-	CONTROL VOLTAGE / MOTOR VOLTAGE INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	G03	12	DC VOLTAGE SUPPLY 2L+/2L-	2.SHUNT RELEASE OFF - VOLTAGE INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	G04	13	SELF MONITORING	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	L01	14	MOTOR DRIVES	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	M01	15	CONTROL CIRCUIT BREAKER	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	M02	16	2.SHUNT RELEASE OFF / INTERTRIPPING	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	P01	17	SPARE INTERLOCKING CIRCUITS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q01	18	POSITION INDICATIONS	CB/ISOLATOR/EARTHING SWITCH INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q02	19	ALARMS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q03	20	ALARMS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q04	21	ALARMS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q05	22	OUTPUTS REF	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q06	23	TRANSFORMER ALARMS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q07	24	SPARE-OUTPUTS REF	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q08	25	SPARE CONTACTS	CIRCUIT BREAKER INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	Q09	26	SPARE CONTACTS	INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	S01	27	CURRENT TRANSFORMER CIRCUIT	CORE 1 INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	S02	28	CURRENT TRANSFORMER CIRCUIT	CORE 1 // CORE 2 // CORE 3 INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	S03	29	VOLTAGE CONNECTION	WINDING 1 // 2 INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	S04	30	VOLTAGE CONNECTION	WINDING 3 INCOMER	
			1VB2383223	02	01	S	=AA1J00 +J03	S04.A	31	SUPERVISION	INCOMER	
		Date	21.10.2014		ABB SA		T.P. AEOLIAN		DYNAMICS LTD.		S =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL							+J03 /B01
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		242658248		Sheet5
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	INCOMER		1VB2383223		75Sh.

Copyright as per ISO 16016 to be observed!

RUPLAN 4.61

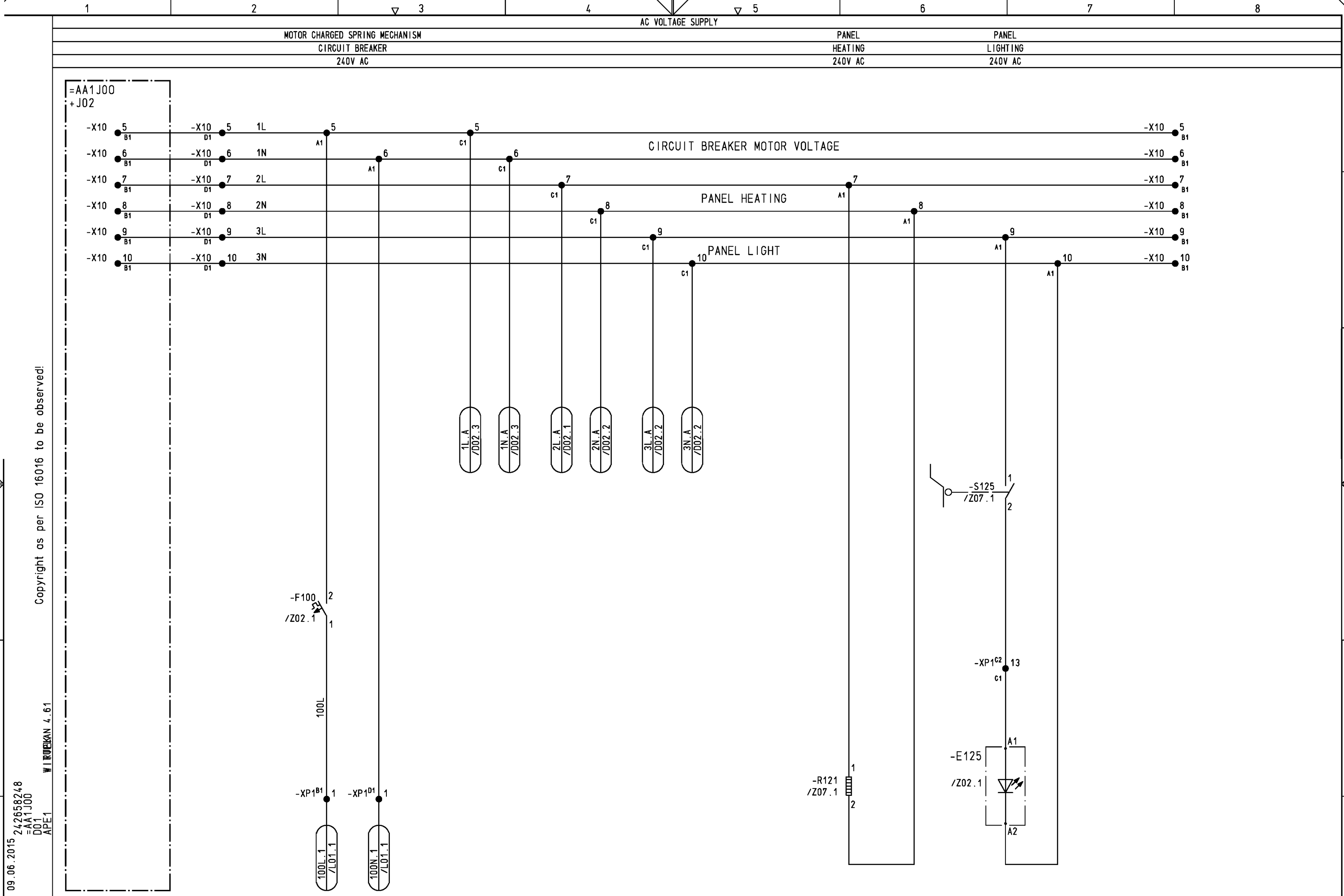
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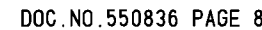
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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.		DEPARTMENT AP	242658248	INCOMER	+J03 /D01	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB					1VB2383223	
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:			75Sh.		
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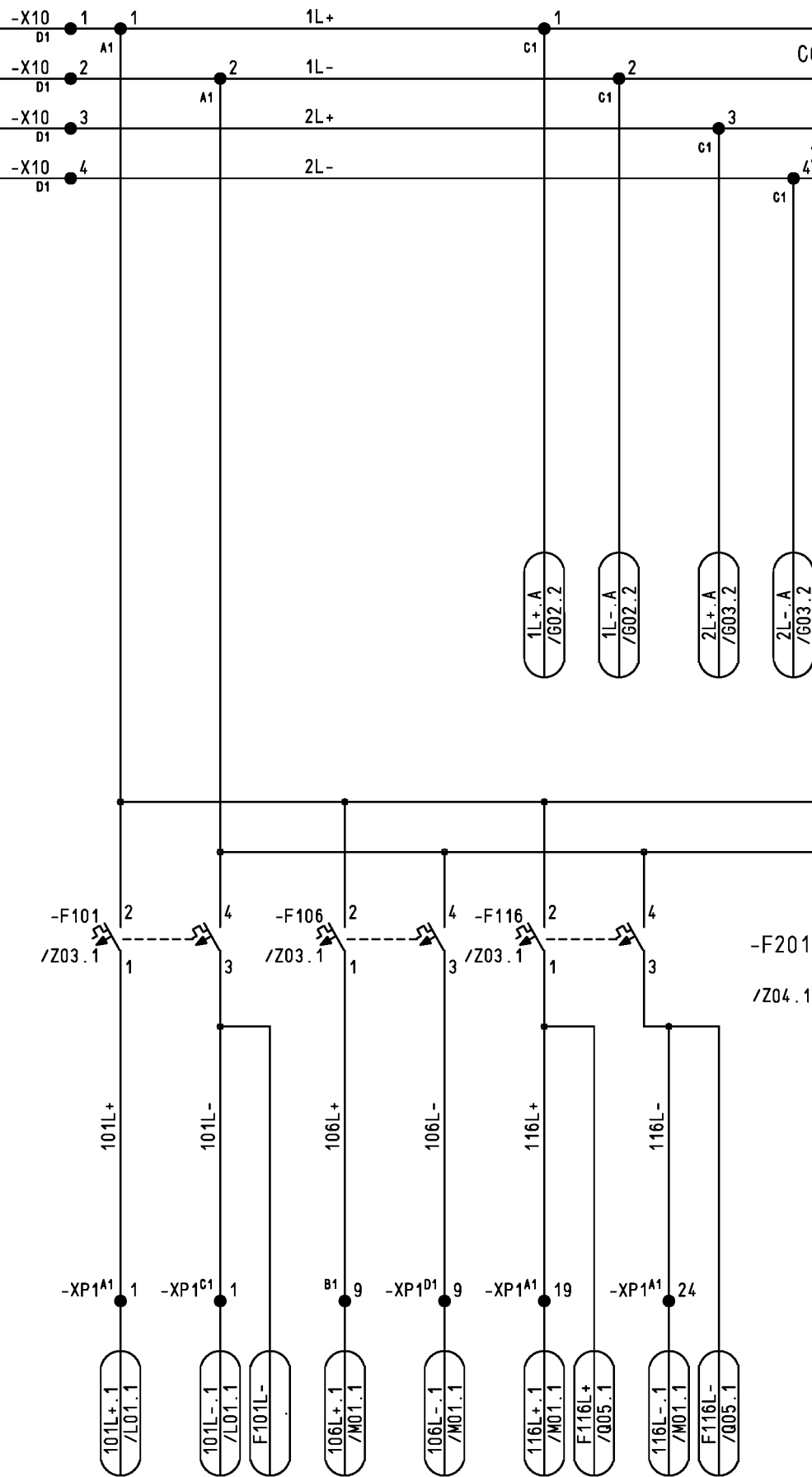
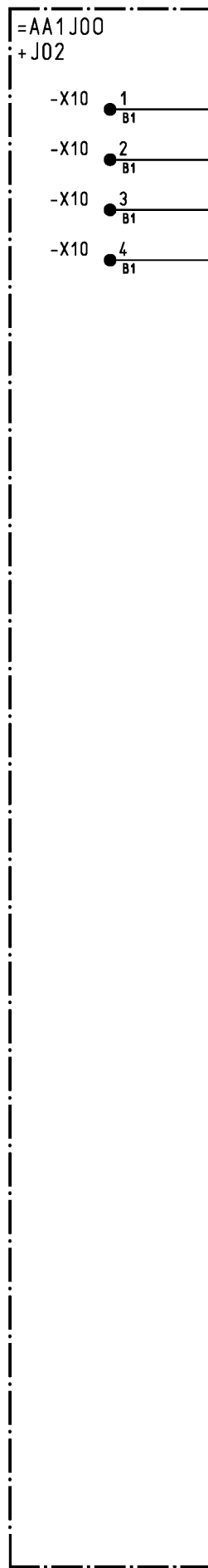


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Copyright as per ISO 16016 to be observed!

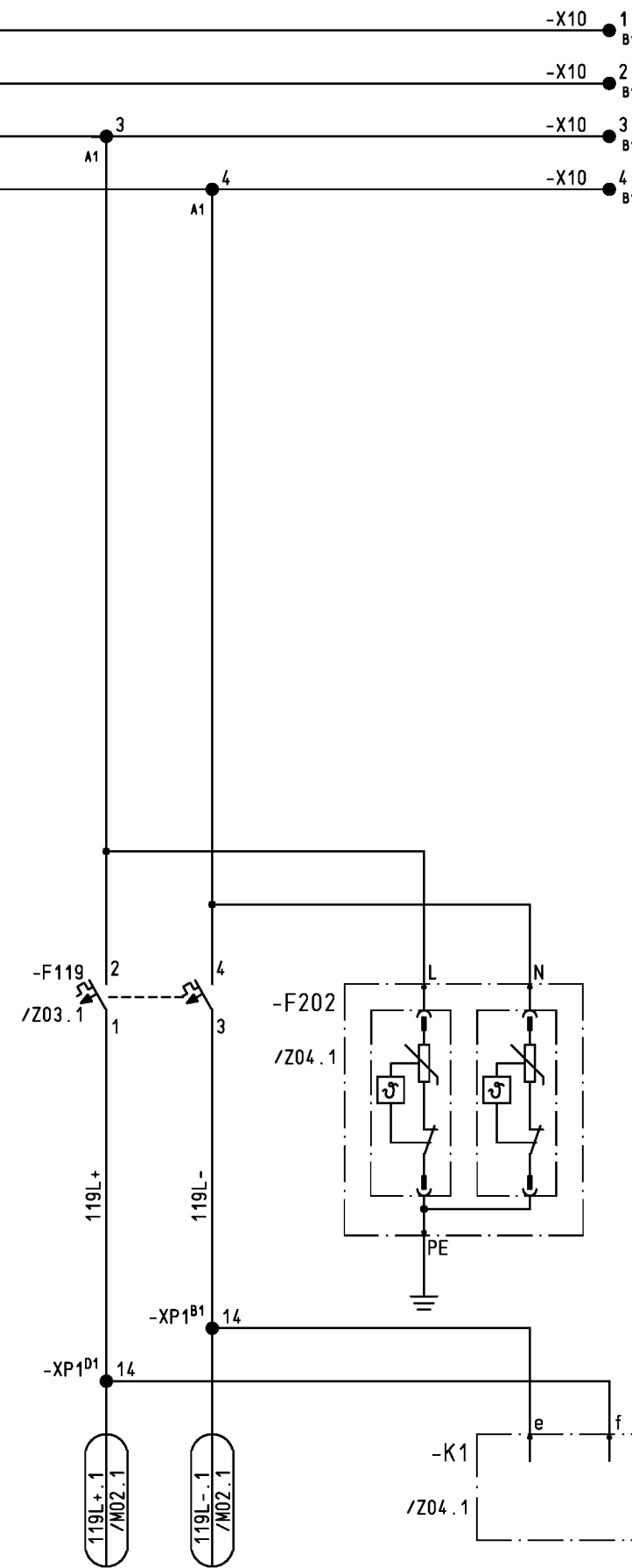
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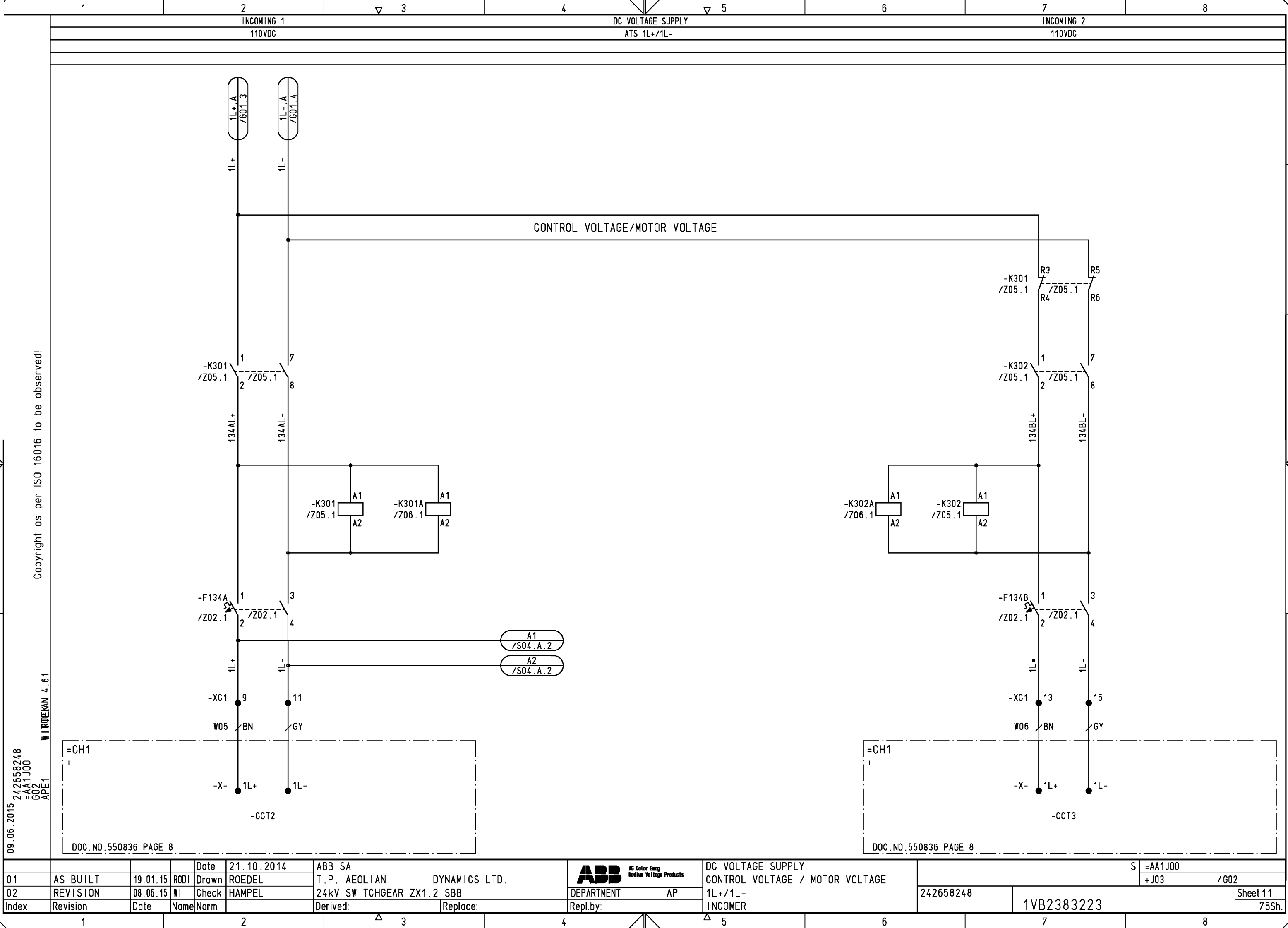


CONTROL VOLTAGE / MOTOR VOLTAGE

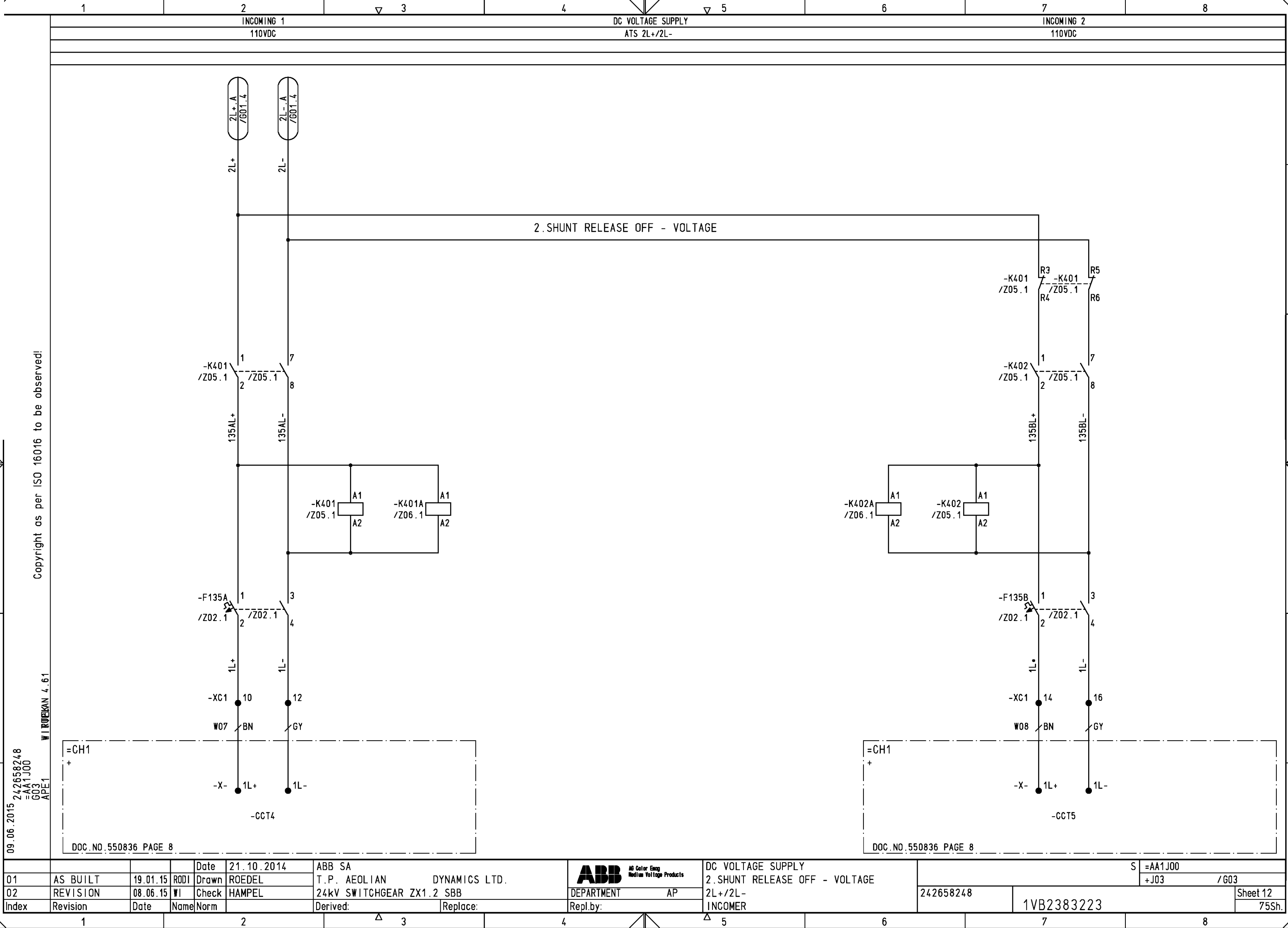
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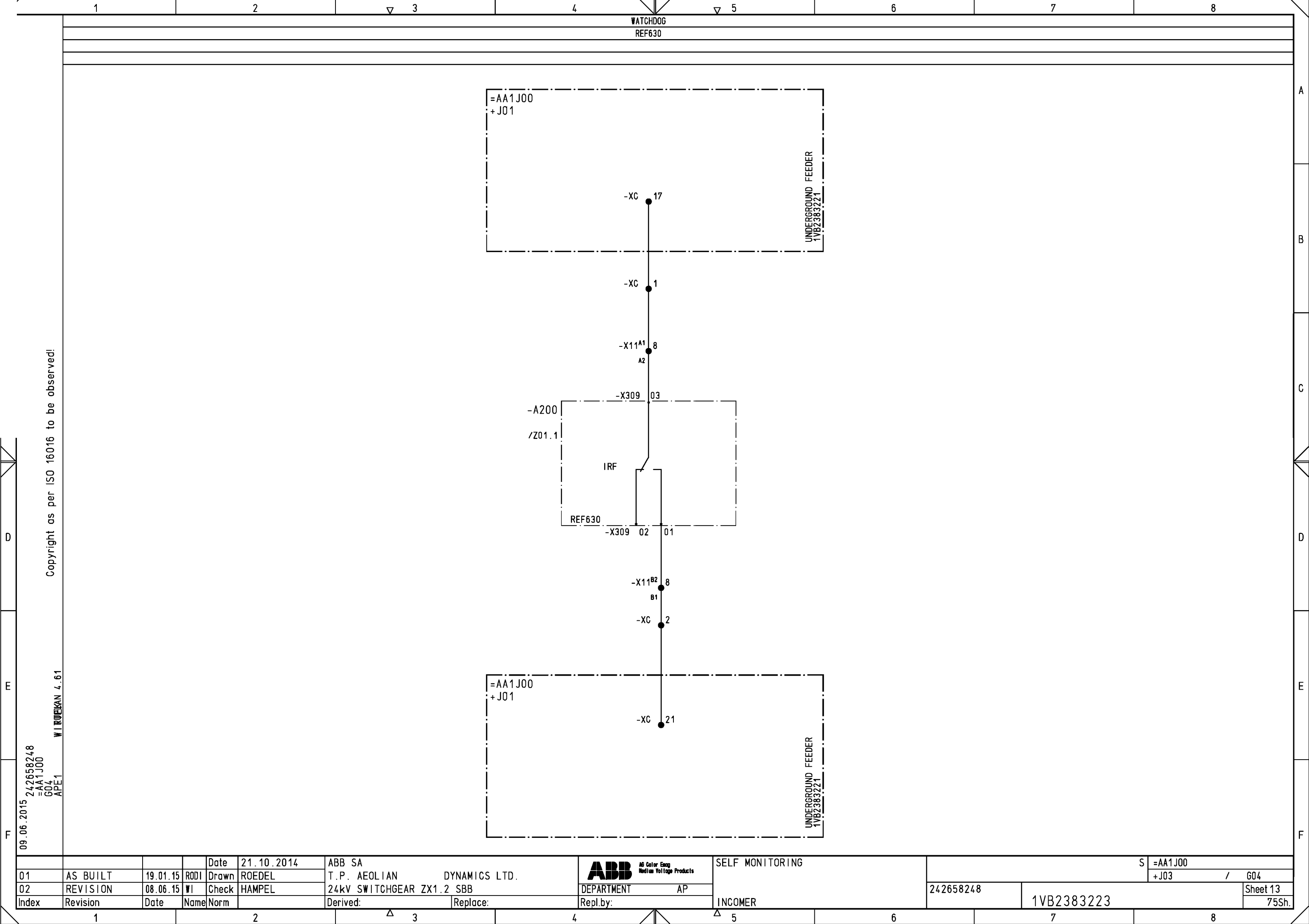


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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB	DEPARTMENT AP		242658248	Sheet 10
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	INCOMER	75Sh.



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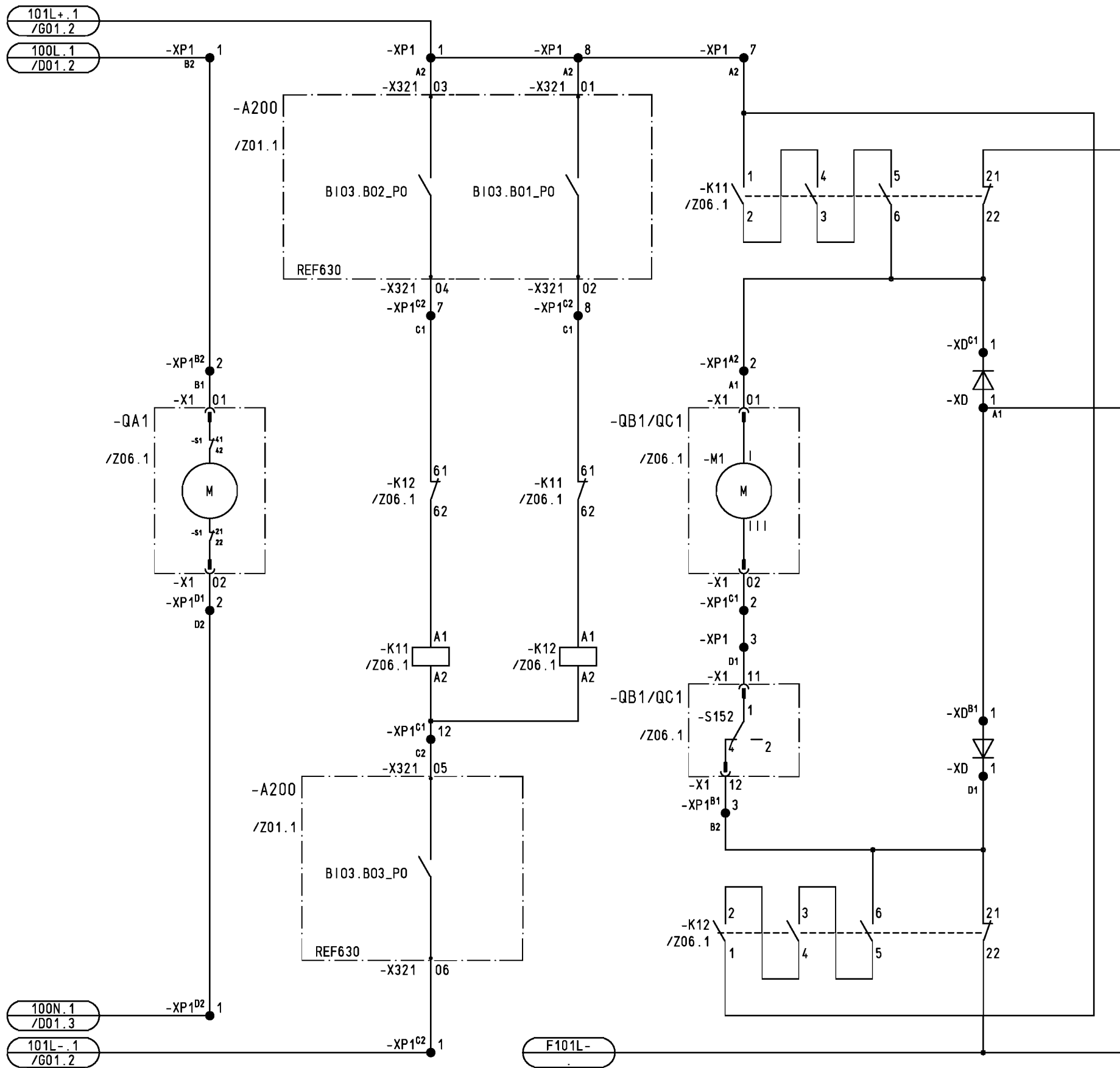
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Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	INCOMER	242658248	1VB2383223
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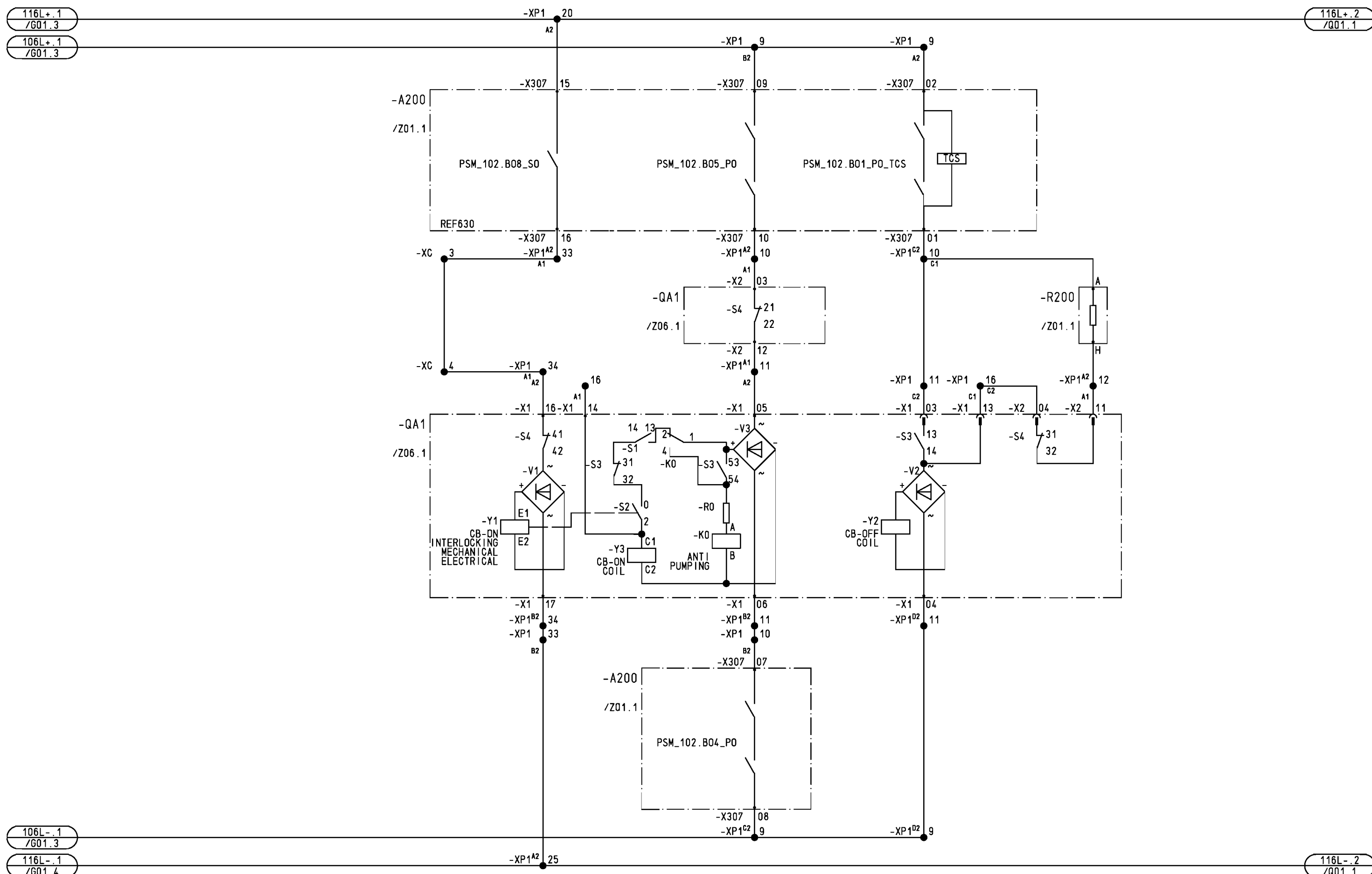
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PROTECTION TRIPPING



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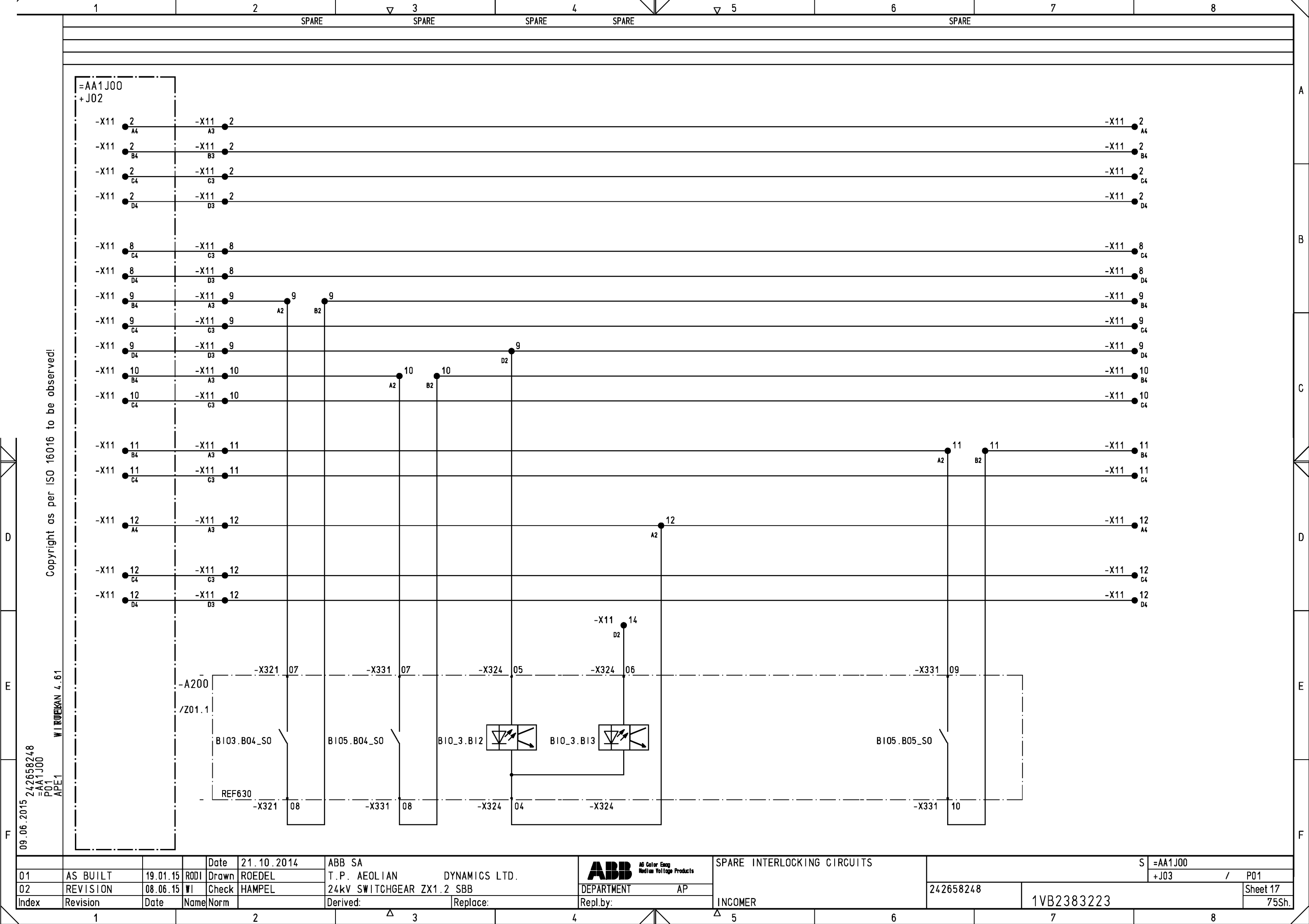
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Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	INCOMER	1VB2383223		75Sh.

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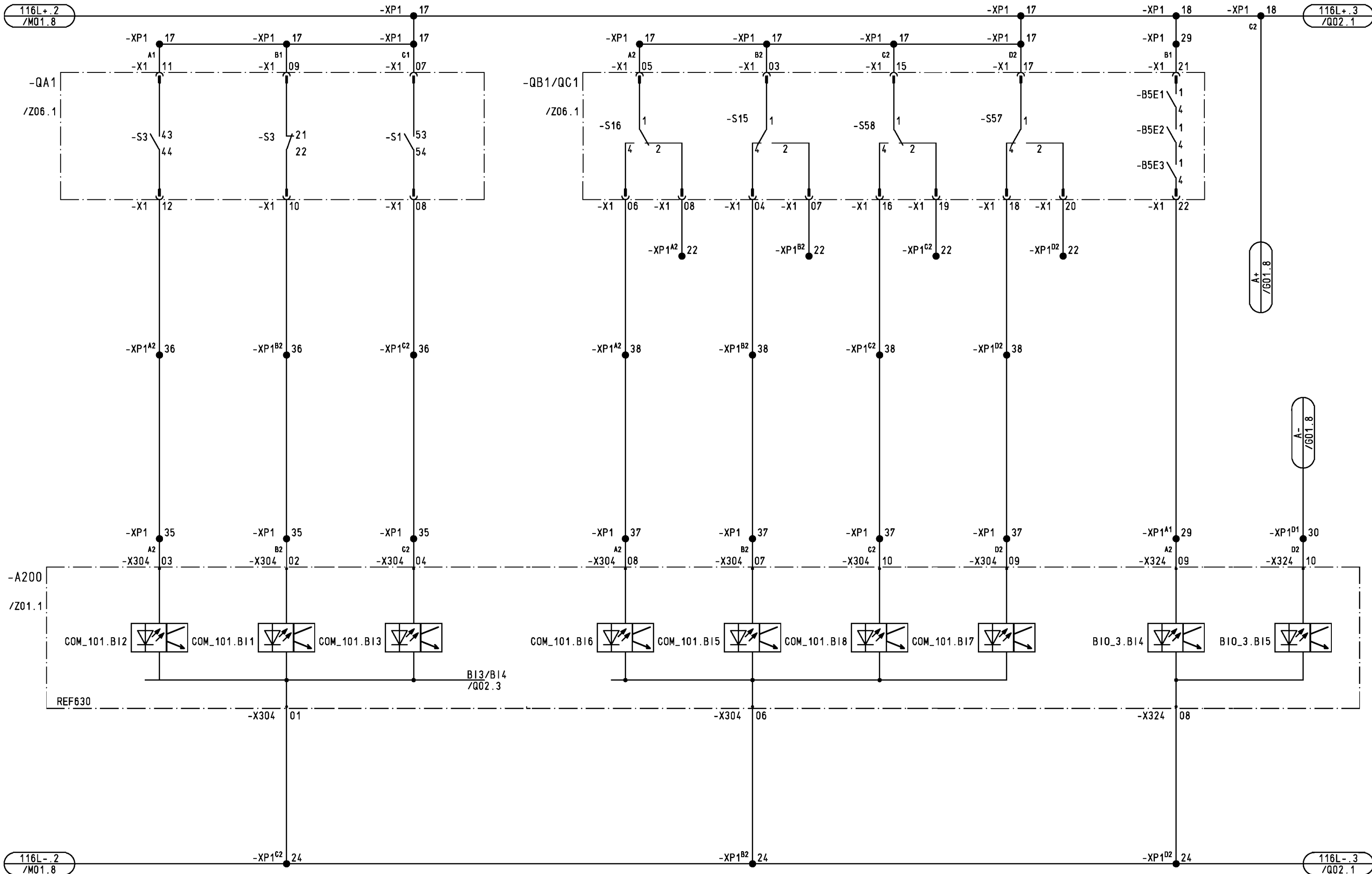
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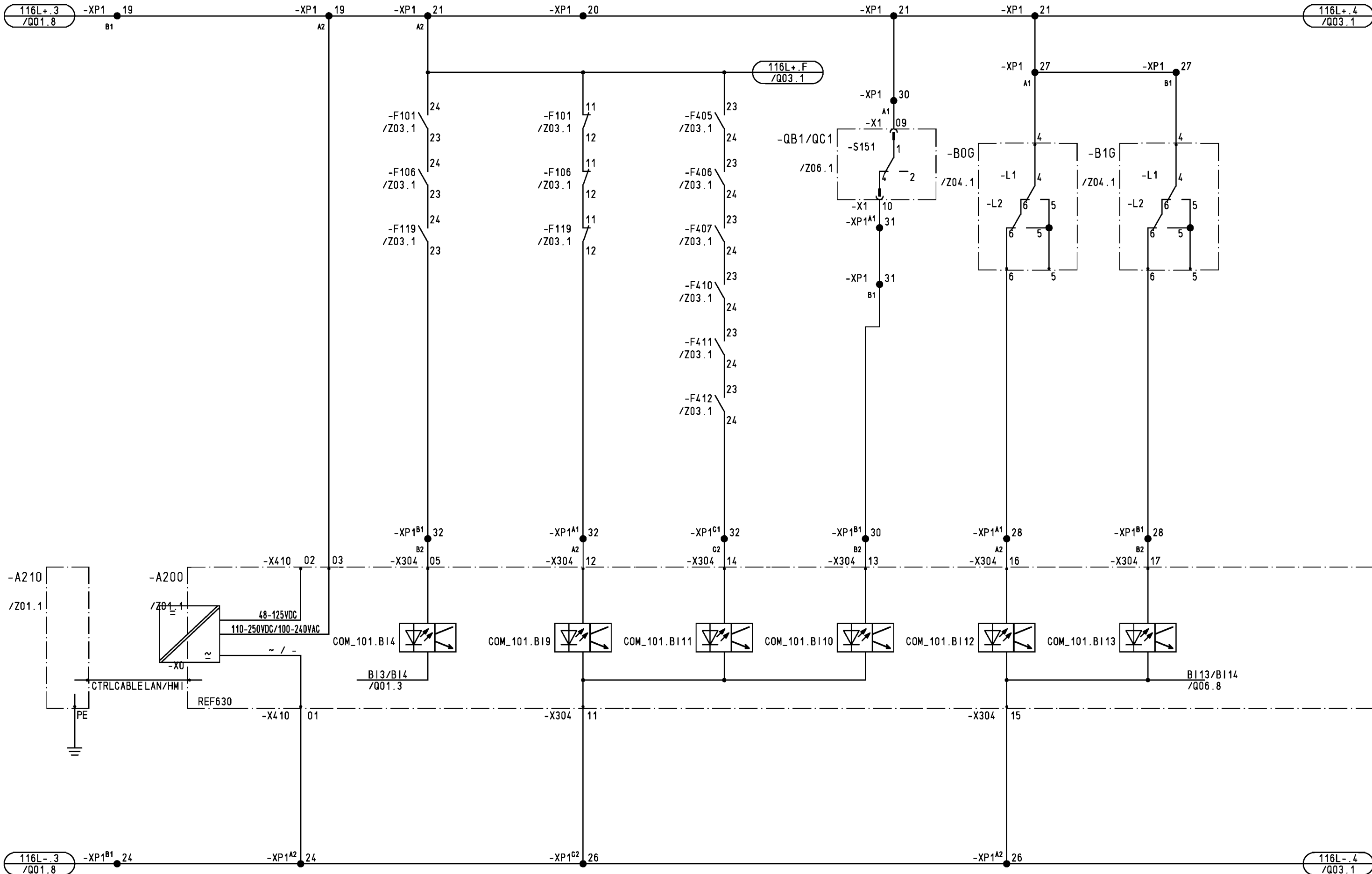


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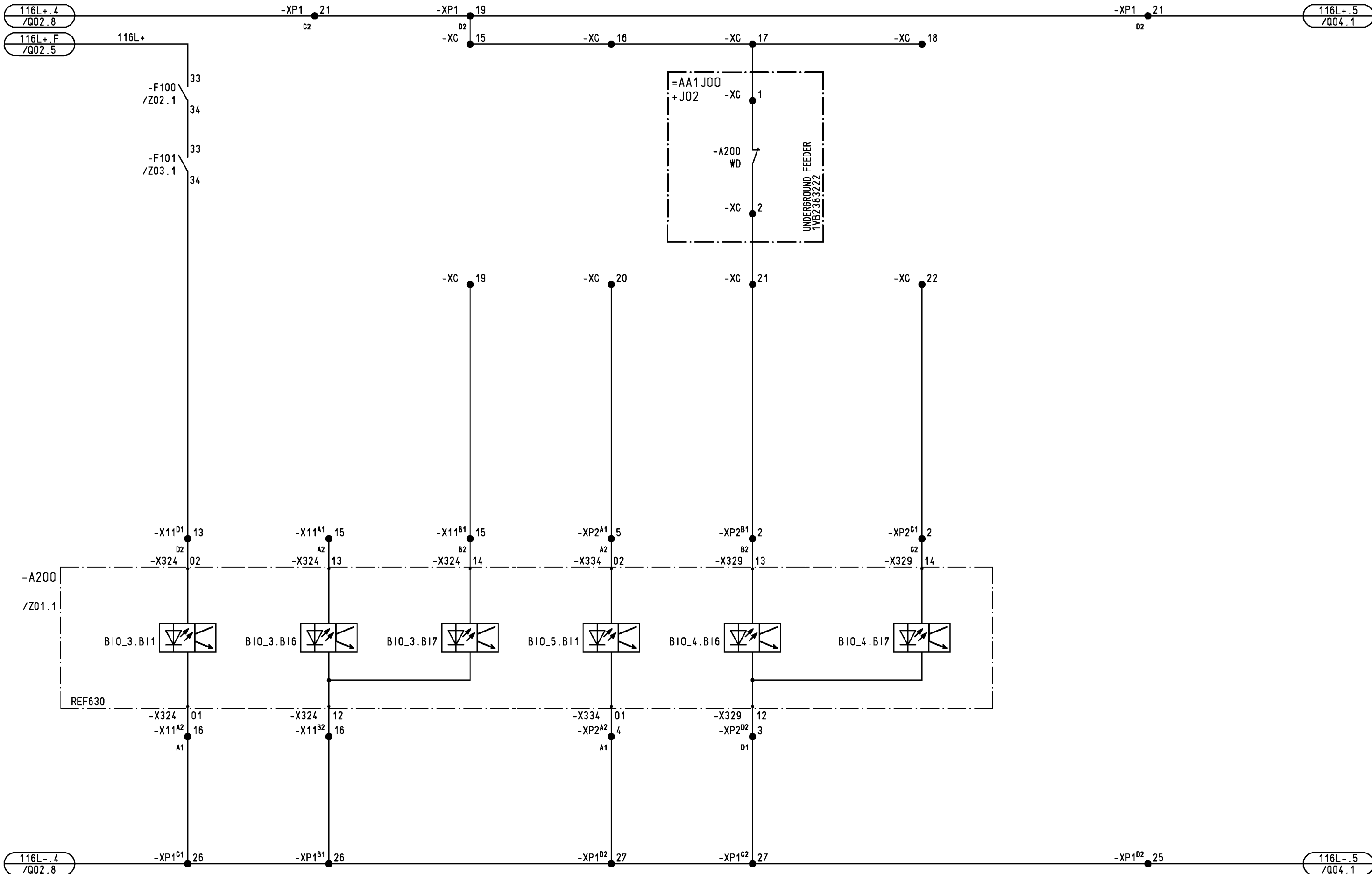


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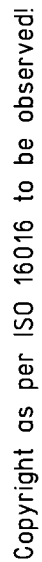
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**AG Color Enag
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87	87
88	88
89	89
90	90
91	91
92	92
93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

S	=AA1J00
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+J03	/	Q05
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242658248

1VB2383223

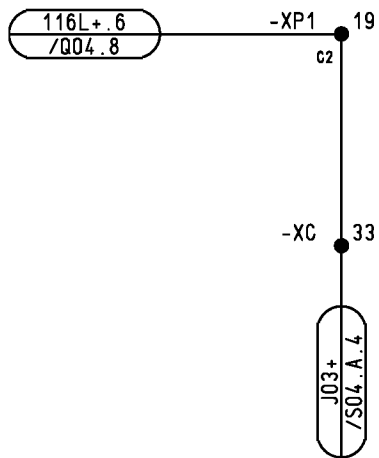
Sheet 22

75Sh.

Copyright as per ISO 16016 to be observed!

09.06.2015 242658248
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Q06
APE 1

WIRUPEKAN 4.61



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NOT IN USE

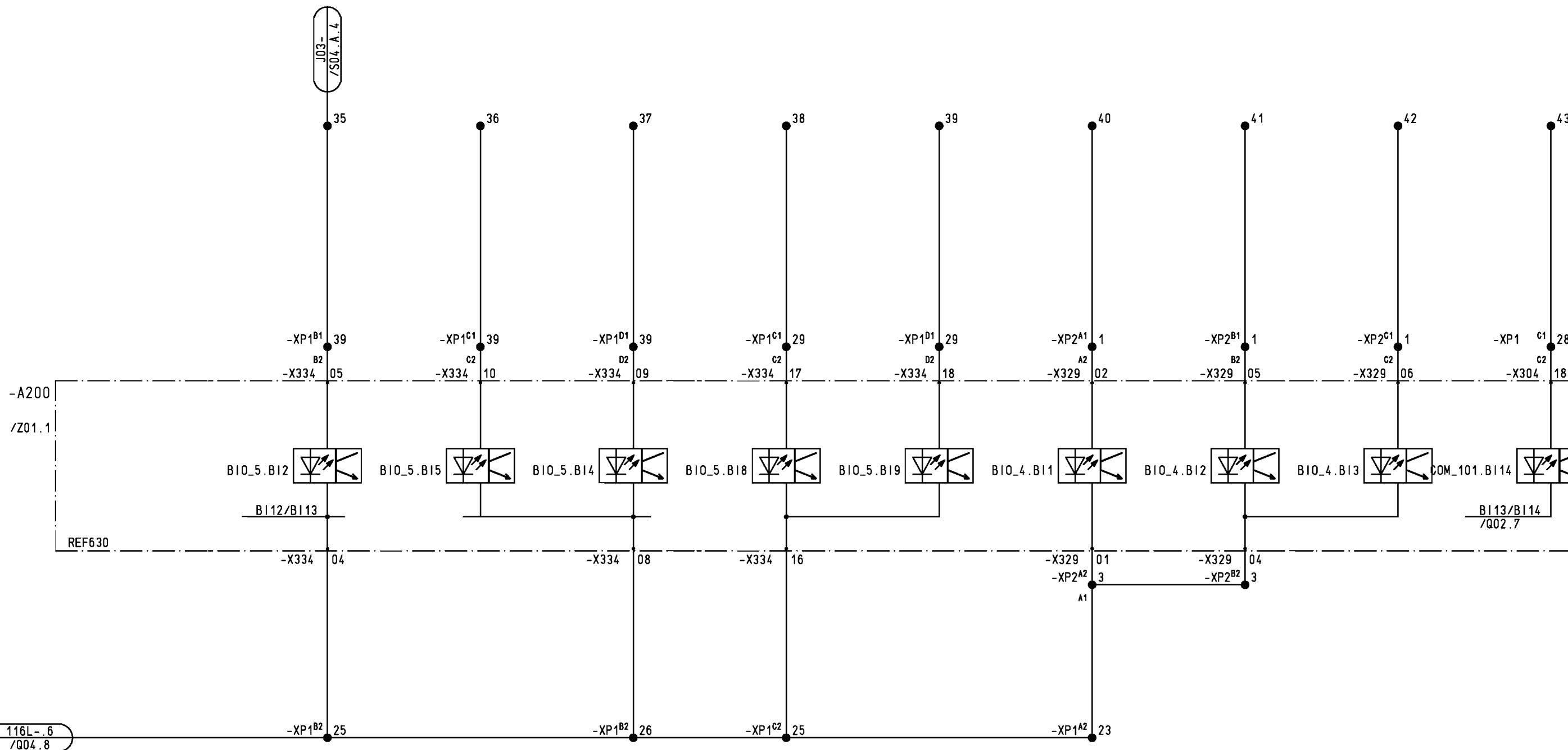
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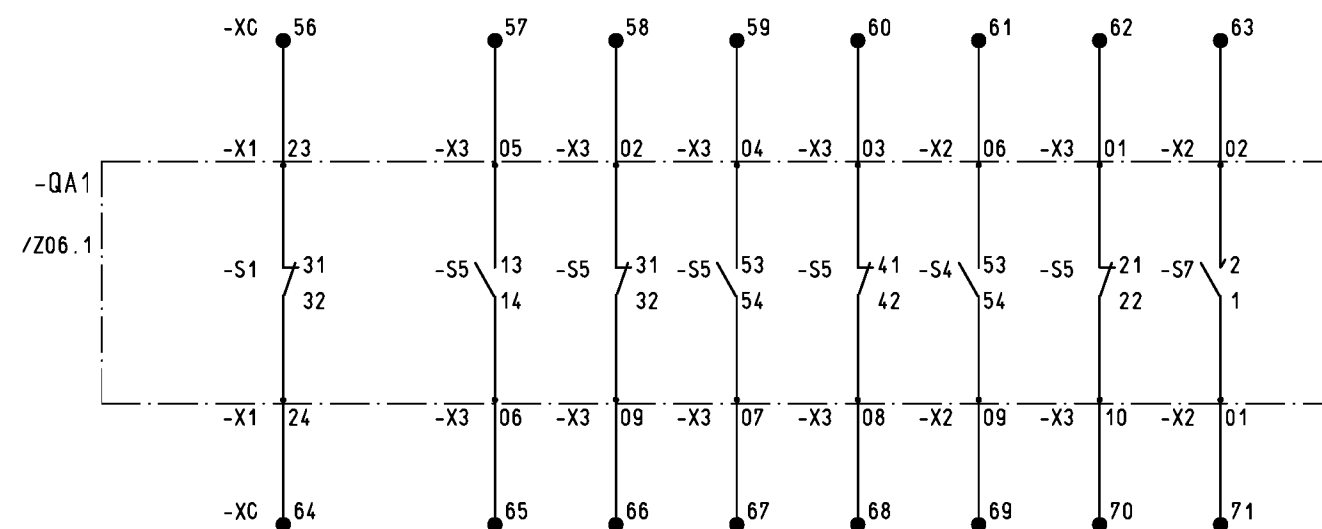
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02	REVISION	08.06.15	WI	Check	HAMPEL	DYNAMICS LTD.		DEPARTMENT AP		242658248		
Index		Revision	Date	Name	Norm	Derived:		Repl.by:		1VB2383223		
						Replace:		Repl.by:		Sheet 23		
								INCOMER		75Sh.		

01
02
Index



Sheet 24
75Sh.

1	2	3	4	5	6	7	8
MOTOR CHARGED	SPRING MECHANISM	CIRCUIT BREAKER	CIRCUIT BREAKER	CIRCUIT BREAKER			
DISCHARGED	CLOSE	OPEN	CLOSE	OPEN	CLOSE	OPEN	CB TRIPPED



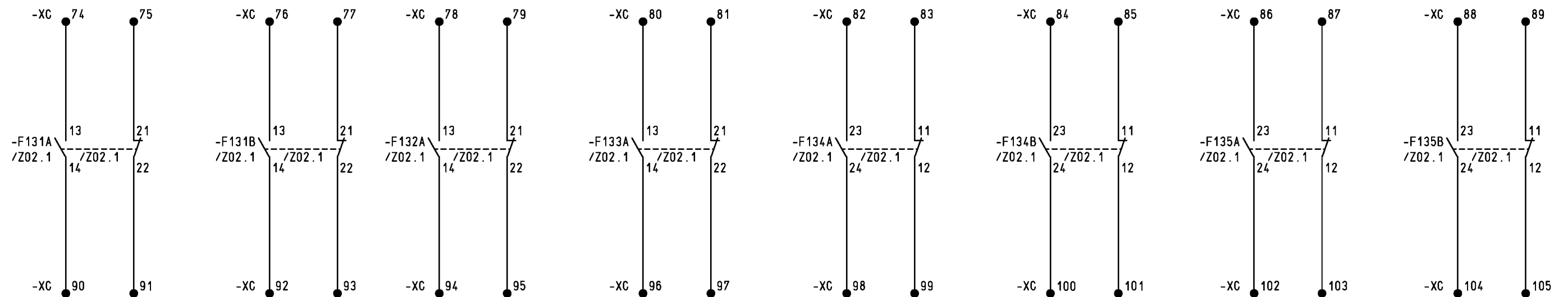
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WIRUPEKAN 4.61

09.06.2015 242658248
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Q08
APE1

				Date	21.10.2014	ABB SA		 AG Color Engr. Medium Voltage Products	SPARE CONTACTS CIRCUIT BREAKER	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J03 / Q08		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
						DEPARTMENT AP						
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	INCOMER	242658248	1VB2383223	Sheet 25 75Sh.

1	2	3	4	5	6	7	8
MCB TRIPPED	MCB TRIPPED	MCB TRIPPED	MCB TRIPPED	MCB TRIPPED	MCB TRIPPED	MCB TRIPPED	MCB TRIPPED
CIRCUIT BREAKER MOTOR VOLTAGE	CIRCUIT BREAKER MOTOR VOLTAGE	PANEL HEATING	PANEL LIGHTING	CONTROL VOLTAGE/MOTOR VOLTAGE	CONTROL VOLTAGE/MOTOR VOLTAGE	2.SHUNT RELEASE OFF	2.SHUNT RELEASE OFF
240V AC	240V AC	240V AC	240V AC	110V DC	110V DC	110V DC	110V DC



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WIRUEKAN 4.61

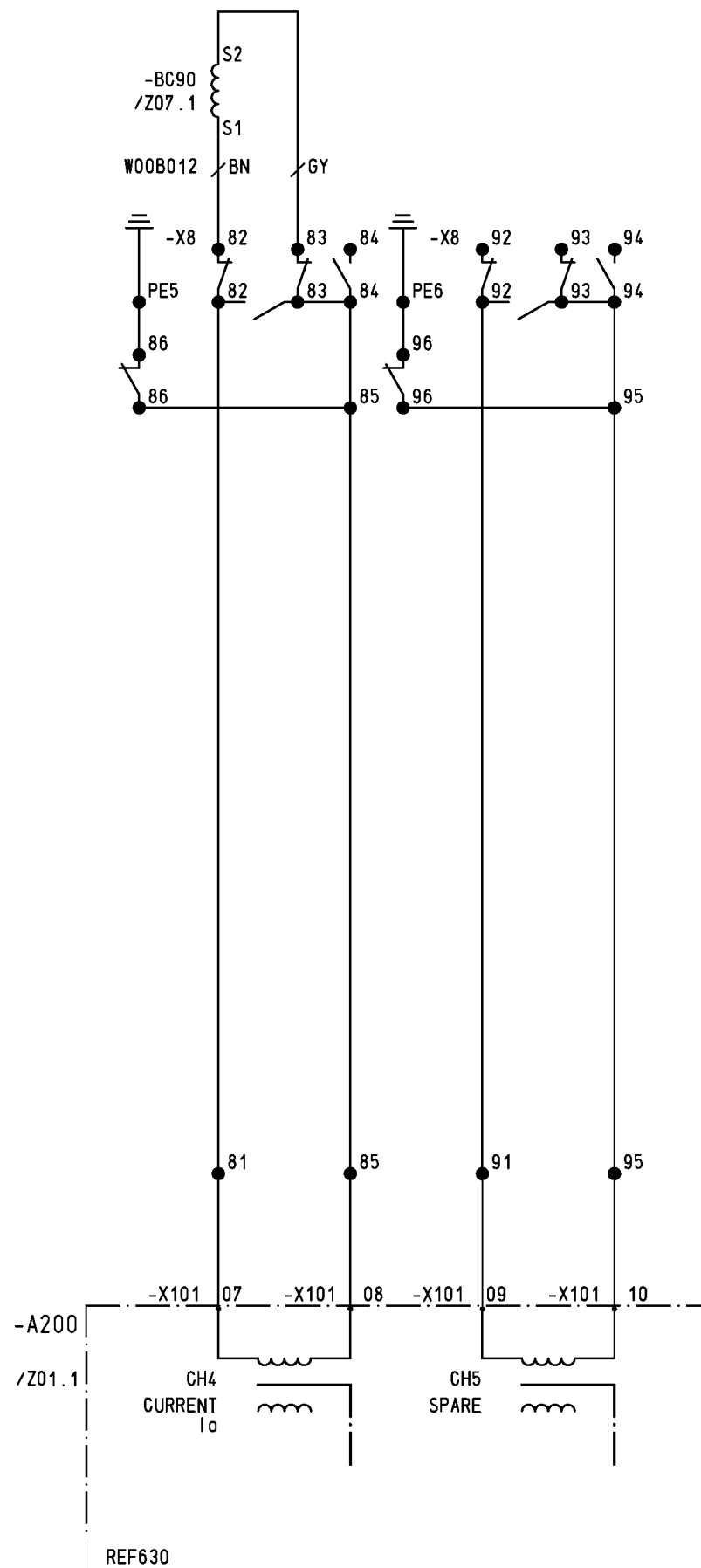
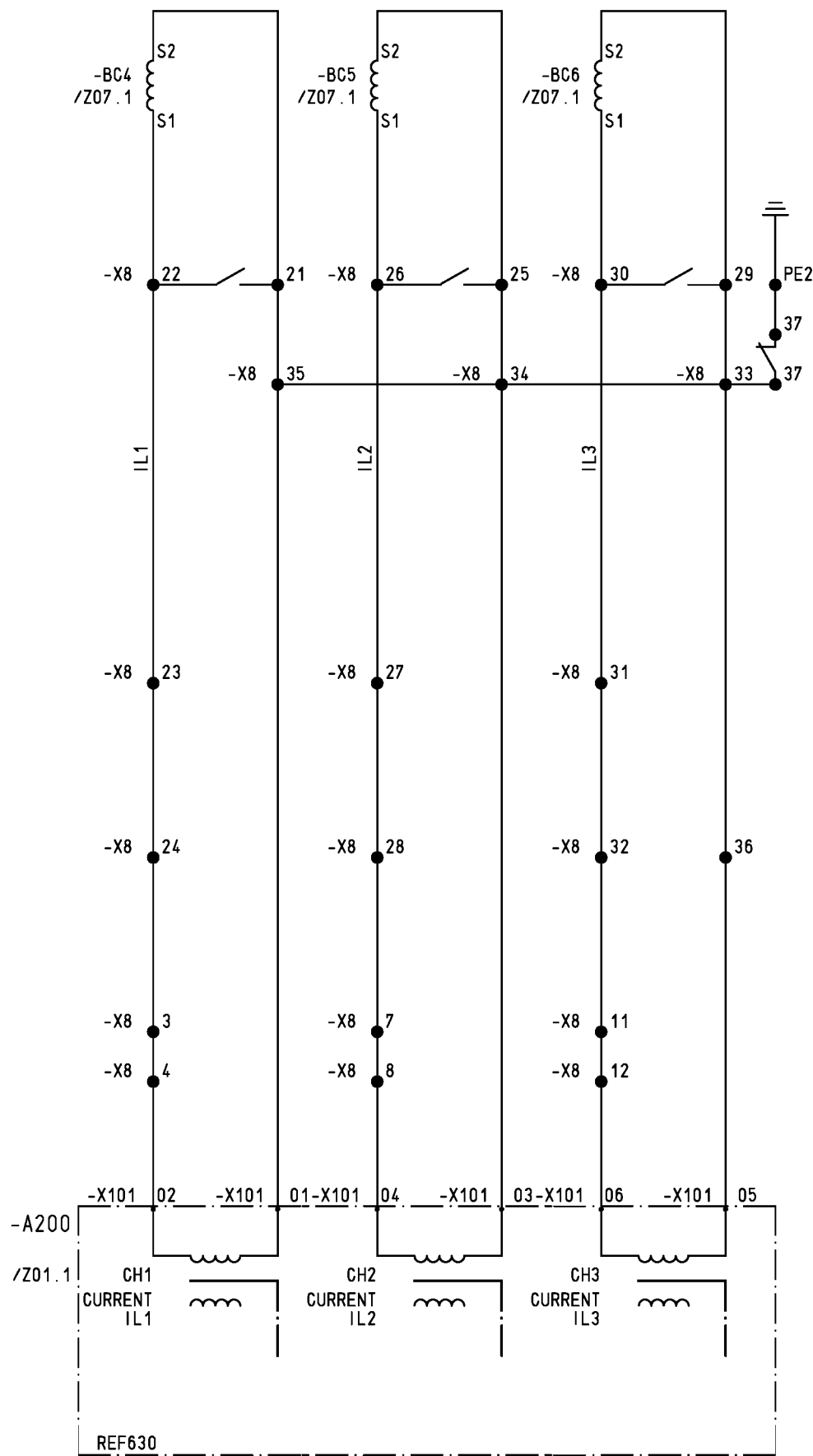
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02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			DEPARTMENT	AP	242658248	1VB2383223	Sheet 26
Index	Revision	Date	Name	Norm		Derived:			Replace:	Repl.by:	INCOMER		75Sh.

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S01
APE1

WIRTEKAN 4.61

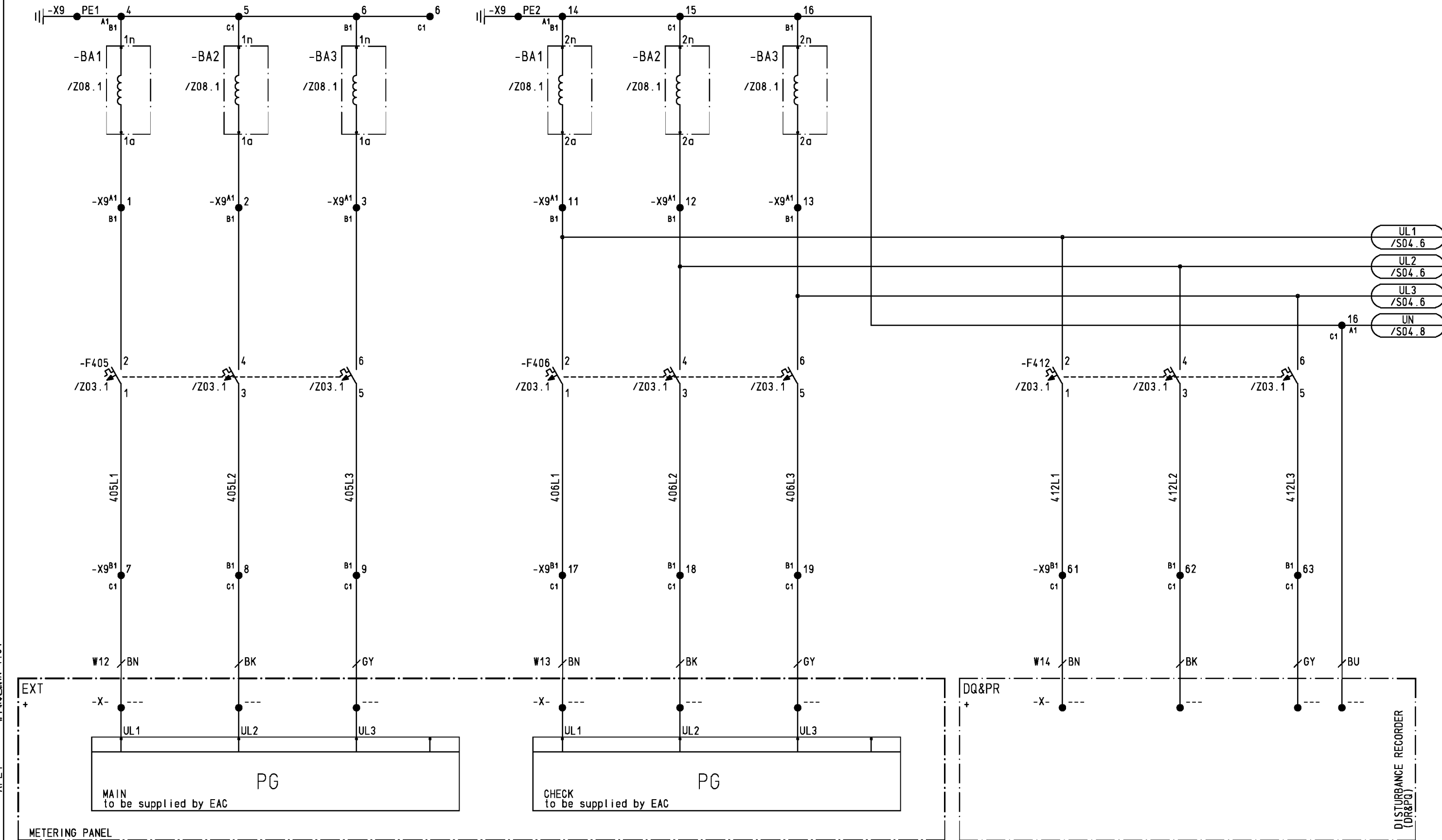


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02		REVISION	08.06.15	WI	Check	HAMPEL								
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:							

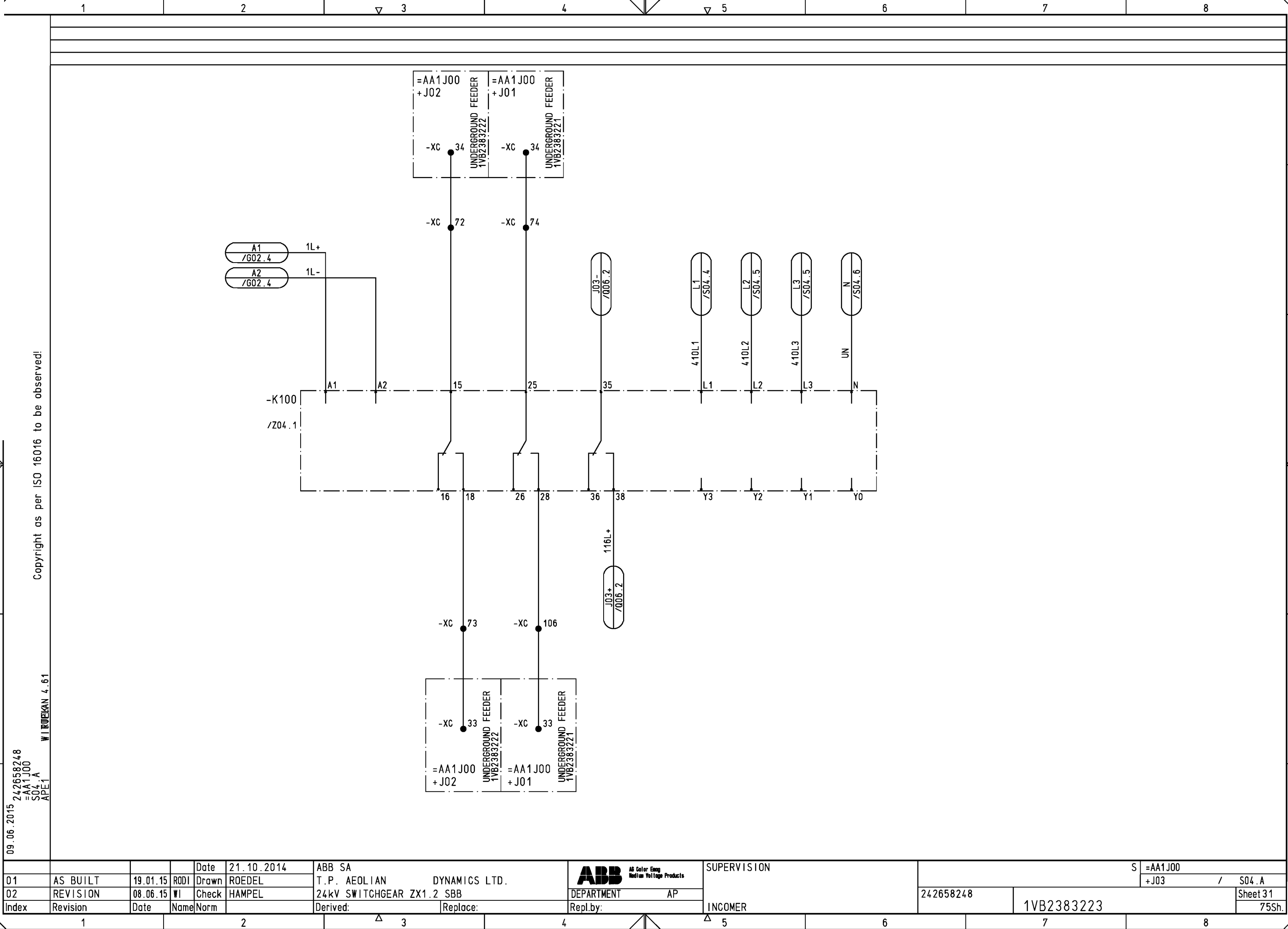
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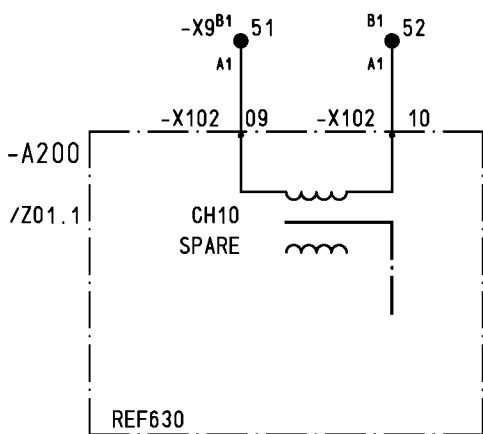
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S03
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WIRTEKAN 4.61



01		AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA	VOLTAGE CONNECTION		S =AA1J00	
02		REVISION	08.06.15	WI	Check	HAMPEL	T.P. AEOLIAN	WINDING 1 // 2		+J03 / S03	
Index		Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:		242658248	Sheet 29
								INCOMER		1VB2383223	75Sh.



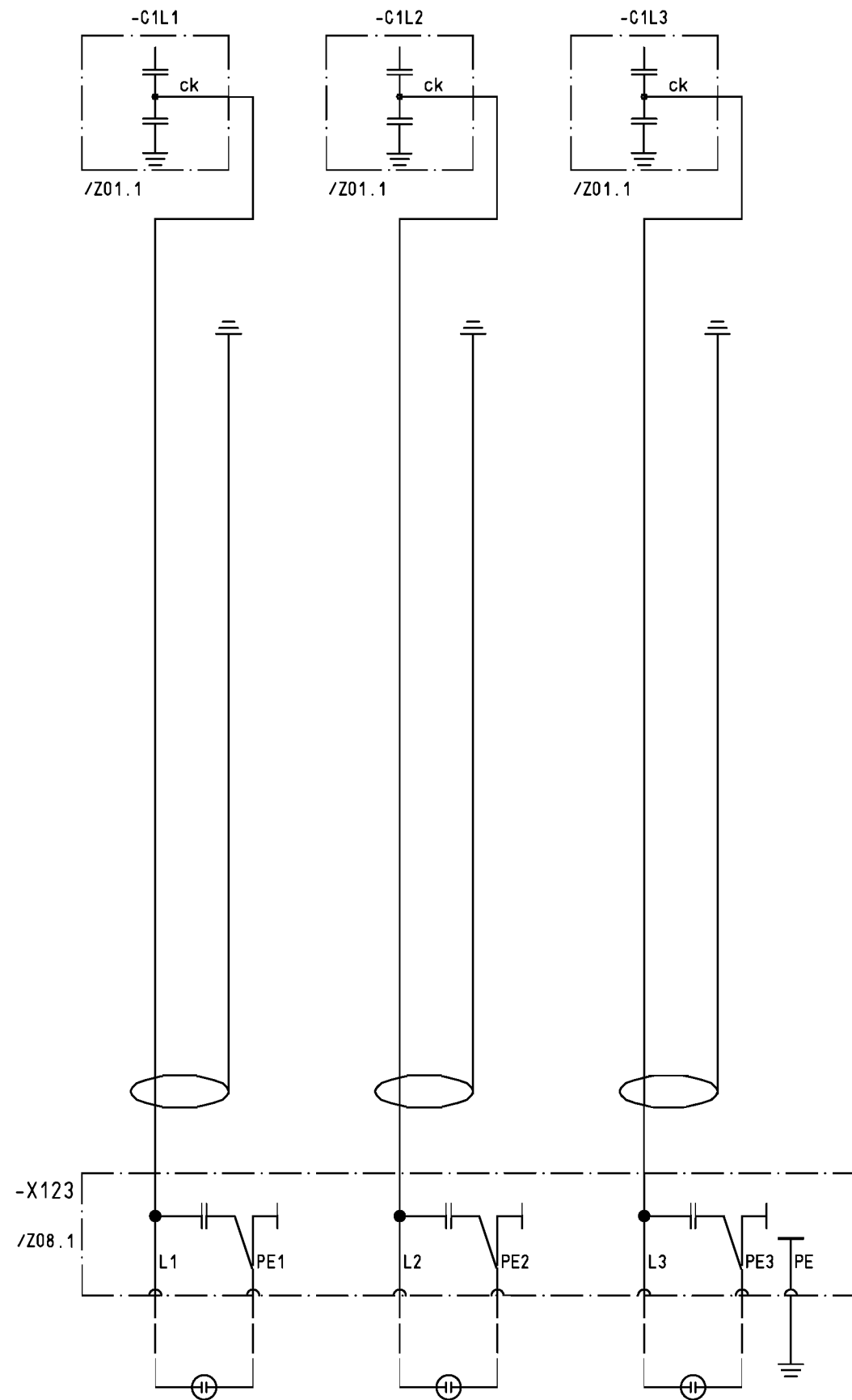
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APE1

WIRPLAN 4.61

CAPACITIVE VOLTAGE-
INDICATION



				Date	21.10.2014	ABB SA	 AG Color Emag Medium Voltage Products	CAPACITIVE VOLTAGE INDICATION	S =AA1J00			
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.			+J03 / S06			
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	INCOMER	242658248	1VB2383223	Sheet 33 75Sh.

				Date	21.10.2014	ABB SA		 ABB AS Color Engr Medium Voltage Products	SPARE TERMINALS	S =AA1J00		
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.				+J03 / X01		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB						
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:	INCOMER	242658248	1VB2383223	Sheet 34 75Sh.

1		2		3		4		5		6		7		8	
1		BAUGRUPPE SUBASSEMBLY		MAKE : ABB-STOTZ ORDER-NO: 1SNA 114 533 R0000		TYPE:EBS/A									
2		BESTUECKUNG COMPONENT PARTS : WIDERSTAND		ANZAHL PIECES : 1											
1		ABSCHLUSSPLATTE WIDERSTAND		33 KOHM / 4W		VITROHM 591-0									
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION							
						+.NS		-R200							
1		FELDLITGERAET BAY CONTROL UNIT		MAKE : ABB OY ORDER-NO: TBFNABADAAAZECNBXC		TYPE:REF630 1.2		REFERENCE TO GROUP DESCRIPTION IN SCHEMATIC DIAGRAM REFERENCE TO SEPARATE DATA WITH DESCRIPTION DETAILS REFERENCE TO DESCRIPTION FOR APPLIANCE CONNECTIONS							
1		41BI 36B0		4CT 1CT0,1/0,5A 4VT ETH.(LC) + SGF(ST)		AUFBAU 6U,DISPL. EXT									
1		41BI 36B0		4CT 1CT0,1/0,5A 4VT ETH.(LC) + SGF(ST)		BUILT-ON 6U, D. EXT									
1		GRUNDEINHEIT BASIC UNIT		PS 110-250=/100-240~1/0 BOARD		: BIO 15-221V DC									
1		STECKERSATZ		F. 4I+5U/5I+4U (6U)		2RCA021736									
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION							
						+.NS		-A200							
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/M01.3		/M01.3		/M01.3		/M01.3		/M01.3		/M01.3		/M01.3		/M01.3	
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/Q02.2		/Q02.2		/Q02.2		/Q02.2		/Q02.2		/Q02.2		/Q02.2		/Q02.2	
/Q03.1		/Q03.1		/Q03.1		/Q03.1		/Q03.1		/Q03.1		/Q03.1		/Q03.1	
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1		UEBERWACHUNG SUPERVISION		MAKE : ABB OY ORDER-NO:		TYPE:HMI X									
1		BEDIENEINHEIT FUER DISPLAY FOR REF 630		REF630											
1		HILFSSPANNUNG AUX.VOLTAGE													
1		TECHNICAL COMPONENT		TYPE		ORDER-NO.									
1															
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION							
						+.NT		-A210							
3		KAP.SPG.ANZEIGE CAP.VOLT.INDICATION		MAKE : ABB Color Emag ORDER-NO:		TYPE: X									
3		KAP. ABGRIFF CAP. TAP		KABELSTECKBUCHSE CABLE PLUG-IN											
3		SYSTEM SYSTEM													
3		TECHNICAL COMPONENT		TYPE		ORDER-NO.									
3															
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION							
						+.WR		-C1L1							
						+.WR		-C1L2							
						+.WR		-C1L3							
1		Date		21.10.2014		ABB SA		LIST OF ITEMS		S =AA1J00					
AS BUILT		19.01.15		RODI Drawn		ROEDEL		T.P. AEOLIAN DYNAMICS LTD.		+J03 / Z01					
REVISION		08.06.15		WI Check		HAMPEL		24kV SWITCHGEAR ZX1.2 SBB		242658248		1VB2383223		Sheet 35	
Revision		Date		Name		Norm		Derived:		Repl.by:		INCOMER		75Sh.	

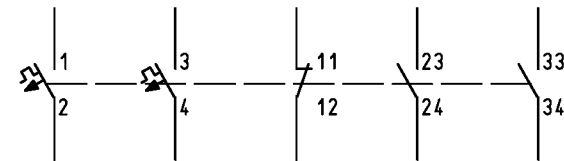
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RUPLAN 4.61

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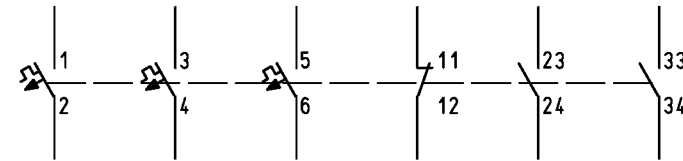
4	SICHERUNGSAUTOMAT		MAKE	: ABB-STOTZ	TYPE: S282 UC-K1
	MINIATURE C.B.		ORDER-NO:	GHS 282 0164 R0217	
	2-POLIG				
	TWO-POLE				
1	NENNSTROM		HILFSSCHALTER		
	RATED CURRENT		: 1A	AUXILIARY SWITCH	: 2S/10E
	TECHNICAL COMPONENT		TYPE		ORDER-NO.
	HILFSSCHALTER		2S/10E	S2-H21	GHS 270 1936 R0001



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+ .NS	-F101
		+ .NS	-F106
		+ .NS	-F116
		+ .NS	-F119

/G01.2	/G01.2	/Q02.4	/Q02.3	/Q03.2												
/G01.3	/G01.3	/Q02.4	/Q02.3													
/G01.3	/G01.4															
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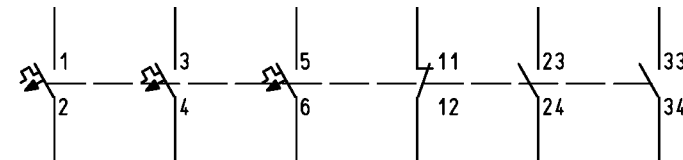
3	SICHERUNGSAUTOMAT MINIATURE C.B.		MAKE : ABB-STOTZ ORDER-NO: GHS 283 0164 R0377		TYPE:S283 UC-K6	
	3-POLIG THREE-POLE					
	NENNSTROM RATED CURRENT		: 6A		HILFSSCHALTER AUXILIARY SWITCH	
					: 2S/10E	
1	TECHNICAL COMPONENT		TYPE		ORDER-NO.	
	HILFSSCHALTER		2S/10E		S2-H21 GHS 270 1936 R0001	



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+ .NS	-F405
		+ .NS	-F406
		+ .NS	-F407

/S03.1	/S03.2	/S03.3		/Q02.5												
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/S04.1	/S04.2	/S04.3		/Q02.5												

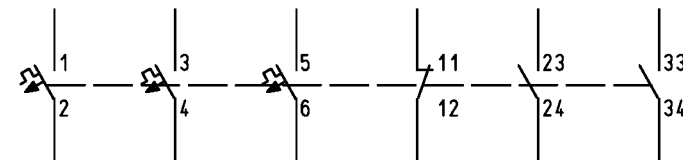
2	SICHERUNGSAUTOMAT		MAKE	: ABB-STQTZ	TYPE:S283 UC-K0,5
	MINIATURE C.B.		ORDER-NO:	GHS 283 0164 R0157	
	3-POLIG				
	THREE-POLE				
1	NENNSTROM		HILFSSCHALTER		
	RATED CURRENT		: 0,5A	AUXILIARY SWITCH	
	TECHNICAL COMPONENT		TYPE		ORDER-NO.
	HILFSSCHALTER		2S/10E	S2-H21	GHS 270 1936 R0001



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+ .NS	-F410
		+ .NS	-F411

/S04.4	/S04.4	/S04.5		/Q02.5												
/S04.6	/S04.7	/S04.8		/Q02.5												

1	SICHERUNGSAUTOMAT MINIATURE C.B.		MAKE : ABB-STOTZ ORDER-NO: GHS 283 0164 R0217	TYPE:S283 UC-K1
	3-POLIG THREE-POLE			
	NENNSTROM RATED CURRENT : 1A		HILFSSCHALTER AUXILIARY SWITCH : 2S/10E	
1	TECHNICAL COMPONENT		TYPE	ORDER-NO.
	HILFSSCHALTER 2S/10E		S2-H21	GHS 270 1936 R0001



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+ .NS	-F412

/S03.6	/S03.7	/S03.8		/Q02.5												
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DEPARTMENT AP

LIST OF ITEMS

INCOMER

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+J03 /Z03

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1VB2383223

Sheet 37

75Sh.

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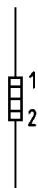
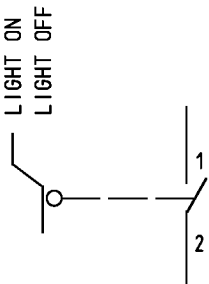
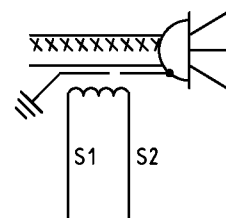
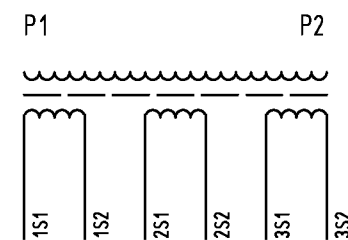
1		2		3		4		5		6		7		8					
2		ZEITRELAIS TIME-LAG-RELAY MULTIFUNKTIONAL MULTIFUNKTIONAL NENNSPANNUNG RATED VOLTAGE : 24-240V AC/DC EINSTELLBEREICH SETTING RANGE : 0,05s-300h TECHNICAL COMPONENT TYPE ORDER-NO.																	
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION											
						+.NS		-K201											
						+.NS		-K202											
4		SCHUETZ RELAY 50/60Hz 50/60Hz NENNSPANNUNG RATED VOLTAGE : 48-130V~/48-130V= KONTAKTE CONTACTS : 2S/20E TECHNICAL COMPONENT TYPE ORDER-NO.																	
1		ABDECKUNG F. 1-ETAGEN SCHUETZE BX4 1SBN 110 108 T1000																	
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION											
						+.NS		-K301											
						+.NS		-K302											
						+.NS		-K401											
						+.NS		-K402											
1		HILFSSCHUETZ AUXILIARY RELAY MAKE : ABB-STOTZ ORDER-NO: GJH 121 3001 R0404 TYPE: KC6-40E NENNSPANNUNG RATED VOLTAGE : 110-125V DC KONTAKTE CONTACTS : 4S TECHNICAL COMPONENT TYPE ORDER-NO.																	
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION											
						+.NS		-K101											
2		HILFSSCHUETZ AUXILIARY RELAY MAKE : ABB-STOTZ ORDER-NO: GJH 121 1001 R8220 TYPE: K6-22Z 40...450Hz 40...450Hz NENNSPANNUNG RATED VOLTAGE : 220-240V AC KONTAKTE CONTACTS : 2S/20E TECHNICAL COMPONENT TYPE ORDER-NO.																	
		APPLICATION		SETTING RANGE		LOCATION		ITEM-DESIGNATION											
						+.NS		-K201A											
						+.NS		-K202A											
01		AS BUILT		19.01.15		RODI		Date		21.10.2014		ABB SA		T.P. AEOLIAN DYNAMICS LTD.					
02		REVISION		08.06.15		WI		Check		HAMPEL		24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP					
Index		Revision		Date		Name		Norm		Derived:		Replce:		Repl.by:					
1		2		3		4		5		6		7		8					

The diagram shows a 4x4 network. On the left, there is a 2x2 sub-network represented by a rectangle. The top input is labeled A1 and the bottom input is labeled A2. The output of this sub-network is a single line that connects to the first input of a larger 4x4 network. This larger network has four inputs labeled 13, 21, 31, and 41, and four outputs labeled 14, 22, 32, and 42. The connections are as follows: the output of the 2x2 sub-network connects to input 13; input 14 connects to output 14; input 21 connects to output 22; input 31 connects to output 32; and input 41 connects to output 42.

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1	HEIZUNG HEATER OHNE BEGRENZUNG WITHOUT LIMITATION NENNSPANNUNG RATED VOLTAGE : 240V TECHNICAL COMPONENT				MAKE : RICA ORDER-NO: 1FZEK833B Pos.001 NENNLEISTUNG RATED POWER : 50W TYPE ORDER-NO.				TYPE:1FZEK X																						
	APPLICATION				SETTING RANGE				LOCATION				ITEM-DESIGNATION																		
									+.NS				-R121				/D01.6														
1	STEUERSCHALTER CONTROL SWITCH TUERKONTAKT ZELLENTUER DOOR-CONTACT PANEL-DOOR SCHALTELEMENT SWITCHING ELEMENT : TECHNICAL COMPONENT				MAKE : CONRAD ORDER-NO: 707201-77 SCHALTSTUFEN SWITCHING STEPS : TYPE ORDER-NO.				TYPE:C.3005CB X																						
	APPLICATION				SETTING RANGE				LOCATION				ITEM-DESIGNATION																		
									+.NS				-S125				/D01.6														
4	KABELUMBAUWANDLER CABLE CURRENT TRANSF IEC/2 IEC/2 PRIMAER PRIMARY : TECHNICAL COMPONENT				MAKE : ORDER-NO: TECHNISCHE DATEN SIEHE UEBERSICHTSSCHALTPLAN TECHNICAL DATA SEE SINGLE LINE SEKUNDAER SECONDARY : TYPE ORDER-NO.				TYPE: X																						
	APPLICATION				SETTING RANGE				LOCATION				ITEM-DESIGNATION																		
									+.KR				-BC4				/S01.3														
									+.KR				-BC5				/S01.3														
								+.KR				-BC6				/S01.4															
								+.KR				-BC90				/S01.6															
3	STROMWANDLER CURRENT TRANSFORMER 2/IEC/2/2/2 2/IEC/2/2/2 NENNSPANNUNG RATED VOLTAGE : TECHNICAL COMPONENT				MAKE : ORDER-NO: TECHNISCHE DATEN SIEHE UEBERSICHTSSCHALTPLAN TECHNICAL DATA SEE SINGLE LINE BETRIEBSSPANNUNG OPERATING VOLTAGE : TYPE ORDER-NO.				TYPE: X																						
	APPLICATION				SETTING RANGE				LOCATION				ITEM-DESIGNATION																		
									+.WR				-BC1				/S02.1 /S02.4 /S02.6														
									+.WR				-BC2				/S02.2 /S02.5 /S02.7														
								+.WR				-BC3				/S02.3 /S02.5 /S02.8															
																</															

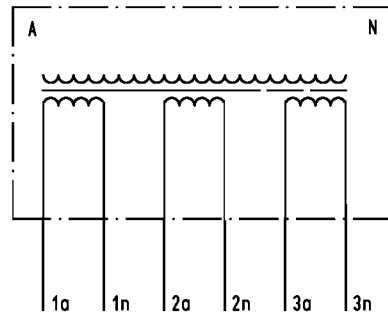
				Date	21.10.2014	ABB SA	 AG Color Emag Medium Voltage Products	LIST OF ITEMS	S =AA1J00			
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	T.P. AEOLIAN DYNAMICS LTD.						
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB			DEPARTMENT AP	242658248	1VB2383223	Sheet 41
Index	Revision	Date	Name	Norm		Derived:			Replace:	Repl.by:	INCOMER	75Sh.

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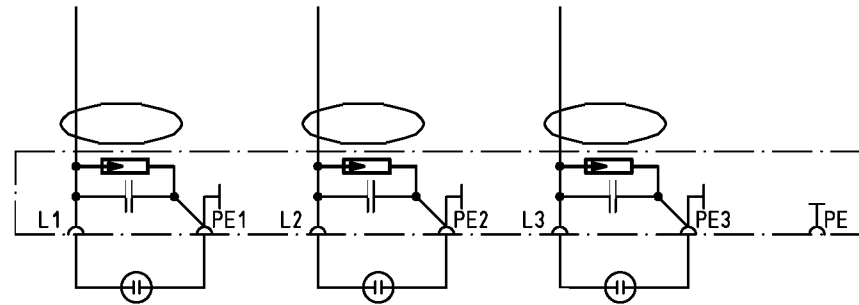
3 SPANNUNGSWANDLER VOLTAGE TRANSFORMER MAKE : ORDER-NO: TYPE: X
1/IEC/2/2/2 1/IEC/2/2/2 TECHN. DATEN SIEHE UEBERSICHTSSCHALTPLAN
NENNSPANNUNG BETRIEBSSPANNUNG
RATED VOLTAGE : OPERATING VOLTAGE :
TECHNICAL COMPONENT TYPE ORDER-NO.



APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.WR	-BA1
		+.WR	-BA2
		+.WR	-BA3

/S03.1	/S03.4	/S04.1													
/S03.2	/S03.4	/S04.2													
/S03.3	/S03.5	/S04.3													

1 KAP.SPG.ANZEIGE CAP.VOLT.INDICATION MAKE : JORDAN TYPE: X
3-POLIG 3-POLE ORDER-NO:
SYSTEM : LRM : FUER ZX
TECHNICAL COMPONENT TYPE ORDER-NO.



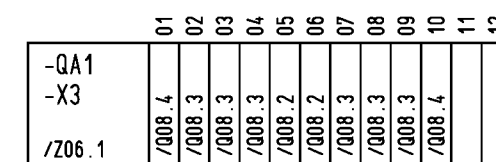
APPLICATION	SETTING RANGE	LOCATION	ITEM-DESIGNATION
		+.ZR	-X123

/S06.4	/S06.4	/S06.4	/S06.5	/S06.5	/S06.6	/S06.6									
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01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	21.10.2014	ABB SA
02	REVISION	08.06.15	WI	Check	HAMPEL		T.P. AEOLIAN DYNAMICS LTD.
Index	Revision	Date	Name	Norm	Derived:	Replace:	

ABB AG Color Emag Medium Voltage Products	
DEPARTMENT	AP
Repl.by:	

LIST OF ITEMS	S =AA1J00	
	+J03	/Z08
INCOMER	242658248	1VB2383223
		Sheet 42
		75Sh.



1	2	Δ 3	4	Δ 5	6	7	8
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WIRTEK 4.61

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-QB1/QC1	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
-X1	/L01.4	/L01.4	/L01.5	/L01.5	/L01.4	/L01.4	/L01.5	/L01.5	/L02.6	/L02.6	/L01.4	/L01.4			/L01.6	/L01.6	/L01.6	/L01.6	/L01.6	/L01.7	/L01.7	/L01.7		
/Z06.1																								

01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA T.P. AEOLIAN DYNAMICS LTD.	DEPARTMENT	AP	LIST OF ITEMS THREE-POSITION SWITCH	S	=AA1J00		
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB	Repl.by:			+J03	/	ZZ02	
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	INCOMER	242658248	1VB2383223		Sheet 44	75Sh.

Timing diagram for PSM_102 showing signals B01_P0_TCS, B02_P0_TCS, B03_P0_TCS, B04_P0, B05_P0, B06_P0, B07_S0, B08_S0, and B09_S0. The diagram shows the relationship between these signals and a clock source. B01_P0_TCS, B02_P0_TCS, and B03_P0_TCS are active-low signals with a TCS (Time-to-Close) delay. B04_P0, B05_P0, B06_P0, B07_S0, B08_S0, and B09_S0 are active-low signals. The clock source is shown as a series of pulses.

Figure 1: Connection diagram of the power supply unit. The diagram shows a vertical bus with various connection points. On the left, there are three groups of terminals: a 3-pin group (01, 02, 03), a 4-pin group (01, 03, 02, 01), and a 4-pin group (01, 02, 03, 04). On the right, there are labels for 'TX', 'RX', 'FIBER OPTIC', 'FIBER OPTIC', 'GND', 'NC', 'IRIG-B-', 'IRIG-B+', 'LAN 1', and 'LAN/HMI'. A power supply unit is shown connected to the bus, with inputs for '~ / -', '110-240VAC', and '48-125VDC'. The unit has a transformer symbol and a ground connection.

ABB AG Color Enag
Medium Voltage Products

DEPARTMENT	AP
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INCOMER

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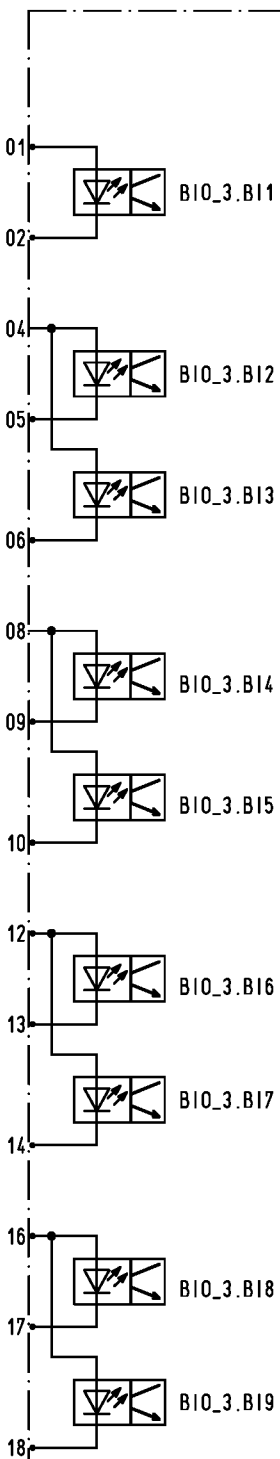
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		+J03	/ ZZ05
242658248	1VB2383223		Sheet 46
			75Sh.

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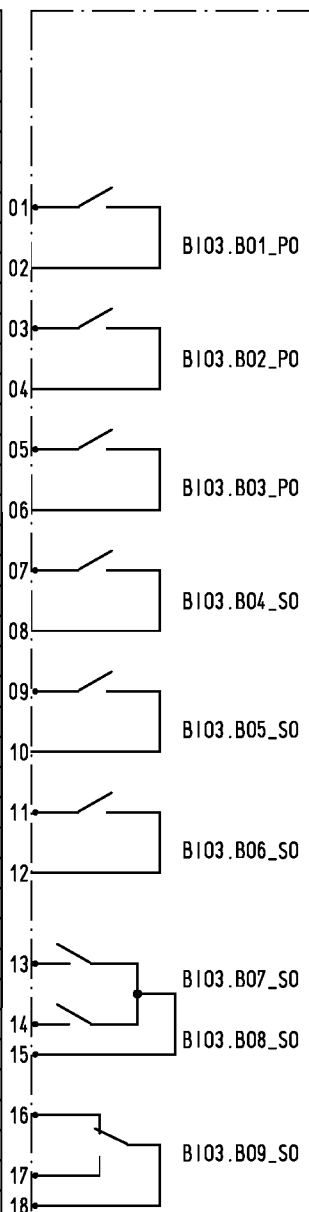
09.06.2015 242658248
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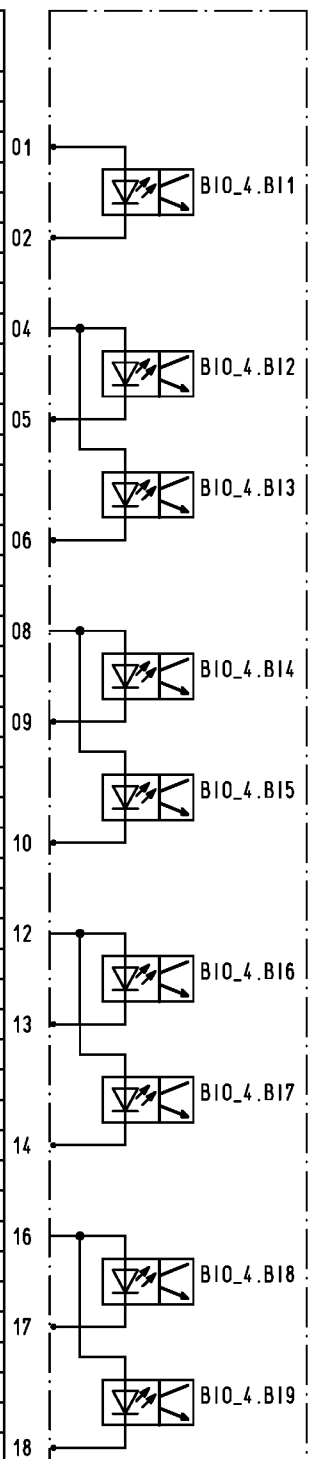
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X324	/Q03.2	
MEANING		
MCB TRIPPED		
MOTOR VOLTAGE		
SPARE		
SPARE		
SPARE		
REEDCONTACTS		
EARTHING SWITCH CLOSED		
DC VOLT		
FAIL		
SPARE		
RELEASE		
ISOLATOR -QB1		
MAIN INCOMING 1		
DC-PRESENT		
SPARE		



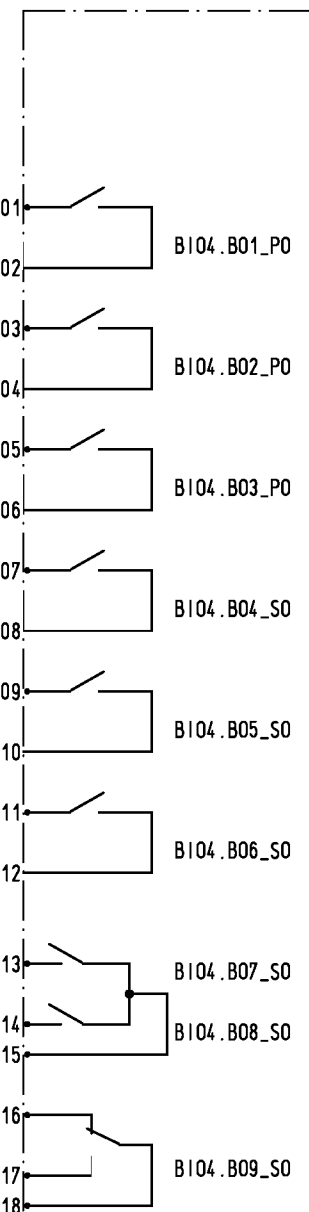
LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X321	/Q03.2	
MEANING		
COMMAND		
ISOLATOR OPEN/EARTHING SWITCH CLOSED		
COMMAND		
ISOLATOR CLOSE/EARTHING SWITCH OPEN		
COMMAND		
DISCONNECTOR CONTROL/OPEN		
SPARE		
RELEASE FOR		
EARTHING SWITCH		
RELEASE FOR		
ISOLATOR		
SPARE		
SPARE		
SPARE		



LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X329	/Q06.6	
MEANING		
PROTECTION		
TRIPPING		
SPARE		
SELF MONITORING		
REF630		
PRESSURE RELIEF		
TRIP		
MAIN INCOMING 1		
AC-PRESENT		
INCOMING 2		
AC-PRESENT		



LOCATION	ITEM-REFERENCE	STATUS
ITEM-DES. -A200	/Z01.1	REFERENCE
TERMINAL -X326	/Q07.2	
MEANING		
COMMAND		
CB-DN		
SPARE		
SPARE		

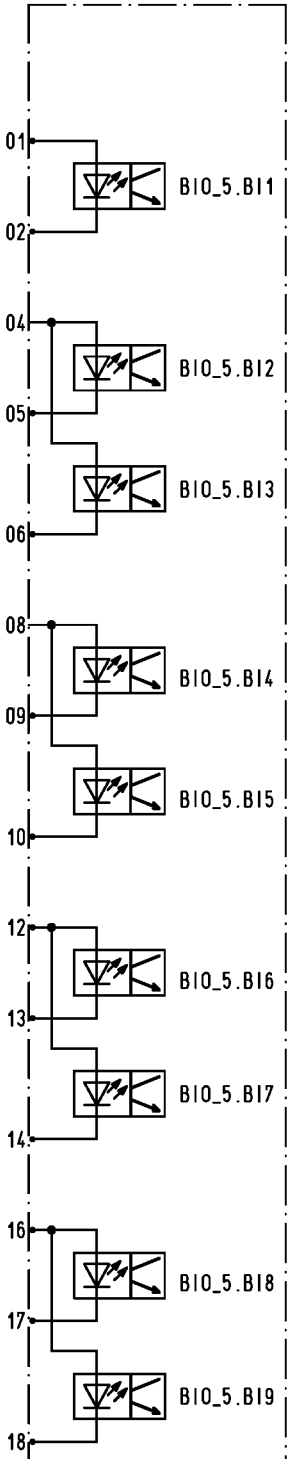


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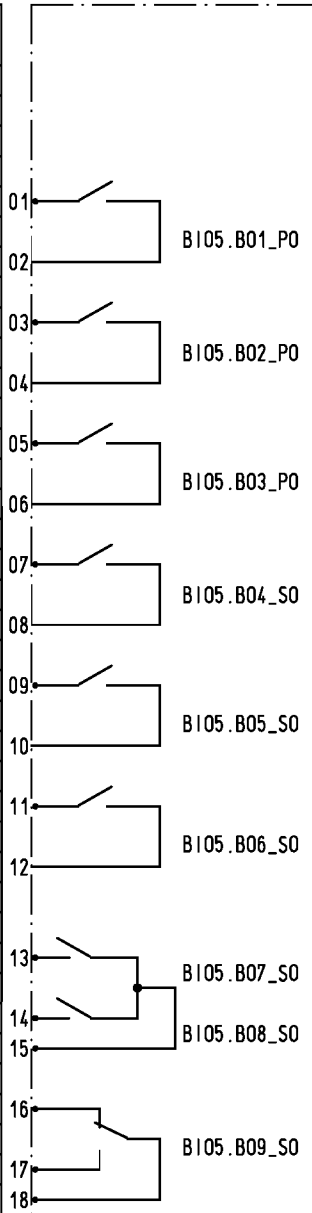
WIRTEKAN 4.61

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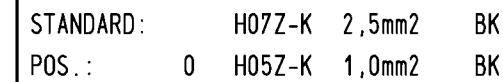
LOCATION		STATUS	ITEM-REFERENCE
ITEM-DES.	-A200		/Z01.1
TERMINAL	-X334		REFERENCE
MEANING			/Q03.4
RELEASE EARTHING SWITCH -QC1			/Q03.4
			/Q03.4
			/Q04.6
INCOMING 2 DC-PRESENT			/Q04.6
			/Q04.6
			/Q04.6
LOSS OF MAIN TRIP			/Q04.6
			/Q04.6
			/Q06.4
MAIN INCOMING 1 DC-PRESENT			/Q06.4
			/Q06.4
			/Q06.3
INCOMING 2 DC-PRESENT			/Q04.4
			/Q04.4
			/Q04.4
			/Q06.5
			/Q06.5
			/Q06.5



LOCATION		STATUS	ITEM-REFERENCE
ITEM-DES.	-A200		/Z01.1
TERMINAL	-X331		REFERENCE
MEANING			/Q07.6
SPARE			/Q07.6
			/Q07.6
			/Q07.5
SPARE			/Q07.5
			/Q07.5
			/Q07.4
SPARE			/Q07.4
			/Q07.4
			/P01.3
SPARE			/P01.3
			/P01.3
			/P01.6
			/P01.6
			/P01.6



[illegible]



STANDARD: WEIDM. ZRV2,5/4

75Sh.

[illegible]

[illegible]

1	Circuit Diagram	2	3	4	5	6	7	8																																		
Installation	Wiring Material	Ø Colour	Terminal Type																																							
Standard: H05Z-K 1,0mm2 BK	Standard: WEIDM. ZRV2,5/4																																									
Reference	Line	Pot	q	A1	Line	Pot	q	B1	Line	Pot	q	C1	Line	Pot	q	D1	D2	C2	B2	A2	D1	C1	B1	A1	Nr	D2	q	Pot	Line	C2	q	Pot	Line	B2	q	Pot	Line	A2	q	Pot	Line	
/Q01.2	116L+			-QA1 -X1 11	116L+			-QA1 -X1 09	116L+			-QA1 -X1 07														17	-QB1/QC1 -X1 17	116L+			-QB1/QC1 -X1 15	116L+			-QB1/QC1 -X1 03	116L+			-QB1/QC1 -X1 05	116L+		
/Q01.8																									18					-K1 11	116L+											
/G01.3	116L+	q		-F116 1																					19	-XC 15	116L+			-XC 33	116L+					-A200 -X410 03	116L+					
/M01.4																									20	-K201A 13	q	116L+							-A200 -X307 15	116L+						
/Q02.3																									21										-F101 11	q	116L+					
/Q01.5																									22	-QB1/QC1 -X1 20			-QB1/QC1 -X1 19			-QB1/QC1 -X1 07			-QB1/QC1 -X1 08							
/Q06.6																									23								-XP2 4C1	116L-		-XP2 3A1	116L-					
/G01.4	116L-	q		-F116 3																					24	-A200 -X324 08	116L-		-A200 -X304 01	116L-		-A200 -X304 06	116L-		-A200 -X410 01	116L-						
/M01.4																									25					-A200 -X334 16	116L-		-A200 -X334 04	116L-		-XP1 33B2	116L-					
/Q04.5	116L-			-X11 16C1	116L-			-X11 16	116L-			-X11 16 A1	116L-												26					-A200 -X304 11	116L-		-A200 -X334 08	116L-		-A200 -X304 15	116L-					
/Q02.7	116L+			-B0G 4	116L+			-B1G 4																	27	-XP2 4 A1	116L-		-XP2 3D1	116L-												
/Q02.6				-B0G 6				-B1G 6				-XC 43													28					-A200 -X304 18			-A200 -X304 17			-A200 -X304 16						
/Q01.7				-QB1/QC1 -X1 22	116L+			-QB1/QC1 -X1 21				-XC 38													29	-A200 -X334 18			-A200 -X334 17						-A200 -X324 09							
/Q02.6	116L+			-QB1/QC1 -X1 09																					30	-A200 -X324 10							-A200 -X304 13									
/Q02.6				-QB1/QC1 -X1 10																					31																	
/Q02.4				-F119 12				-F119 23				-F412 24													32					-A200 -X304 14			-A200 -X304 05			-A200 -X304 12						
09.06.2015 242658248 =AA1J00+J03 007 APE1	RUPLAN 4.61																																									
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	21.10.2014	ABB SA	T.P. AEOLIAN	DYNAMICS LTD.	24kV SWITCHGEAR ZX1.2 SBB	DEPARTMENT	AP	X-PLUG	-XP1	V	=AA1J00	+J03	/007	242658248	1VB2383223	Sheet 55	75Sh.																				
02	REVISION	08.06.15	WI	Check	HAMPEL																																					
Index	Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	INCOMER																																		

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1		2				3		4		5		6		7		8														
NO	CABLE	USED CORES				SHEET\COMPL.		CABLE TYPE Ø		CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE														
1	W14					1	4	XLPE/PVC/SWA/PVC 4x4mm ²			/017	-X9		STANDARD: H07Z-K 1,5mm2 BK		STANDARD: WEIDM. ZDU2,5N/4AN														
2	W13					3	3	XLPE/PVC/SWA/PVC 4x4mm ²																						
ASSEMBLY																														
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS				POT	NO.	DESTINATIONS				POS.	CORES/ LINE	REMARK			
/S03.4														99		-BA1				2a	A L1	11 B	-F406				2	Q	99	
/S03.4														99		-BA2				2a	C	D	-F406				4	Q	99	
/S03.5														99		-BA3				2a	A L2	12 B	-F406				6	Q	99	
/S03.4														87		-X9				PE2	C	D	-BA1				2n		99	
/S03.4														99		-BA2				2n	A L3	13 B								
/S03.5	<-															BU					C	D	-BA3				2n		99	
/S03.4																DQ&PR + -X-				---	A N	14 B	-F406				1			
/S03.4	<-															BN					C	D	-F406				3			
/S03.4																BK					A 406L1	17 B	-F406				5			
/S03.5	<-															GY					C	D								

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1		2		3		4		5		6		7		8										
NO	CABLE	USED CORES	SHEET/COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE												
1	W15		4	4	XLPE/PVC/SWA/PVC 4x4mm²					STANDARD: H07Z-K 1,5mm2 BK POS.: 87 H07Z-K 2,5mm2 GN/YE 99 NSHXAFO 1,5mm2 BK		STANDARD: WEIDM. ZDU2,5N/4AN												
									-X9															
REFERENCE		NO.												CORES/ LINE	POS.	DESTINATIONS		POT	NO.	DESTINATIONS		POS.	CORES/ LINE	REMARK
/S04.1		1	2	3	4	5	6	7	8	9	10	11	12	99		-BA1	3a	A L1	21 B	-F407		2	99	
/S04.2														99		-BA2	3a	C	D					
/S04.3														99		-BA3	3a	A L2	22 B	-F407		4	99	
/S04.1														99		-BA1	3n	C	D					
/S04.2																		A L3	23 B	-F407		6	99	
/S04.3														99		-BA3	3n	C	D					
/S04.3																		A UN	24 B	-X9		PE3	87	
/S04.5																-A200	-X102	06	25 B	-BA2		3n	99	
/S04.8		<-												BU		VESTAS + -X-	---	C	D					
/S04.4																-A200	-X102	03	26 B					
/S04.4																-K100		L1	D					
/S04.4																-A200	-X102	05	27 B	-X11		5D2		
/S04.5																-K100		L2	D	-A200		-X102	08	
/S04.6		<-												BN		VESTAS + -X-	---	C	D	-K100		N		
/S04.7		<-												BK		VESTAS + -X-	---	A 410L1	31 B					
/S04.8		<-												GY		VESTAS + -X-	---	C	D					
																		A 410L2	32 B	-F410		1		
																		C	D	-F410		3		
																		A 410L3	33 B	-F410		5		
																		C	D	-F411		1		
																		A 411L1	41 B	-F411		3		
																		C	D	-F411		5		
																		A 411L2	42 B					
																		C	D					
																		A 411L3	43 B					
																		C	D					

Date 21.10.2014		ABB SA		T.P. AEOLIAN DYNAMICS LTD.		ABB		Terminal Diagram -X9		V =AA1J00	
01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		+J03 /016	
02	REVISION	08.06.15	WI	Check	HAMPEL	Derived:		Repl.by:		Sheet 64	
Index	Revision	Date	Name	Norm		Repl.		INCOMER		75Sh.	

1		2		3		4		5		6		7		8	
NO	CABLE \ USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM	-X9				WIRING MATERIAL		TERMINAL TYPE	
1	W14	3	4	XLPE/PVC/SWA/PVC 4x4mm²								STANDARD: H07Z-K 1,5mm2 BK		STANDARD: WEIDM. ZDU2,5N/4AN	
												POS.: 87 H07Z-K 2,5mm2 GN/YE		PE1-PE3 PH0E. USLK66N	

09.06.2015 242658248
=AA1J00+J03
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1		2		3		4		5		6		7		8							
NO	CABLE	USED CORES	SHEET\COMPL.	CABLE TYPE	Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE									
								-XC		STANDARD: H05Z-K 1,0mm2 BK		STANDARD: WDU1,5/BLZ/5.08/ZQV									
REFERENCE	NO.												CORES/ LINE	POS.	DESTINATIONS	POT	NO.	DESTINATIONS	POS.	CORES/ LINE	REMARK
	1	2	3	4	5	6	7	8	9	10	11	12									
/Q08.3																	69	-QA1	-X2	09	
/Q08.4																	70	-QA1	-X3	10	
/Q08.4																	71	-QA1	-X2	01	
/S04.A.3																	72	-K100		15	
/S04.A.3																	73	-K100		18	
/S04.A.4																	74	-K100		25	
/Q09.2																	75	-F131A		21	
/Q09.2																	76	-F131B		13	
/Q09.3																	77	-F131B		21	
/Q09.3																	78	-F132A		13	
/Q09.3																	79	-F132A		21	
/Q09.4																	80	-F133A		13	
/Q09.4																	81	-F133A		21	
/Q09.5																	82	-F134A		23	
/Q09.5																	83	-F134A		11	
/Q09.6																	84	-F134B		23	
/Q09.6																	85	-F134B		11	
/Q09.7																	86	-F135A		23	
/Q09.7																	87	-F135A		11	
/Q09.8																	88	-F135B		23	
/Q09.8																	89	-F135B		11	
/Q09.1																	90	-F131A		14	
/Q09.2																	91	-F131A		22	
/Q09.2																	92	-F131B		14	
/Q09.3																	93	-F131B		22	
/Q09.3																	94	-F132A		14	
/Q09.3																	95	-F132A		22	
/Q09.4																	96	-F133A		14	
/Q09.4																	97	-F133A		22	
/Q09.5																	98	-F134A		24	
/Q09.5																	99	-F134A		12	
/Q09.6																	100	-F134B		24	
/Q09.6																	101	-F134B		12	
/Q09.7																	102	-F135A		24	

01	AS BUILT	19.01.15	RODI	Drawn	ROEDEL	ABB SA	T.P. AEOLIAN DYNAMICS LTD.		ABB AG Color Emag Medium Voltage Products	TERMINAL DIAGRAM -XC		V =AA1J00	
02	REVISION	08.06.15	WI	Check	HAMPEL	24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		INCOMER		+J03 /020	
Index	Revision	Date	Name	Norm		Derived:	Replace:	Repl.by:			242658248	1VB2383223	Sheet 68 75Sh.

09.06.2015 242658248
=AA1J00+J03
021 APE1
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NO

CABLE

USED CORES

SHEET\COMPL.

CABLE TYPE

Ø

CORES

CAB. ROUTE

SCHEMATIC DIAGRAM

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=AA1J00+J03
022
APE1

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1		2		3		4		5		6		7		8												
NO	CABLE	USED CORES		SHEET\COMPL.		CABLE TYPE		Ø	CORES	CAB. ROUTE	SCHEMATIC DIAGRAM		WIRING MATERIAL		TERMINAL TYPE											
1	W03			2	2	XLPE/PVC/SWA/PVC 2x4mm ²					-XC1		STANDARD: H07Z-K 2,5mm2 BK		STANDARD: WEIDM. WDU6											
2	W01			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
3	W02			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
4	W04			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
5	W05			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
6	W07			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
7	W06			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
8	W08			2	2	XLPE/PVC/SWA/PVC 2x4mm ²																				
REFERENCE		NO.												CORES/ LINE	POS.	DESTINATIONS		POT	NO.	DESTINATIONS		POS.	CORES/ LINE	REMARK		
		1	2	3	4	5	6	7	8	9	10	11	12													
/D02.3		<-												BN		1NE00	+NE0	-X-	1L	1L		1	-F131A	2		
/D02.1			<-											BN		1NE00	+NE0	-X-	2L	2L		2	-F132A	2		
/D02.2				<-										BN		1NE00	+NE0	-X-	3L	3L		3	-F133A	2		
/D02.3		<-												GY		1NE00	+NE0	-X-	1N	1N		4	-F131A	4N		
/D02.2			<-											GY		1NE00	+NE0	-X-	2N	2N		5	-F132A	4N		
/D02.2				<-										GY		1NE00	+NE0	-X-	3N	3N		6	-F133A	4N		
/D02.7					<-									BN		=CH1	+	-X-	1L	1L		7	-F131B	2		
/D02.7					<-									BY		=CH1	+	-X-	1N	1N		8	-F131B	4N		
/G02.2						<-								BN		=CH1	+	-X-	1L+	1L+		9	-F134A	2	Q	
/G03.2							<-							BN		=CH1	+	-X-	1L+	1L+		10	-F135A	2		
/G02.2						<-								GY		=CH1	+	-X-	1L-	1L-		11	-F134A	4	Q	
/G03.2							<-							GY		=CH1	+	-X-	1L-	1L-		12	-F135A	4		
/G02.7								<-						BN		=CH1	+	-X-	1L+	1L•		13	-F134B	2		
/G03.7									<-					BN		=CH1	+	-X-	1L+	1L•		14	-F135B	2		
/G02.7							<-							GY		=CH1	+	-X-	1L-	1L-		15	-F134B	4		
/G03.7								<-						GY		=CH1	+	-X-	1L-	1L-		16	-F135B	4		

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=AA1J00+J03
022
APE 1

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01		AS BUILT	19.01.15	RODI	Drawn	ROEDEL	Date 21.10.2014		ABB SA		T.P. AEOLIAN DYNAMICS LTD.		24kV SWITCHGEAR ZX1.2 SBB		DEPARTMENT AP		TERMINAL DIAGRAM -XC		V =AA1J00		+J03 /022	
02		REVISION	08.06.15	WI	Check	HAMPEL	Date		Derived:		Replace:		Repl.by:		INCOMER		242658248		1VB2383223		Sheet 70	
Index		Revision	Date	Name	Norm																75Sh.	

CROSS CONNECTION											
BACK REFERENCE		POTENTIAL POSITION CROSS SECTION	WIRING MATERIAL COLOR	START TERMINAL	DESTINATION 1	DESTINATION 2	DESTINATION 3	DESTINATION 4	DESTINATION 5	DESTINATION 6	
D01		1L 83 2,5mm2	H07Z-K BK	-X10 5 C1	-K201 18	-K201A 21					
D01		1N 83 2,5mm2	H07Z-K BK	-X10 6 C1	-K201 24/28	-K201A 31					
D02		131AL 83 2,5mm2	H07Z-K BK		-F131A 1	-K201 A1	-K201 15/17	-K201A A1			
D02		131AN 83 2,5mm2	H07Z-K BK		-F131A 3N	-K201 A2	-K201 21/25/27	-K201A A2			
D02		131BL 83 2,5mm2	H07Z-K BK		-F131B 1	-K202 A1	-K202 15/17	-K202A A1			
D02		131BN 83 2,5mm2	H07Z-K BK		-F131B 3N	-K202 A2	-K202 21/25/27	-K202A A2			
D02		83 2,5mm2	H07Z-K BK		-K201A 22	-K202 18					
D02		83 2,5mm2	H07Z-K BK		-K201A 32	-K202 24/28					
G01		88 4,0mm2	H07Z-K GN/YE		-F201 PE	PE PE					
G01		88 4,0mm2	H07Z-K GN/YE		-F202 PE	PE PE					
G01		1L+ 83 2,5mm2	H07Z-K BK	-X10 1 A1	-F101 2	-F106 2	-F116 2	-F201 L			
G01		1L+ 83 2,5mm2	H07Z-K BK	-X10 1 C1	-K301 1	-K301 R3					
G01		1L- 83 2,5mm2	H07Z-K BK	-X10 2 A1	-F101 4	-F106 4	-F116 4	-F201 N			
G01		1L- 83 2,5mm2	H07Z-K BK	-X10 2 C1	-K301 7	-K301 R5					
G01		2L+ 83 2,5mm2	H07Z-K BK	-X10 3 A1	-F119 2	-F202 L					
G01		2L+ 83 2,5mm2	H07Z-K BK	-X10 3 C1	-K401 1	-K401 R3					
G01		2L- 83 2,5mm2	H07Z-K BK	-X10 4 A1	-F119 4	-F202 N					
G01		2L- 83 2,5mm2	H07Z-K BK	-X10 4 C1	-K401 7	-K401 R5					
G01		116L- 0 1,0mm2	H05Z-K BK	-XP1 24 A1	-F116 3	-K101 A2					
G02		134AL+ 83 2,5mm2	H07Z-K BK		-F134A 1	-K301 2	-K301 A1	-K301A A1			
		Date	21.10.2014	ABB SA		TABLE OF CROSS CONNECTION		V =AA1J00			
01	AS BUILT	19.01.15	R0DI	Drawn	ROEDEL	DEPARTMENT AP		+J03 /Q01			
02	REVISION	08.06.15	W1	Check	HAMPEL			242658248			
Index		Revision	Date	Name	Norm	Derived:	Replace:	Repl.by:	INCOMER	1VB2383223	
1		2		3		4		5		6	

1		2		3		4		5		6		7		8									
CROSS CONNECTION																							
BACK REFERENCE		POTENTIAL POSITION CROSS SECTION		WIRING MATERIAL COLOR		START TERMINAL		DESTINATION 1		DESTINATION 2		DESTINATION 3		DESTINATION 4		DESTINATION 5		DESTINATION 6					
G02		134AL-83 2,5mm2		H07Z-K BK				-F134A 3		-K301 8		-K301 A2		-K301A A2									
G02		134BL+83 2,5mm2		H07Z-K BK				-F134B 1		-K302 2		-K302 A1		-K302A A1									
G02		134BL-83 2,5mm2		H07Z-K BK				-F134B 3		-K302 8		-K302 A2		-K302A A2									
G02		83 2,5mm2		H07Z-K BK				-K301 R4		-K302 1													
G02		83 2,5mm2		H07Z-K BK				-K301 R6		-K302 7													
G02		1L+83 2,5mm2		H07Z-K BK		-XC1 9		-F134A 2		-K100 A1													
G02		1L-83 2,5mm2		H07Z-K BK		-XC1 11		-F134A 4		-K100 A2													
G03		135AL+83 2,5mm2		H07Z-K BK				-F135A 1		-K401 2		-K401 A1		-K401A A1									
G03		135AL-83 2,5mm2		H07Z-K BK				-F135A 3		-K401 8		-K401 A2		-K401A A2									
G03		135BL+83 2,5mm2		H07Z-K BK				-F135B 1		-K402 2		-K402 A1		-K402A A1									
G03		135BL-83 2,5mm2		H07Z-K BK				-F135B 3		-K402 8		-K402 A2		-K402A A2									
G03		0 1,0mm2		H05Z-K BK				-K401 R4		-K402 1													
G03		0 1,0mm2		H05Z-K BK				-K401 R6		-K402 7													
L01		0 1,0mm2		H05Z-K BK				-K11 2		-K11 4													
L01		0 1,0mm2		H05Z-K BK				-K11 3		-K11 5													
L01		0 1,0mm2		H05Z-K BK				-K11 62		-K12 A1													
L01		0 1,0mm2		H05Z-K BK				-K11 A1		-K12 62													
L01		0 1,0mm2		H05Z-K BK				-K12 2		-K12 4													
L01		0 1,0mm2		H05Z-K BK				-K12 3		-K12 5													
L01		101L-0 1,0mm2		H05Z-K BK		-XD 1 A1		-K11 21		-K12 22													
				Date		21.10.2014		ABB SA		TABLE OF CROSS CONNECTION						V =AA1J00							
01		AS BUILT		19.01.15		RODI		Drawn ROEDEL		T.P. AEOLIAN DYNAMICS LTD.						+J03 /Q02							
02		REVISION		08.06.15		WI		Check HAMPEL		24kV SWITCHGEAR ZX1.2 SBB													
Index		Revision		Date		Name		Norm		Derived:		Replace:		Repl.by:		INCOMER		242658248		1VB2383223		Sheet73	
1		2		3		4		5		6		7		8									

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T.P. AEOLIAN DYNAMICS LTD.
24kV SWITCHGEAR ZX1.2 SBB

1	2		3		4		5		6		7		8				
CROSS CONNECTION																	
BACK REFERENCE		POTENTIAL POSITION CROSS SECTION	WIRING MATERIAL COLOR	START TERMINAL		DESTINATION 1		DESTINATION 2		DESTINATION 3		DESTINATION 4		DESTINATION 5		DESTINATION 6	
L01		0 1,0mm2	H05Z-K BK	-XD 1 C1		-XP1 2 A2		-K11 6		-K11 22							
L01		0 1,0mm2	H05Z-K BK	-XD 1 D1		-XP1 3 B2		-K12 6		-K12 21							
L01		0 1,0mm2	H05Z-K BK	-XP1 7 A2		-K11 1		-K12 1									
L01		0 1,0mm2	H05Z-K BK	-XP1 12 C1		-K11 A2		-K12 A2									
Q02		PE 88 4,0mm2	H07Z-K GN/YE			-A210 PE		PE PE									
Q02		0 1,0mm2	H05Z-K BK			-F101 12		-F106 11									
Q02		0 1,0mm2	H05Z-K BK			-F101 23		-F106 24									
Q02		0 1,0mm2	H05Z-K BK			-F106 12		-F119 11									
Q02		0 1,0mm2	H05Z-K BK			-F106 23		-F119 24									
Q02		0 1,0mm2	H05Z-K BK			-F406 23		-F405 24									
Q02		0 1,0mm2	H05Z-K BK			-F407 23		-F406 24									
Q02		0 1,0mm2	H05Z-K BK			-F410 23		-F407 24									
Q02		0 1,0mm2	H05Z-K BK			-F411 23		-F410 24									
Q02		0 1,0mm2	H05Z-K BK			-F412 23		-F411 24									
Q02		116L+ 0 1,0mm2	H05Z-K BK	-XP1 21 A2		-F101 11		-F405 23		-F100 33		-F101 24					
Q03		0 1,0mm2	H05Z-K BK			-F100 34		-F101 33									
Q04		116L+ 0 1,0mm2	H05Z-K BK	-XP1 20 D2		-K201A 13		-K202A 13		-K301A 13		-K302A 13		-K401A 13		-K402A 13	
Q05		116L+ 0 1,0mm2	H05Z-K BK	-XP2 9 B1		-XP1 19 A1		-F116 1									
S03		L1 99 1,5mm2	NSHXAFÖ BK	-X9 11 B1		-F406 2		-F411 2		-F412 2							
S03		L2 99 1,5mm2	NSHXAFÖ BK	-X9 12 B1		-F406 4		-F411 4		-F412 4							
		Date	21.10.2014	ABB SA		T.P. AEOLIAN		DYNAMICS LTD.		TABLE OF CROSS CONNECTION		V =AA1J00					
01		AS BUILT	19.01.15	RODI	Drawn	ROEDEL							+J03		/Q03		
02		REVISION	08.06.15	WI	Check	HAMPEL									Sheet74		
Index		Revision	Date	Name	Norm	Derived:		Replace:		Repl.by:		INCOMER		242658248		1VB2383223	
1		2		3		4		5		6		7		8			

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Q03
APE1

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TABLE OF CROSS CONNECTION

ABB SA
T.P. AEOLIAN DYNAMICS LTD.
24kV SWITCHGEAR ZX1.2 SBB

1VB2383223

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